



Federal Office
for Information Security

Technical Guideline TR-03122-3

Conformance Test Specification for BSI TR-03121 Biometrics for Public Sector Applications

Part 3: Test Cases for Function Modules and Processes

Version 7.0



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1 Introduction

This document is part three of the Conformance Test Specification (TR-03122). It is the counterpart of TR-03121-3 describing the test cases for the defined Function Modules.

Not all Function Modules have test cases associated with them.

2 Test Cases Function Modules

The following sections define test cases for Function Modules.

2.1 Test Cases Acquisition Hardware

2.1.1 Test Cases FM AH-ALL-SSS

Test Case ID: TC-AH-ALL-SSS-001

Scope Examination of the surveillance camera systems of the self-service system (SSS) used for to obtain digitised facial images and fingerprints.

Preconditions

- The IUT is in operation, required modules are loaded.
- A surveillance camera system which supervises the area around the SSS is available.
- A surveillance camera system which supervises the fingerprint capture system of the SSS is available.
- The IUT implements a graphical user interface.
- The IUT implements a user interface of fingerprint acquisition shown to the biometric subject.
- The following test resources are necessary:
 - Two test subjects are available to perform the acquisition with.

CTS Mode

- interactive
- no provision of pre-defined input data
- HTTP method: GET
- test case variants:

/TR03122/TC-AH-ALL-SSS-001/1

Table 2.2 TC-AH-ALL-SSS-001 Test Case Variants

Description

Step	Description / Expected Result
1	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • Perform a bona fide fingerprint presentation on the fingerprint scanner of the IUT. • Interact with the graphical user interface of the IUT.

Test Case ID: TC-AH-ALL-SSS-001

Expected Result:

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- A fingerprint image is captured.
- The surveillance camera system captures an image of the surrounding area.
- The image of the surrounding area is made at the moment of the finger capture attempt (i.e. the timestamps of the corresponding images are less than 100ms apart).
- The image captured by the camera system surveying the environment has a resolution of at least 1280 x 720.
- The image of the surrounding area is a coloured image, in which the face of the biometric subject using the fingerprint scanner is visible.
- The surveillance camera system captures a colour image of the fingerprint acquisition area at the moment of the finger capture attempt.
- The image of the fp area is made at the moment of the finger capture attempt (i.e. the timestamps of the corresponding images are less than 100ms apart).
- The image captured by the camera system surveying the fp_area has a resolution of at least 1280 x 720.
- The image of the fp area is a coloured image in which the fingerprint capture system is shown up close.
- The images captured by the surveillance camera systems are cached locally.

2

Description:

- A single biometric subject initiates the SSS process of the IUT.

Expected Result:

- The IUT starts the SSS process and enables interaction via the graphical user interface.

3

Description:

- A second biometric subject moves into the corridor of the SSS (distance of 100cm in front the SSS).

Expected Result:

- The image captured by the surveillance camera system of the surrounding area allows to identify whether that more than one person is in range of the SSS.

4

Description:

- Both biometric subjects leave the corridor of the SSS.

Expected Result:

- The IUT stops the SSS process.

Table 2.1 Test Case ID: TC-AH-ALL-SSS-001

2.1.2 Test Cases FM AH-FI-BCL

Test Case ID: TC-AH-FI-BCL-001

Scope	Examination of technical specifications of the digital camera used to obtain facial biometrics.
Preconditions	• The following test resources are necessary: • Product documentation of IUT is at hand (e.g. data sheet, manual). • Product documentation of camera model is at hand (e.g. data sheet, manual).
CTS Mode	Not supported

Test Case ID: TC-AH-FI-BCL-001

Description	Step	Description / Expected Result
	1	<i>Description:</i> - Consult product documentation of the camera. <i>Expected Result:</i> - The product documentation states that the camera supports the capture of color images.
	2	<i>Description:</i> - Consult product documentation of the camera system. <i>Expected Result:</i> - The product documentation states the support of capturing facial images of a biometric subject with a body height in the range of 140 cm to 200 cm.

Table 2.3 Test Case ID: TC-AH-FI-BCL-001**Test Case ID: TC-AH-FI-BCL-002**

Scope	Examination of image resolution and cropping features of the digital camera used to obtain facial biometrics.
Preconditions	• The IUT is in operation, required modules are loaded. • The hardware used in the specific application scenario is connected • The IUT implements an interface for conformance testing where a single image can be captured. • The IUT is able to return the results for every captured image via the test interface. • The following test resources are necessary: • One test subject is available to perform the acquisition with.
CTS Mode	• interactive • no provision of pre-defined input data • HTTP method: GET • test case variants:
	/TR03122/TC-AH-FI-BCL-002/1

Table 2.5 TC-AH-FI-BCL-002 Test Case Variants

Description	Step	Description / Expected Result
	1	<i>Description:</i> - Initiate test case by calling the IUT via the test interface. - Position the biometric subject in a distance of 60 cm towards the camera system. Note, the distance between camera system and biometric subject is defined as the geometrical optical-path length between the forehead of the biometric subject and the active camera system's optic. The optical-path MAY, for example, follow a straight line from forehead to optic, or be re-routed by using mirrors so that the biometric subject can stand closer to the device - Acquire a photo of the biometric subject using the IUT's camera system. - Examine the response returned by the IUT via the test interface.

Test Case ID: TC-AH-FI-BCL-002*Expected Result:*

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response contains the correct number of elements based on the XPath expression //bio:FaceAcquisition. Exactly 1 element is expected in accordance to the acquisition process.
- The response contains the correct number of elements based on the XPath expression //bio:FaceAcquisition//bio:Records/bio:BinaryRecord/tradd:BinaryData/tradd:Value/node(). Exactly 1 element is expected in accordance to the acquisition process.
- The image contained in //bio:FaceAcquisition//bio:Records/bio:BinaryRecord/tradd:BinaryData/tradd:Value/node() has a resolution of at least 1080 x 1440.
- The image contained in //bio:FaceAcquisition//bio:Records/bio:BinaryRecord/tradd:BinaryData/tradd:Value/node() has a resolution of at most 1200 x 1600.
- The image depicts the testperson (i.e. their fingers or face) in the moment of the acquisition.

Table 2.4 Test Case ID: TC-AH-FI-BCL-002**Test Case ID: TC-AH-FI-BCL-003**

<i>Scope</i>	Examination of image processing features of the digital camera or provided/integrated software used to obtain facial biometrics.
<i>Preconditions</i>	• The IUT is in operation, required modules are loaded. • The hardware used in the specific application scenario is connected • The IUT implements an interface for conformance testing where a single image can be captured. • The IUT is able to return the results for every captured image via the test interface. • The following test resources are necessary: • Two test subjects are available to perform the acquisition with.
<i>CTS Mode</i>	• interactive • no provision of pre-defined input data • HTTP method: GET • test case variants:

/TR03122/TC-AH-FI-BCL-003/1

Table 2.7 TC-AH-FI-BCL-003 Test Case Variants*Description*

Step	Description / Expected Result
------	-------------------------------

Test Case ID: TC-AH-FI-BCL-003

1

Description:

- Initiate test case by calling the IUT via the test interface.
- Position one biometric subject in front of the camera system. Verify that this biometric subject is standing in a distance above 40 cm and below 100 cm to the camera system. Note, the distance between camera system and biometric subject is defined as the geometrical optical-path length between the forehead of the biometric subject and the active camera system's optic. The optical-path MAY, for example, follow a straight line from forehead to optic, or be rerouted by using mirrors so that the biometric subject can stand closer to the device
- Position the second biometric subject in front of the camera system. Verify that the biometric subject is standing in a distance below 40 cm to the camera system.
- Attempt to acquire an image with the camera system by using the CTS.
- Examine the response returned by the IUT via the test interface.

Expected Result:

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response returned via the test interface (bio:FaceAcquisition) does not contain facial image data.

Table 2.6 Test Case ID: TC-AH-FI-BCL-003

Test Case ID: TC-AH-FI-BCL-004

Scope

Examination of image processing features of the digital camera or provided/integrated software used to obtain facial biometrics.

Preconditions

- The IUT is in operation, required modules are loaded.
- The hardware used in the specific application scenario is connected
- The IUT implements an interface for conformance testing where a single image can be captured.
- The IUT is able to return the results for every captured image via the test interface.
- The following test resources are necessary:
 - Two test subjects are available to perform the acquisition with.

CTS Mode

- interactive
- no provision of pre-defined input data
- HTTP method: GET
- test case variants:

/TR03122/TC-AH-FI-BCL-004/1

Table 2.9 TC-AH-FI-BCL-004 Test Case Variants

Description

Step	Description / Expected Result
------	-------------------------------

Test Case ID: TC-AH-FI-BCL-004

1

Description:

- Initiate test case by calling the IUT via the test interface.
- Position one biometric subject in front of the camera system. Verify that this biometric subject is standing in a distance above 40 cm and below 100 cm to the camera system. Note, the distance between camera system and biometric subject is defined as the geometrical optical-path length between the forehead of the biometric subject and the active camera system's optic. The optical-path MAY, for example, follow a straight line from forehead to optic, or be rerouted by using mirrors so that the biometric subject can stand closer to the device.
- Position the second biometric subject in front of the camera system. Verify that the biometric subject is standing in a distance above 100 cm to the camera system.
- Acquire an image with the camera system by using the CTS.
- Examine the response returned by the IUT via the test interface.

Expected Result:

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- A message conforming to TR-03121 XML containing the encoded facial image is returned via the test interface (bio:FaceAcquisition).
- The returned image depicts the biometric subject which was positioned in a distance above 40 cm and below 100 cm to the camera system.

Table 2.8 Test Case ID: TC-AH-FI-BCL-004

2.1.3 Test Cases FM AH-FI-DC

Test Case ID: TC-AH-FI-DC-001

<i>Scope</i>	Examination of technical specifications of the facial image camera used to obtain facial biometrics.	
<i>Applicability</i>	This test is optional for scenarios where only a facial verification is performed.	
<i>Preconditions</i>	• The following test resources are necessary: • Product documentation of IUT is at hand (e.g. data sheet, manual). • Product documentation of camera model is at hand (e.g. data sheet, manual).	
<i>CTS Mode</i>	Not supported	
<i>Description</i>	Step	Description / Expected Result
	1	<i>Description:</i> • Consult product documentation of the camera. • Verify that the sensor of the camera provides a physical resolution of at least 1200 x 1600 pixels. <i>Expected Result:</i> • The product documentation states that the physical, native resolution is at least 1200 x 1600 pixels.
	2	<i>Description:</i> • Consult product documentation of the camera or options in the cameras firmware. • Verify that the active camera setting (e.g. configurable via camera firmware) used for the specific application provides an image resolution of at least 1200 x 1600 pixels.

Test Case ID: TC-AH-FI-DC-001*Expected Result:*

- The active setting provides a native image resolution of at least 1200 x 1600 pixels.

Table 2.10 Test Case ID: TC-AH-FI-DC-001**Test Case ID: TC-AH-FI-DC-002**

Scope Examination of technical specifications of the digital camera used to obtain digitised facial images. This test case covers the colour space of the captured facial image.

Applicability This test is optional for scenarios where only a facial verification is performed.

Preconditions

- The following test resources are necessary:
 - Product documentation of camera model is at hand (e.g. data sheet, manual).

CTS Mode Not supported

Description

Step	Description / Expected Result
1	<i>Description:</i> • Consult product documentation of the digital camera. • Verify that the digital camera provides a 24 bit sRGB mode. <i>Expected Result:</i> • The product documentation states that the digital camera is able to capture images in 24 bit sRGB.

Table 2.11 Test Case ID: TC-AH-FI-DC-002

2.1.4 Test Cases FM AH-FI-DC2

Test Case ID: TC-AH-FI-DC2-001

Scope Examination of technical specifications of the digital camera used to obtain facial biometrics.

Applicability This test is optional for scenarios where only a facial verification is performed.

Preconditions

- The following test resources are necessary:
 - Product documentation of camera model is at hand (e.g. data sheet, manual).

CTS Mode Not supported

Description

Step	Description / Expected Result
1	<i>Description:</i> • Consult product documentation of the camera. • Verify that the sensor of the camera provides a physical resolution of at least 1244 x 1600 pixels. <i>Expected Result:</i> • The product documentation states that the physical, native resolution is at least 1244 x 1600 pixels.
2	<i>Description:</i> • Consult product documentation of the camera or options in the cameras firmware. • Verify that the active camera setting (e.g. configurable via camera firmware) used for the specific application provides an image resolution of at least 1244 x 1600 pixels.

Test Case ID: TC-AH-FI-DC2-001*Expected Result:*

- The active setting provides a native image resolution of at least 1244 x 1600 pixels.

Table 2.12 Test Case ID: TC-AH-FI-DC2-001**Test Case ID: TC-AH-FI-DC2-002**

Scope Examination of technical specifications of the digital camera used to obtain digitised facial images. This test case covers the colour space of the captured facial image.

Applicability This test is optional for scenarios where only a facial verification is performed.

Preconditions

- The following test resources are necessary:
 - Product documentation of camera model is at hand (e.g. data sheet, manual).

CTS Mode Not supported

Description

Step Description / Expected Result

- | Step | Description / Expected Result |
|------|--|
| 1 | <p><i>Description:</i></p> <ul style="list-style-type: none"> • Consult product documentation of the digital camera. • Verify that the digital camera provides a 24 bit sRGB mode. <p><i>Expected Result:</i></p> <ul style="list-style-type: none"> • The product documentation states that the digital camera is able to capture images in 24 bit sRGB. |

Table 2.13 Test Case ID: TC-AH-FI-DC2-002**Test Case ID: TC-AH-FI-DC2-003**

Scope Examination of the documents describing the image capturing features of the digital camera used to obtain facial biometrics. This test case covers the requirements for focal length (depending on the size of the camera sensor) as described in [BIB_ICAO_TR_Portrait_Quality] (chapters 5.2.1 and 5.2.2).

Applicability This TC applies to IUT which use a smartphone or a similar device instead of the DSLR or system camera originally intended by the FM. Note: This TC does not extend to digital cameras which only support a wide-angle lens (e.g. webcams).

Preconditions

- Product documentation of the camera model (i.e. smartphone or similar device) provides reliable information regarding focal length and sensor size.
- The following test resources are necessary:
 - Product documentation of IUT is at hand (e.g. data sheet, manual).
 - Product documentation of camera model is at hand (e.g. data sheet, manual).

CTS Mode Not supported

Description

Step Description / Expected Result

- | Step | Description / Expected Result |
|------|---|
| 1 | <p><i>Description:</i></p> <ul style="list-style-type: none"> • Examination of the provided product documentation regarding permitted camera-to-subject-distance. <p><i>Expected Result:</i></p> <ul style="list-style-type: none"> • The permitted placement options of the biometric subject are limited in a way that the optical path of the image capturing system is at minimum 1 m and at maximum 4 m. |

Test Case ID: TC-AH-FI-DC2-003

2

Description:

- Examination of the provided product documentation regarding effective focal length f_{eff} . If the document does not explicitly specify the used effective focal length (compared to a 35 mm full frame), calculate the used effective focal length via $f_{eff} = f/c_f$ where f is the used focal length and c_f the crop factor (the crop factor may be calculated from the sensor's physical dimensions). If the camera model uses different lenses, additional evidence on which lense is actually being used shall be collected. Note: In some cases the effective focal length may also be extracted from the EXIF data.

Expected Result:

- The used effective focal length $f_{eff} = f/c_f$ is above 50 mm and below 130 mm.

Table 2.14 Test Case ID: TC-AH-FI-DC2-003

Test Case ID: TC-AH-FI-DC2-004

Scope

Examination of the image quality of the digital camera used to obtain facial biometrics. This test case covers the requirements for focal length (depending on the size of the camera sensor) as described in [BIB_ICAO_TR_Portrait_Quality] (chapters 5.2.1 and 5.2.2).

Applicability

This TC applies to IUT which use a smartphone or a similar device instead of the DSLR or system camera originally intended by the FM. Note: This TC does not extend to digital cameras which only support a wide-angle lens (e.g. webcams).

Preconditions

- The IUT is in operation, required modules are loaded.
- Product documentation of the camera model provides information regarding permitted camera-to-subject distances and zoom settings.
- The IUT implements an interface for conformance testing where a single image of the target (cf. [BIB_ICAO_TR_Portrait_Quality]) can be captured.
- The target is lit evenly from the front. A strong lightsource (i.e. softbox) is placed approximately 35° atop of the imaginary line between camera and target.
- The following test resources are necessary:
 - A conformant test-target.
 - An image-processing programm which allows to rotate a picture and crop it without quality loss.
 - A script which allows to measure the size of a square in a picture.
 - Product documentation of IUT is at hand (e.g. data sheet, manual).

CTS Mode

- interactive
- no provision of pre-defined input data
- HTTP method: GET
- test case variants:

/TR03122/TC-AH-FI-DC2-004/1

Table 2.16 TC-AH-FI-DC2-004 Test Case Variants

Description

Step	Description / Expected Result
------	-------------------------------

Test Case ID: TC-AH-FI-DC2-004

1	<i>Description:</i>
	• The target is positioned directly in front of the camera at the desired working distance (cf. Evaluation Step). Camera and target are at the same height. • The IUT including camera system is configured according to the documentation. • Initiate test case by calling the IUT via the test interface. • A single image of the target is acquired.
	<i>Expected Result:</i>
2	<i>Description:</i>
	• Examine the response returned by the IUT via the test interface.
	<i>Expected Result:</i>
3	<i>Description:</i>
	• The acquired images are prepared in a suitable way using the image processing program: If necessary, the images may be rotated slightly to appear vertically upright in the picture. • The acquired image is horizontally divided into four lines of three pictures containing a single pattern s.t. each of the patterns is depicted. Each line is then divided into two groups of images: One group containing the leftmost and center pattern, the other containing the center and rightmost pattern. • For each of the groups: measure the magnification distortion, i.e. the differences in height of the left/rightmost pattern in comparison to the center pattern. Note: To ensure comparability between different measurements a script shall be used for support.
	<i>Expected Result:</i>
Evaluation	<i>Description:</i>
	• If the product documentation permits several different CSD or zoom settings, the TC shall be repeated for all of them. Alternatively, if this is not practicable, the TC shall be repeated for each combination of the respective minimum, medium and maximum values.
	<i>Expected Result:</i>
1	<i>Description:</i>
	• For each combination of CSD and zoom setting the expected results are met.
	<i>Expected Result:</i>

Table 2.15 Test Case ID: TC-AH-FI-DC2-004

2.1.5 Test Cases FM AH-FI-EGT

Test Case ID: TC-AH-FI-EGT-001							
Scope	Examination of technical specifications of the camera system used to obtain digitised facial images in an e-gate scenario.						
Preconditions	- The following test resources are necessary: - Product documentation of IUT is at hand (e.g. data sheet, manual). - Product documentation of camera model is at hand (e.g. data sheet, manual).						
CTS Mode	Not supported						
Description	<table> <tr> <th>Step</th><th>Description / Expected Result</th></tr> <tr> <td>1</td><td> <i>Description:</i> - Consult product documentation of the camera system. <i>Expected Result:</i> - The product documentation states the support of capturing facial images of a biometric subject with a body height in the range of 140 cm to 200 cm standing in front of the camera system. - The product documentation states a physical resolution of at least 600x800 pixels without application of upscaling. </td></tr> <tr> <td>2</td><td> <i>Description:</i> - Assess the design of the camera system. <i>Expected Result:</i> - The camera system is designed to be placed in the moving direction of the biometric subject. - The camera system obviously does not require a rotation of the biometric subject for acquisition. </td></tr> </table>	Step	Description / Expected Result	1	<i>Description:</i> - Consult product documentation of the camera system. <i>Expected Result:</i> - The product documentation states the support of capturing facial images of a biometric subject with a body height in the range of 140 cm to 200 cm standing in front of the camera system. - The product documentation states a physical resolution of at least 600x800 pixels without application of upscaling.	2	<i>Description:</i> - Assess the design of the camera system. <i>Expected Result:</i> - The camera system is designed to be placed in the moving direction of the biometric subject. - The camera system obviously does not require a rotation of the biometric subject for acquisition.
Step	Description / Expected Result						
1	<i>Description:</i> - Consult product documentation of the camera system. <i>Expected Result:</i> - The product documentation states the support of capturing facial images of a biometric subject with a body height in the range of 140 cm to 200 cm standing in front of the camera system. - The product documentation states a physical resolution of at least 600x800 pixels without application of upscaling.						
2	<i>Description:</i> - Assess the design of the camera system. <i>Expected Result:</i> - The camera system is designed to be placed in the moving direction of the biometric subject. - The camera system obviously does not require a rotation of the biometric subject for acquisition.						

Table 2.17 Test Case ID: TC-AH-FI-EGT-001

Test Case ID: TC-AH-FI-EGT-002			
Scope	Examination of properties of digitised facial images captured by the camera system in an e-gate scenario.		
Preconditions	• The IUT is in operation, required modules are loaded. • The digital camera used in the specific application scenario is connected • The IUT implements an interface for conformance testing where a single image of the target (cf. [BIB_ICAO_TR_Portrait_Quality]) can be captured. • The IUT is able to return the results for every captured image via the test interface. • The following test resources are necessary: • One test subject is available to perform the acquisition with.		
CTS Mode	• interactive • no provision of pre-defined input data • HTTP method: GET • test case variants:<table><tr><td>/TR03122/TC-AH-FI-EGT-002/1</td></tr></table>	/TR03122/TC-AH-FI-EGT-002/1	
/TR03122/TC-AH-FI-EGT-002/1			
Table 2.19 TC-AH-FI-EGT-002 Test Case Variants			
Description	<table><tr><th>Step</th><th>Description / Expected Result</th></tr></table>	Step	Description / Expected Result
Step	Description / Expected Result		

Test Case ID: TC-AH-FI-EGT-002

1

Description:

- Initiate test case by calling the IUT via the test interface.
- Position the biometric subject in a distance of 50 cm towards the camera system.
- Acquire an image with the camera system.
- Examine the response returned by the IUT via the test interface.

Expected Result:

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response contains the correct number of elements based on the XPath expression `//bio:FaceAcquisition`. Exactly 1 element is expected in accordance to the acquisition process.
- The response contains the correct number of elements based on the XPath expression `./bio:Records/bio:BinaryRecord/tradd:BinaryData/tradd:Value`. Exactly 1 element is expected in accordance to the acquisition process.
- The image returned by the IUT has a resolution of at least 600 x 800.
- The image has sufficient sharpness for visual identification of the subject and does not obviously distort the captured face.

2

Description:

- Initiate test case by calling the IUT via the test interface.
- Position the biometric subject in a distance of 100 cm towards the camera system.
- Acquire an image with the camera system.
- Examine the response returned by the IUT via the test interface.

Expected Result:

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response contains the correct number of elements based on the XPath expression `//bio:FaceAcquisition`. Exactly 1 element is expected in accordance to the acquisition process.
- The response contains the correct number of elements based on the XPath expression `./bio:Records/bio:BinaryRecord/tradd:BinaryData/tradd:Value`. Exactly 1 element is expected in accordance to the acquisition process.
- The image returned by the IUT has a resolution of at least 600 x 800.
- The image has sufficient sharpness for visual identification of the subject and does not obviously distort the captured face.

3

Description:

- Initiate test case by calling the IUT via the test interface.
- Position the biometric subject in a distance of 200 cm towards the camera system.
- Acquire an image with the camera system.
- Examine the response returned by the IUT via the test interface.

Test Case ID: TC-AH-FI-EGT-002

Expected Result:

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response contains the correct number of elements based on the XPath expression `//bio:FaceAcquisition`. Exactly 1 element is expected in accordance to the acquisition process.
- The response contains the correct number of elements based on the XPath expression `./bio:Records/bio:BinaryRecord/tradd:BinaryData/tradd:Value`. Exactly 1 element is expected in accordance to the acquisition process.
- The image returned by the IUT has a resolution of at least 600 x 800.
- The image has sufficient sharpness for visual identification of the subject and does not obviously distort the captured face.

Table 2.18 Test Case ID: TC-AH-FI-EGT-002

2.1.6 Test Cases FM AH-FI-FBS

Test Case ID: TC-AH-FI-FBS-001

Scope	Examination of capable image resolution of flat bed scanner used to scan photos for enrolment purposes.				
Preconditions	• The following test resources are necessary: • Product documentation of flat bed scanner model is at hand (e.g. data sheet, manual).				
CTS Mode	Not supported				
Description	<table border="1"> <thead> <tr> <th>Step</th><th>Description / Expected Result</th></tr> </thead> <tbody> <tr> <td>1</td><td> <i>Description:</i> • Consult product documentation of the scanner. • Verify that the scanner provides a physical resolution of at least 600 dpi. <i>Expected Result:</i> • The product documentation states that the physical/native resolution is at least 600 dpi. </td></tr> </tbody> </table>	Step	Description / Expected Result	1	<i>Description:</i> • Consult product documentation of the scanner. • Verify that the scanner provides a physical resolution of at least 600 dpi. <i>Expected Result:</i> • The product documentation states that the physical/native resolution is at least 600 dpi.
Step	Description / Expected Result				
1	<i>Description:</i> • Consult product documentation of the scanner. • Verify that the scanner provides a physical resolution of at least 600 dpi. <i>Expected Result:</i> • The product documentation states that the physical/native resolution is at least 600 dpi.				

Table 2.20 Test Case ID: TC-AH-FI-FBS-001

Test Case ID: TC-AH-FI-FBS-002

Scope	Examination of image quality of flat bed scanner used to scan photos for enrolment purposes.
Preconditions	• The IUT is in operation, required modules are loaded. • The flat bed scanner used in the specific application scenario is connected • The IUT is able to return the results for every captured image via the test interface. • The following test resources are necessary: • A photo with at least 36mm x 45mm.
CTS Mode	• interactive • no provision of pre-defined input data • HTTP method: GET • test case variants:

/TR03122/TC-AH-FI-FBS-002/1

Table 2.22 TC-AH-FI-FBS-002 Test Case Variants

Test Case ID: TC-AH-FI-FBS-002

Description	Step	Description / Expected Result
	1	<i>Description:</i> - Initiate test case by calling the IUT via the test interface. - Place the photo on the scan area of the FBS - Finish the scanning process as required by the IUT. - Examine the response returned by the IUT via the test interface. <i>Expected Result:</i> - The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. - The response contains the correct number of elements based on the XPath expression <code>//bio:Records/bio:BinaryRecord[@type='jpeg' and not(@copyReasons)]/tradd:BinaryData/tradd:Value/node()</code>. Exactly 1 element is expected in accordance to the acquisition process. - The image provided in <code>//bio:Records/bio:BinaryRecord[@type='jpeg' and not(@copyReasons)]/tradd:BinaryData/tradd:Value/node()</code> is of file format type JPEG. - The image captured by the scanner has a resolution of 622 x 800.

Table 2.21 Test Case ID: TC-AH-FI-FBS-002

2.1.7 Test Cases FM AH-FI-ICS

Test Case ID: TC-AH-FI-ICS-001

Scope	Examination of technical specifications of the integrated camera systems used to obtain digitised facial images. This test case covers the minimum physical resolution of the captured facial image.	
Applicability	This test is optional for scenarios where only a facial verification is performed.	
Preconditions	- The following test resources are necessary: - Product documentation of camera model is at hand (e.g. data sheet, manual).	
CTS Mode	Not supported	
Description	Step	Description / Expected Result
	1	<i>Description:</i> - Consult product documentation of the camera. - Verify that the sensor of the camera provides a physical resolution of at least 1200 x 1600 pixels without any up-scaling. <i>Expected Result:</i> - The product documentation states that the physical, native resolution is at least 1200 x 1600 pixels.

Table 2.23 Test Case ID: TC-AH-FI-ICS-001**Test Case ID: TC-AH-FI-ICS-002**

Scope	Examination of technical specifications of the integrated camera systems used to obtain digitised facial images. This test case covers the illumination.	
Preconditions	- The following test resources are necessary: - Product documentation of IUT is at hand (e.g. data sheet, manual). - Product documentation of camera model is at hand (e.g. data sheet, manual).	
CTS Mode	Not supported	
Description	Step	Description / Expected Result

Test Case ID: TC-AH-FI-ICS-002

1

Description:

- Consult product documentation of the integrated camera system and of the IUT.
- Verify that the integrated camera system provides diffuse lighting adapting to the environmental light conditions.
- Verify that the integrated camera system allows high quality acquisitions independently from the environmental light situation.

Expected Result:

- The product documentation states that the camera system uses diffuse lighting adapting to the environmental light conditions.
- The product documentation states that the camera system allows high quality acquisitions independently from the environmental light situation that can usually be found in the environment in question.

Table 2.24 Test Case ID: TC-AH-FI-ICS-002

Test Case ID: TC-AH-FI-ICS-003

Scope Examination of technical specifications of the integrated camera systems used to obtain digitised facial images. This test case covers the colour space of the captured facial image.

Applicability This test is optional for scenarios where only a facial verification is performed.

Preconditions

- The following test resources are necessary:
 - Product documentation of camera model is at hand (e.g. data sheet, manual).

CTS Mode Not supported

Description

Step

Description / Expected Result

1

Description:

- Consult product documentation of the integrated camera system.
- Verify that the integrated camera system provides a 24 bit sRGB mode.

Expected Result:

- The product documentation states that the camera system is able to capture images in 24 bit sRGB.

Table 2.25 Test Case ID: TC-AH-FI-ICS-003

Test Case ID: TC-AH-FI-ICS-004

Scope Examination of technical specifications of the integrated camera systems used to obtain digitised facial images. This test case covers the presence of a digital mirror.

Preconditions

- The following test resources are necessary:
 - Product documentation of IUT is at hand (e.g. data sheet, manual).
 - Product documentation of camera model is at hand (e.g. data sheet, manual).

CTS Mode Not supported

Description

Step

Description / Expected Result

1

Description:

- Verify that the integrated camera system provides a digital mirror.
- Consult product documentation of the integrated camera system and of the IUT.

Test Case ID: TC-AH-FI-ICS-004*Expected Result:*

- The product documentation states that the camera system provides a digital mirror.

Table 2.26 Test Case ID: TC-AH-FI-ICS-004**2.1.8 Test Cases FM AH-FI-ICS2****Test Case ID: TC-AH-FI-ICS2-001**

<i>Scope</i>	Examination of technical specifications of the integrated camera systems used to obtain digitised facial images. This test case covers the minimum physical resolution of the captured facial image.	
<i>Applicability</i>	This test is optional for scenarios where only a facial verification is performed.	
<i>Preconditions</i>	• The following test resources are necessary: • Product documentation of camera model is at hand (e.g. data sheet, manual).	
<i>CTS Mode</i>	Not supported	
<i>Description</i>	Step	Description / Expected Result
	1	<i>Description:</i> • Consult product documentation of the camera. • Verify that the sensor of the camera provides a physical resolution of at least 1244 x 1600 pixels without any up-scaling. <i>Expected Result:</i> • The product documentation states that the physical, native resolution is at least 1244 x 1600 pixels.

Table 2.27 Test Case ID: TC-AH-FI-ICS2-001**Test Case ID: TC-AH-FI-ICS2-002**

<i>Scope</i>	Examination of technical specifications of the integrated camera systems used to obtain digitised facial images. This test case covers the illumination.	
<i>Preconditions</i>	• The following test resources are necessary: • Product documentation of IUT is at hand (e.g. data sheet, manual). • Product documentation of camera model is at hand (e.g. data sheet, manual).	
<i>CTS Mode</i>	Not supported	
<i>Description</i>	Step	Description / Expected Result
	1	<i>Description:</i> • Consult product documentation of the integrated camera system and of the IUT. • Verify that the integrated camera system provides diffuse lighting adapting to the environmental light conditions. • Verify that the integrated camera system allows high quality acquisitions independently from the environmental light situation, provided that the system is set up in an environment that can usually be found in agencies and that offers normal lighting conditions (no direct light from windows etc.). <i>Expected Result:</i> • The product documentation states that the camera system uses diffuse lighting adapting to the environmental light conditions. • The product documentation states that the camera system allows high quality acquisitions independently from the environmental light situation that can usually be found in the environment in question.

Table 2.28 Test Case ID: TC-AH-FI-ICS2-002

Test Case ID: TC-AH-FI-ICS2-003

<i>Scope</i>	Examination of technical specifications of the integrated camera systems used to obtain digitised facial images. This test case covers the colour space of the captured facial image.				
<i>Applicability</i>	This test is optional for scenarios where only a facial verification is performed.				
<i>Preconditions</i>	- The following test resources are necessary: - Product documentation of camera model is at hand (e.g. data sheet, manual).				
<i>CTS Mode</i>	Not supported				
<i>Description</i>	<table border="1"> <thead> <tr> <th>Step</th><th>Description / Expected Result</th></tr> </thead> <tbody> <tr> <td>1</td><td> <i>Description:</i> - Consult product documentation of the integrated camera system. - Verify that the integrated camera system provides a 24 bit sRGB mode. <i>Expected Result:</i> - The product documentation states that the camera system is able to capture images in 24 bit sRGB. </td></tr> </tbody> </table>	Step	Description / Expected Result	1	<i>Description:</i> - Consult product documentation of the integrated camera system. - Verify that the integrated camera system provides a 24 bit sRGB mode. <i>Expected Result:</i> - The product documentation states that the camera system is able to capture images in 24 bit sRGB.
Step	Description / Expected Result				
1	<i>Description:</i> - Consult product documentation of the integrated camera system. - Verify that the integrated camera system provides a 24 bit sRGB mode. <i>Expected Result:</i> - The product documentation states that the camera system is able to capture images in 24 bit sRGB.				

Table 2.29 Test Case ID: TC-AH-FI-ICS2-003**Test Case ID: TC-AH-FI-ICS2-004**

<i>Scope</i>	Examination of technical specifications of the integrated camera systems used to obtain digitised facial images regarding required range for body height for systems that capture facial images of biometric subjects in a standing position.								
<i>Applicability</i>	This test case is an alternative for test case TC-AH-FI-ICS2-005. Systems can either use a standing position to obtain facial images (than TC-AH-FI-ICS2-004 is required) or a sitting position (then TC-AH-FI-ICS2-005 is required).								
<i>Preconditions</i>	- The IUT is in operation, required modules are loaded. - An Integrated Camera System (ICS) is connected. - The following test resources are necessary: - Product documentation of IUT is at hand (e.g. data sheet, manual). - Product documentation of camera model is at hand (e.g. data sheet, manual). - One test subject is available to perform the acquisition with.								
<i>CTS Mode</i>	Not supported								
<i>Description</i>	<table border="1"> <thead> <tr> <th>Step</th><th>Description / Expected Result</th></tr> </thead> <tbody> <tr> <td>1</td><td> <i>Description:</i> - Consult the product documentation of the camera system and of the IUT. <i>Expected Result:</i> - The product documentation states the support of capturing facial images of a biometric subject with a body height at least in the range of 140 cm to 200 cm if standing upright in front of the system. </td></tr> <tr> <td>2</td><td> <i>Description:</i> - Position a face in front of the camera system in such a way that the upper edge of the head is at the minimum height of 140 cm. Capture a facial image. <i>Expected Result:</i> - The entire face is visible on the image, in the required resolution. </td></tr> <tr> <td>3</td><td> <i>Description:</i> - Position a face in front of the camera system in such a way that the upper edge of the head is at the height of 170 cm. Capture a facial image. </td></tr> </tbody> </table>	Step	Description / Expected Result	1	<i>Description:</i> - Consult the product documentation of the camera system and of the IUT. <i>Expected Result:</i> - The product documentation states the support of capturing facial images of a biometric subject with a body height at least in the range of 140 cm to 200 cm if standing upright in front of the system.	2	<i>Description:</i> - Position a face in front of the camera system in such a way that the upper edge of the head is at the minimum height of 140 cm. Capture a facial image. <i>Expected Result:</i> - The entire face is visible on the image, in the required resolution.	3	<i>Description:</i> Position a face in front of the camera system in such a way that the upper edge of the head is at the height of 170 cm. Capture a facial image.
Step	Description / Expected Result								
1	<i>Description:</i> - Consult the product documentation of the camera system and of the IUT. <i>Expected Result:</i> - The product documentation states the support of capturing facial images of a biometric subject with a body height at least in the range of 140 cm to 200 cm if standing upright in front of the system.								
2	<i>Description:</i> - Position a face in front of the camera system in such a way that the upper edge of the head is at the minimum height of 140 cm. Capture a facial image. <i>Expected Result:</i> - The entire face is visible on the image, in the required resolution.								
3	<i>Description:</i> Position a face in front of the camera system in such a way that the upper edge of the head is at the height of 170 cm. Capture a facial image.								

Test Case ID: TC-AH-FI-ICS2-004

4	<i>Expected Result:</i>
	The entire face is visible on the image, in the required resolution.
	<i>Description:</i>
	Position a face in front of the camera system in such a way that the upper edge of the head is at the maximum height of 200 cm. Capture a facial image.
	<i>Expected Result:</i>
	The entire face is visible on the image, in the required resolution.

Table 2.30 Test Case ID: TC-AH-FI-ICS2-004

Test Case ID: TC-AH-FI-ICS2-005

<i>Scope</i>	Examination of technical specifications of the integrated camera systems used to obtain digitised facial images regarding required range for body height for systems that capture facial images of biometric subjects in a sitting position.	
<i>Applicability</i>	This test case is an alternative for test case TC-AH-FI-ICS2-004. Systems can either use a standing position to obtain facial images (than TC-AH-FI-ICS2-004 is required) or a sitting position (then TC-AH-FI-ICS2-005 is required).	
<i>Preconditions</i>	- The IUT is in operation, required modules are loaded. - An Integrated Camera System (ICS) is connected. - The following test resources are necessary: - Product documentation of IUT is at hand (e.g. data sheet, manual). - Product documentation of camera model is at hand (e.g. data sheet, manual). 3 test subjects are available to perform the acquisition with.	
<i>CTS Mode</i>	Not supported	
<i>Description</i>	Step	Description / Expected Result
	1	<i>Description:</i> - Consult the product documentation of the camera system and of the IUT. <i>Expected Result:</i> - The product documentation states the support of capturing facial images of a biometric subject with a body height at least in the range of 140 cm to 200 cm if sitting upright in front of the system.
	2	<i>Description:</i> - Position a biometric subject with a body height of 140 cm looking frontally into the camera and sitting upright in front of the system. Capture a facial image. <i>Expected Result:</i> - The entire face is visible on the image, in the required resolution.
	3	<i>Description:</i> - Position a biometric subject with a body height of 170 cm looking frontally into the camera and sitting upright in front of the system. Capture a facial image. <i>Expected Result:</i> - The entire face is visible on the image, in the required resolution.
	4	<i>Description:</i> Position a biometric subject with a body height of 200 cm looking frontally into the camera and sitting upright in front of the system. Capture a facial image.

Test Case ID: TC-AH-FI-ICS2-005*Expected Result:*

- The entire face is visible on the image, in the required resolution.

Table 2.31 Test Case ID: TC-AH-FI-ICS2-005

2.1.9 Test Cases FM AH-FI-SSS

Test Case ID: TC-AH-FI-SSS-001

<i>Scope</i>	Examination of SSS scenarios where a digitised facial image is obtained.	
<i>Preconditions</i>	• The IUT is in operation, required modules are loaded. • The SSS implements a graphical user interface and an interface where a single image can be captured. • The following test resources are necessary: • One test subject is available to perform the acquisition with.	
<i>CTS Mode</i>	Not supported	
<i>Description</i>	Step	Description / Expected Result
	1	<i>Description:</i> • The biometric subject initiates the SSS process of the IUT. • Acquire an image with the camera system of the IUT. • Interact with the graphical user interface of the IUT. <i>Expected Result:</i> • A facial image is captured. • During the interaction with the IUT's graphical user interface the biometric subject did not have to rotate his/her head towards the camera system of the IUT for capturing the facial image.

Table 2.32 Test Case ID: TC-AH-FI-SSS-001

2.1.10 Test Cases FM AH-FI-SSS2

Test Case ID: TC-AH-FI-SSS2-001

<i>Scope</i>	Examination of technical specifications of the digital camera used to obtain facial biometrics.	
<i>Preconditions</i>	• The following test resources are necessary: • Product documentation of IUT is at hand (e.g. data sheet, manual). • Product documentation of camera model is at hand (e.g. data sheet, manual).	
<i>CTS Mode</i>	Not supported	
<i>Description</i>	Step	Description / Expected Result
	1	<i>Description:</i> • Consult the product documentation of the camera system and of the IUT. <i>Expected Result:</i> • The product documentation states the support of capturing facial images of a biometric subject with a body height at least in the range of 140 cm to 200 cm either standing or sitting upright in front of the camera.

Table 2.33 Test Case ID: TC-AH-FI-SSS2-001**Test Case ID: TC-AH-FI-SSS2-002**

<i>Scope</i>	Examination of SSS scenarios where a digitised facial image is obtained.
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Test Case ID: TC-AH-FI-SSS2-002

- Preconditions*
- The IUT is in operation, required modules are loaded.
 - The SSS implements a graphical user interface and an interface where a single image can be captured.
 - The following test resources are necessary:
 - One test subject is available to perform the acquisition with.

CTS Mode Not supported

Description

Step	Description / Expected Result
1	<i>Description:</i> • The biometric subject initiates the SSS process of the IUT. • Acquire a facial image with the camera system of the IUT. • Interact with the graphical user interface of the IUT. <i>Expected Result:</i> • A facial image is captured. • During the interaction with the IUT's graphical user interface the biometric subject did not have to rotate his/her head towards the camera system of the IUT for capturing the facial image.

Table 2.34 Test Case ID: TC-AH-FI-SSS2-002

Test Case ID: TC-AH-FI-SSS2-003

Scope Examination of technical specifications of the digital camera used to obtain facial biometrics.

- Preconditions*
- The following test resources are necessary:
 - Product documentation of camera model is at hand (e.g. data sheet, manual).

CTS Mode Not supported

Description

Step	Description / Expected Result
1	<i>Description:</i> • Consult the product documentation of the camera system. <i>Expected Result:</i> • A facial image is captured. • The product documentation states the support of capturing facial images of a biometric subject is at least in the range from at least of 70 cm up to 100 cm. In case the product documentation states that it can operate in a greater range than 100 cm, the exact range is listed. This distances are in general to be understood as the length of the optical path between the forehead and the camera optics, not between the forehead and the front glass of the device. Therefore, this interval must be shifted according to the length of the internal optical path in order to obtain the correct distances from the device.

Table 2.35 Test Case ID: TC-AH-FI-SSS2-003

Test Case ID: TC-AH-FI-SSS2-004

Scope Examination of image resolution and cropping features of the integrated camera system used to obtain facial image

Test Case ID: TC-AH-FI-SSS2-004*Preconditions*

- The IUT is in operation, required modules are loaded.
- The integrated camera system used in the specific application scenario is connected
- The IUT implements an interface for conformance testing where a single image can be captured.
- The IUT is able to return the results for every captured image via the test interface.
- The following test resources are necessary:
 - One test subject is available to perform the acquisition with.
 - Product documentation of IUT is at hand (e.g. data sheet, manual).

CTS Mode

- interactive
- no provision of pre-defined input data
- HTTP method: GET
- test case variants:

/TR03122/TC-AH-FI-SSS2-004/1

Table 2.37 TC-AH-FI-SSS2-004 Test Case Variants*Description*

Step	Description / Expected Result
1	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • Position the biometric subject looking frontally into the camera at a distance of 70 cm from the camera system. This distance is to be understood as the length of the optical path. • Acquire a facial image with the camera system. • Examine the response returned by the IUT via the test interface. <i>Expected Result:</i> • The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • The response contains the correct number of elements based on the XPath expression <code>//bio:FaceAcquisition/bio:Records/bio:BinaryRecord[@type='bmp' or @type='png' or @type='jpeg' or @type='jpeg2000']/tradd:BinaryData/tradd:Value/node()</code>. Exactly 1 element is expected in accordance to the acquisition process. • The image captured by the camera has a resolution of at least 1244 x 1600. • The image is a coloured image with a colour space of 24 bit sRGB. • The response contains a facial image according to the definition of "full frontal" (see [ISO_FACE]).
2	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • Position the biometric subject looking frontally into the camera at a distance of 100 cm from the camera system. This distance is to be understood as the length of the optical path. • Acquire a facial image with the camera system. • Examine the response returned by the IUT via the test interface.

Test Case ID: TC-AH-FI-SSS2-004

Expected Result:

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response contains the correct number of elements based on the XPath expression `//bio:FaceAcquisition/bio:Records/bio:BinaryRecord[@type='bmp' or @type='png' or @type='jpeg' or @type='jpeg2000']/tradd:BinaryData/tradd:Value/node()`. Exactly 1 element is expected in accordance to the acquisition process.
- The image captured by the camera has a resolution of at least 1244 x 1600.
- The image is a coloured image with a colour space of 24 bit sRGB.
- The CTS shows a facial image according to the definition of "full frontal" (see [ISO_FACE_2011]).

3

Description:

- Initiate test case by calling the IUT via the test interface.
- Position the biometric subject looking frontally into the camera at the maximal possible distance (according to the product documentation) from the camera system. This distance is to be understood as the length of the optical path.
- Acquire a facial image with the camera system.
- Examine the response returned by the IUT via the test interface.

Expected Result:

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response contains the correct number of elements based on the XPath expression `//bio:FaceAcquisition/bio:Records/bio:BinaryRecord[@type='bmp' or @type='png' or @type='jpeg' or @type='jpeg2000']/tradd:BinaryData/tradd:Value/node()`. Exactly 1 element is expected in accordance to the acquisition process.
- The image captured by the camera has a resolution of at least 1244 x 1600.
- The image is a coloured image with a colour space of 24 bit sRGB.
- The CTS shows a facial image according to the definition of "full frontal" (see [ISO_FACE_2011]).

Table 2.36 Test Case ID: TC-AH-FI-SSS2-004

2.1.11 Test Cases FM AH-FP-ALL

Test Case ID: TC-AH-FP-ALL-001

Scope Check of the image sensors FBI certification.

Preconditions • The vendor of the image sensor provides the FBI certificate of the successful FBI certification.

CTS Mode Not supported

Description

Step Description / Expected Result

1

Description:

- Examination of the FBI certificate provided by the vendor.

- In case of multi finger scanners: The FBI certificate shows, that the device under test is certified according to [EBTS/F].
- In case of single finger scanners: The FBI certificate shows, that the device under test is certified according to [EBTS/F] or [PIV].
- The successful FBI certification is not older than 10 years.

2.1.12 Test Cases FM AH-FP-OPT

Scope	Test of image sensor linearity.		
Applicability	Depending on the following conditions method A, B, C or D is chosen for the test: Method A: It is possible to control the integration time of the scanner's image sensor. Method B: It is possible to control the intensity of the scanner's illumination. Method C: It is only possible to turn the scanner's illumination on and off. Method D: Neither of the above can be controlled.		
Preconditions	• The IUT is in operation, required modules are loaded. • The scanner is connected. • The scanner software is set to output unprocessed images. • Captured images can be saved for subsequent evaluation. • The following test resources are necessary: • Method A and B: A diffuse reflecting uniform white target (e.g. Munsell N9). • Method C: An external extended uniform light source (e.g. LED backlight or similar). • Method B and C: Measurement equipment for measuring the luminance of the illumination. • Method D: A diffuse reflecting target with a grey scale step tablet or a set of uniform neutral grey targets cards with various known reflectance. • Image processing software to determine the average grey value in a region of interest in the image.		
CTS Mode	Not supported		
Description	<table border="1"> <thead> <tr> <th>Step</th><th>Description / Expected Result</th></tr> </thead> </table>	Step	Description / Expected Result
Step	Description / Expected Result		

Test Case ID: TC-AH-FP-OPT-001

1

Description:

- Method A: The white target is placed on the scanner platen. The scanner's illumination is turned on and left unchanged during the test. Multiple images are taken while varying the integration time of the image sensor. The setting of the integration time is recorded with each image.
- Method B: The white target is placed on the scanner platen. Multiple images are taken while varying the intensity of the illumination of the scanner. The luminance of the illumination is monitored at the same time and recorded with each image.
- Method C: The scanner's illumination is turned off. The external light source is placed on the scanner platen. Multiple images are taken while varying the intensity of the external illumination. The luminance of the external illumination is monitored at the same time and recorded with each image.
- Method D: The grey scale target is placed on the scanner platen and an image is taken. In case of a set of uniform targets images of each target are taken sequentially and the reflectance of the target is recorded with each image.
- All methods: At least nine images/individual grey values are required that substantially cover the scanners dynamic range.

Expected Result:

- Set of images with a uniform area covered by the target or a single image of a target with a grey scale step tablet.

2

Description:

- For each of the images the average grey value is determined in the area covered by the colour test card. The average grey value is plotted against the previously recorded integration time (Method A), luminance (Method B, C) or reflectance (Method D).

Expected Result:

- The resulting plot should be close to a linear function.
- A linear regression is performed over the value pairs (integration time, average grey value), (luminance, average grey value) or (reflectance, grey value). The difference between the average grey value and the grey value from regression is calculated for each measurement.
- None of the calculated differences exceed 7.65 grey levels.

3

Description:

- A linear regression is performed over the value pairs (integration time, average grey value), (luminance, average grey value) or (reflectance, grey value). The difference between the average grey value and the grey value from regression is calculated for each measurement.

Expected Result:

- None of the calculated differences exceed 7.65 grey levels.

Table 2.39 Test Case ID: TC-AH-FP-OPT-001

Test Case ID: TC-AH-FP-OPT-002

Scope

Test of scanner resolution and geometrical accuracy.

Test Case ID: TC-AH-FP-OPT-002

Preconditions

- The IUT is in operation, required modules are loaded.
- The scanner is connected.
- The scanner software is set to output unprocessed images.
- Captured images can be saved for subsequent evaluation.
- The following test resources are necessary:
 - A Ronchi bar pattern target, that consists of an equal-width bar and space square wave pattern at 1.0 cy/mm, preferably on a flexible, diffuse white reflecting substrate. The target pattern shall cover at least 70% of the scanner's capture area.
 - Test software.

CTS Mode

Not supported

Description

Step	Description / Expected Result
1	<i>Description:</i> • The target is placed on the scanner platen and images are taken with the lines of the Ronchi pattern parallel to the pixel columns (vertical bars) and rows (horizontal bars) of the image sensor, respectively. A flexible target has to be pressed against the scanner platen to make sure the target is in contact with the platen over the entire pattern area. To discern properties of the scanner and the target the procedure may be repeated with the target being rotated by 180 degrees. If accuracy failures in the images remain at the same location they are most probably due to the scanner. <i>Expected Result:</i> • Set of images with the lines of the Ronchi pattern in vertical and horizontal direction, respectively.
2	<i>Description:</i> • The obtained image files are converted to raw image data (pixel grey values without header) and are processed with the software WinGeo. The program calculates the 1-bar and 6-bar distances of the pattern in the image (across-bar accuracy) as well as the along-bar accuracy. Also the resolution is calculated from the 6-bar distances. <i>Expected Result:</i> • Resolution: The scanner's output resolution in both sensor detector row and column directions is $(R \pm 0.01R)$, with R being 500 pixels per inch or 1000 pixels per inch. • If a 500-ppi scanner is tested: Across-bar accuracy: The difference $D = Y - X$ between the distance of the bars on the target (X) and the corresponding distance measured in the image (Y) does not exceed in at least 99% of the tested cases in horizontal and vertical direction, respectively: $D \leq 0.0007$ for $0.00 < X \leq 0.07$ and $D \leq 0.01X$ for $0.07 \leq X \leq 1.5$. D, X, Y are in inches. • If a 1000-ppi scanner is tested: Across-bar accuracy: The difference $D = Y - X$ between the distance of the bars on the target (X) and the corresponding distance measured in the image (Y) does not exceed in at least 99% of the tested cases in horizontal and vertical direction, respectively: $D \leq 0.0005$ for $0.00 < X \leq 0.07$ and $D \leq 0.0071X$ for $0.07 \leq X \leq 1.5$. D, X, Y are in inches. • Along-bar accuracy: In both vertical bar and horizontal bar orientations, the maximum difference in the horizontal or vertical direction, respectively, between the locations of any two points within a 1.5-inch segment of a given bar image does not exceed 0.016 inches for at least 99.0 percent of the tested cases.

Table 2.40 Test Case ID: TC-AH-FP-OPT-002

Test Case ID: TC-AH-FP-OPT-003

Scope Test of spatial frequency response.

- Preconditions
- The IUT is in operation, required modules are loaded.
 - The scanner is connected.
 - The scanner software is set to output unprocessed images.
 - Captured images can be saved for evaluation.
 - The following test resources are necessary:
 - A test chart containing Ronchi test patterns meeting or exceeding the minimum requirements given in ▶Table 2.42 for 500 ppi scanners or in ▶Table 2.43 for 1000 ppi scanners. The target must also contain large black and white patches for normalizing the contrast. Preferably the target should consist of dark patterns on a flexible, diffuse white reflecting substrate material. The material shall be optically opaque and exhibit no significant volume scattering as this degrades the contrast of the patterns. Rigid materials should not be used, if possible, to avoid damage to the image sensor.
 - A file describing the type, position and size of the different patterns on the test chart. This is required for processing the captured images with WinMTF.
 - Test software.

CTS Mode Not supported

Description

Step	Description / Expected Result
1	<i>Description:</i> • The target is placed on the scanner platen and images are taken with the lines of the Ronchi pattern parallel to the pixel columns (vertical bars) and rows (horizontal bars) of the image sensor, respectively. A flexible target has to be pressed against the scanner platen to make sure the target is in contact with the platen over the entire pattern area. If the target pattern covers less than 25% of the scanners capture area, the procedure is applied to the right and the left half of the capture area, respectively. The bars of the Ronchi patterns have to be aligned with the pixel rows and columns to an angle less than 0.5°. If the images from the scanner are inverted they have to be inverted back to show dark patterns on a bright background. <i>Expected Result:</i> • Set of images with the bars of the Ronchi pattern in vertical and horizontal direction, showing dark patterns on a bright background.
2	<i>Description:</i> • The resulting images files are converted to TIFF format and are processed with the tool WinMTF. This program determines the modulation in each of the Ronchi test patterns. These modulation values are absolute with respect to the dynamic range of the image and do not represent the contrast transfer function (CTF). <i>Expected Result:</i> • Absolute modulation values for each of the Ronchi patterns on the test chart.
3	<i>Description:</i> • The absolute modulation values have to be normalized by the modulation of the large black and white patches on the test chart, representing a spatial frequency of approximately zero cycles per millimetre. The average grey level in the black and white patches is calculated and the modulation is calculated as follows: $M_0 = \frac{G_{white} - G_{black}}{G_{white} + G_{black}}$ with Modulation for 0 cy/mm (M_0), average grey level in the white patch (G_{white}), and average grey level in the black patch (G_{black}). Having determined M_0 the value of the CTF can be calculated from the modulation of the Ronchi pattern $CTF(R) = \frac{M(R)}{M_0}$ with $M(R)$ Modulation at spatial frequency R.

Test Case ID: TC-AH-FP-OPT-003

Expected Result:

- The values of the CTF for each of the spatial frequencies specified in ▶Table 2.42 (500 ppi scanners) or ▶Table 2.43 (1000 ppi scanners) must be greater than the corresponding minimum CTF and less than the corresponding maximum CTF specified in the same table.
- The resulting image is free from significant effects of aliasing.

Table 2.41 Test Case ID: TC-AH-FP-OPT-003

Spatial Frequency R [cycles/mm]	R/Rnyquist 500ppi	Min. Number of Bars	Bar width [mm]	Min. Length of Bars [mm]	Min. CTF 500ppi	Max. CTF
<0,3	3%	1	>1.7	1.7	-	-
1	10%	4	0.500	2.5	0.948	1.05
2	20%	5	0.250	2.5	0.869	1.05
3	30%	5	0.167	2.5	0.791	1.05
4	40%	5	0.125	2.5	0.713	1.05
5	50%	10	0.100	2.5	0.636	1.05
6	60%	10	0.083	2.5	0.559	1.05
7	70%	10	0.071	2.5	0.483	1.05
8	80%	10	0.063	2.5	0.408	1.05
9	90%	10	0.056	2.5	0.333	1.05
10	100%	10	0.050	2.5	0.259	1.05

Table 2.42 Requirements for CTF test charts for 500 ppi scanners

Spatial Frequency R [cycles/mm]	R/Rnyquist 500ppi	Min. Number of Bars	Bar width [mm]	Min. Length of Bars [mm]	Min. CTF 500ppi	Max. CTF
<0,3	2%	1	>1.7	1.7	-	-
1	5%	4	0.500	2.5	0.957	1.05
2	10%	4	0.250	2.5	0.904	1.05
3	15%	5	0.167	2.5	0.854	1.05
4	20%	5	0.125	2.5	0.805	1.05
5	25%	5	0.100	2.5	0.760	1.05
6	30%	5	0.083	2.5	0.716	1.05
7	35%	5	0.071	2.5	0.675	1.05
8	40%	5	0.063	2.5	0.636	1.05
9	45%	10	0.056	2.5	0.598	1.05
10	50%	10	0.050	2.5	0.563	1.05
12	60%	10	0.042	2.5	0.497	1.05
14	70%	10	0.036	2.5	0.437	1.05
16	80%	10	0.031	2.5	0.382	1.05
18	90%	10	0.028	2.5	0.332	1.05
20	100%	10	0.025	2.5	0.284	1.05

Table 2.43 Requirements for CTF test charts for 1000 ppi scanners

Test Case ID: TC-AH-FP-OPT-004

Scope Test of signal to noise ratio and grey level uniformity.

- Preconditions*
- The IUT is in operation, required modules are loaded.
 - The scanner is connected.
 - The scanner software is set to output unprocessed images.
 - Captured images can be saved for evaluation.
 - The following test resources are necessary:
 - A uniform diffuse reflecting neutral grey target of high reflectance that covers the scanner's entire capture area (e.g Munsell N9).
 - A uniform diffuse reflecting neutral grey target of low reflectance that covers the scanner's entire capture area (e.g Munsell N2).
 - Test software.

CTS Mode Not supported

Description

Step	Description / Expected Result
1	<i>Description:</i> • The black target is placed on the scanner platen and an image is taken. After that the procedure is repeated with the white target. The definition of "black image" and "white image" for the subsequent analyses depends on the grey value that the target would produce in the final fingerprint image. If high reflectance produces a high grey value in the final fingerprint image then the image obtained with the high reflectance target is considered the "white image". If low reflectance produces a high grey value (inverting behaviour) then the image obtained with the low reflectance target is considered the "white image". The other image is then considered the "black image". <i>Expected Result:</i> • A black image with the average grey value in the image being at least 4 grey levels above the scanner's minimum output grey level. • A white image with the average grey value in the image being at least 4 grey levels below the scanner's maximum output grey level.
2	<i>Description:</i> • The black and white images are converted to raw image data (pixel grey values without header) and are processed with the software signal-to-noise ratio (SNR). The program divides the images in blocks of 6.35 mm x 6.35 mm and determines the average grey level, the standard deviation, the grey level averages in the columns and rows, respectively, and the number of pixels with a higher-than-allowed difference of the grey level from the average in each of the blocks. With these quantities the required tests regarding signal-noise-ratio and grey level uniformity are performed.

Test Case ID: TC-AH-FP-OPT-004

Expected Result:

- Signal to noise ratio: Signal is defined as the difference between the average grey levels of corresponding blocks in the white and the black image. Noise is defined as the standard deviation of the grey levels in each block of the white and black image, respectively. The signal-noise ratio is calculated as $SNR_{black} = \frac{\overline{G_{white}} - \overline{G_{black}}}{\sigma_{black}}$, $SNR_{white} = \frac{\overline{G_{white}} - \overline{G_{black}}}{\sigma_{white}}$ with $\overline{G_{white}}$, $\overline{G_{black}}$ average grey level in a block of the black and white image, respectively and σ_{white} , σ_{black} standard deviation of the grey levels in a block of the black and white image, respectively. The signal to noise ratio exceeds 125 for at least 97% of the tested blocks in both the black and the white image.
- Grey level uniformity requirement #1 (adjacent row, column uniformity): At least 99% of the average grey levels between every two adjacent 6.35 mm long rows and 99% between every two adjacent 6.35 mm long columns, do not differ by more than 1.0 grey levels in the black and do not differ by more than 2.0 grey levels in the white image.
- Grey level uniformity requirement #2 (pixel to pixel uniformity): For at least 99.9% of all pixels within every independent 6.35 mm x 6.35 mm block no individual pixel's grey level varies from the average by more than 8.0 grey levels in the black image, and does not vary from the average by more than 22.0 grey levels in the white image.
- Grey level uniformity requirement #3 (small area uniformity): For every two independent 6.35 mm x 6.35 mm blocks the average grey levels of the two blocks do not differ by more than 3.0 grey levels in the black image, and do not differ by more than 12.0 grey levels in the white image.

Table 2.44 Test Case ID: TC-AH-FP-OPT-004

Test Case ID: TC-AH-FP-OPT-005

Scope Test of fingerprint image quality.

- Preconditions
- The IUT is in operation, required modules are loaded.
 - The scanner is connected.
 - The scanner software is set to output completely processed fingerprint images.
 - Captured images can be saved for evaluation.
 - The following test resources are necessary:
 - 10 test subjects are available to perform the acquisition with.
 - Image processing software for histogram analysis.

CTS Mode Not supported

Description

Step	Description / Expected Result
------	-------------------------------

- | | |
|---|--|
| 1 | <p><i>Description:</i></p> <ul style="list-style-type: none"> • Images of every finger of each test person are taken with the scanner. If the scanner is capable of capturing 4-finger images additional 4-finger images of each hand are taken. <p><i>Expected Result:</i></p> <ul style="list-style-type: none"> • Images of every finger of each test person. In case of a 4-finger scanner also 4-finger images of every hand of each test person. |
| 2 | <p><i>Description:</i></p> <ul style="list-style-type: none"> • For each image a histogram of the grey levels is produced. The histogram is tested for gaps and the number of grey levels with at least 5 counts in the image are counted. |

Test Case ID: TC-AH-FP-OPT-005*Expected Result:*

- The number of grey levels present with at least 5 pixels in the image is at least 128 for at least 99% of the tested images.
- The number of grey levels present with at least 5 pixels in the image is at least 200 for at least 80% of the tested images.
- There are no gaps in the histogram.

Table 2.45 Test Case ID: TC-AH-FP-OPT-005**2.1.13 Test Cases FM AH-FP-SSS****Test Case ID: TC-AH-FP-SSS-001**

<i>Scope</i>	Examination of SSS scenarios where digitised fingerprint images are obtained.
<i>Preconditions</i>	• The IUT is in operation, required modules are loaded. • The IUT's SSS is ready for operation. • The SSS implements a graphical user interface and an interface where fingerprint images can be captured. • The IUT is able to return the results for every captured image via the test interface. • The following test resources are necessary: • Two test subjects are available to perform the acquisition with.
<i>CTS Mode</i>	• interactive • no provision of pre-defined input data • HTTP method: GET • test case variants:

/TR03122/TC-AH-FP-SSS-001/1

Table 2.47 TC-AH-FP-SSS-001 Test Case Variants

<i>Description</i>	Step	Description / Expected Result
	1	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • The biometric subject interacts with the GUI of the SSS to acquire fingerprint images of the right hand. • Acquire fingerprint images of the biometric subject with the SSS of the IUT using the right hand and finish the process at the SSS. • Examine the response returned by the IUT via the test interface. <i>Expected Result:</i> • The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • Fingerprint images of the right hand were captured. • The image depicts the testperson (i.e. their fingers or face) in the moment of the acquisition.

Test Case ID: TC-AH-FP-SSS-001

2

Description:

- Initiate test case by calling the IUT via the test interface.
- The biometric subject interacts with the GUI of the SSS to acquire fingerprint images of the left hand.
- Acquire fingerprint images of the biometric subject with the SSS of the IUT using the left hand and finish the process at the SSS.
- Examine the response returned by the IUT via the test interface.

Expected Result:

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- Fingerprint images of the left hand were captured.
- The image depicts the testperson (i.e. their fingers or face) in the moment of the acquisition.

3

Description:

- Initiate test case by calling the IUT via the test interface.
- The biometric subject interacts with the GUI of the SSS to acquire fingerprint images while another subject is present within the reach of the fingerprint acquisition system of the SSS. .
- Acquire fingerprint images of the biometric subject with the SSS of the IUT using the right or left hand and finish the process at the SSS.
- Examine the response returned by the IUT via the test interface.

Expected Result:

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- During the fingerprint acquisition process multiple persons are detected by the SSS.
- Fingerprint images of the left or right hand were captured.
- The multiple person detection result is cached locally on the SSS.

Table 2.46 Test Case ID: TC-AH-FP-SSS-001

2.2 Test Cases Acquisition Software

2.2.1 Test Cases FM AS-FI-DC

Test Case ID: TC-AS-FI-DC-001

Scope Examination of the software module used for acquisition of digitised facial images

Preconditions

- The IUT is in operation, required modules are loaded.
- The digital camera used in the specific application scenario is connected
- IUT is configured to perform the acquisition of a facial image.
- A testperson is available to test the IUT's behaviour.

CTS Mode

- interactive
- no provision of pre-defined input data
- HTTP method: GET
- test case variants:

/TR03122/TC-AS-FI-DC-001/1

Table 2.49 TC-AS-FI-DC-001 Test Case Variants

Test Case ID: TC-AS-FI-DC-001

Description

Step

Description / Expected Result

1

Description:

- Initiate test case by calling the IUT via the test interface.
- Capture an image with the camera
- Examine encoding, format and resolution of the returned digitised image data
- Check the resulting image for visible compression artefacts

Expected Result:

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response contains the correct number of elements based on the XPath expression `./bio:Records/bio:BinaryRecord[@type='bmp' and not(@copyReasons)]/tradd:BinaryData/tradd:Value`. Exactly 1 element is expected in accordance to the acquisition process.
- The data embedded in the element `./bio:Records/bio:BinaryRecord[@type='bmp' and not(@copyReasons)]/tradd:BinaryData/tradd:Value` of the response XML is base64-encoded BMP.
- The image captured by the camera has a resolution of at least 1200 x 1600.
- The image has a colour space of 24bit RGB.
- The image depicts the testperson in the moment of the acquisition.
- The inter eye distance is at least 120 pixels.
- If the digital camera was not a webcam: The image does not contain compression artefacts.
- If the digital camera was a webcam: The image does not contain severe compression artefacts.

Table 2.48 Test Case ID: TC-AS-FI-DC-001

2.2.2 Test Cases FM AS-FI-DC2

Test Case ID: TC-AS-FI-DC2-001

Scope

Examination of the software module used for acquisition of digitised facial images

Preconditions

- The IUT is in operation, required modules are loaded.
- The digital camera used in the specific application scenario is connected
- IUT is configured to perform the acquisition of a facial image.
- A testperson is available to test the IUT's behaviour.

CTS Mode

- interactive
- no provision of pre-defined input data
- HTTP method: GET
- test case variants:

/TR03122/TC-AS-FI-DC2-001/1

Table 2.51 TC-AS-FI-DC2-001 Test Case Variants

Description

Step

Description / Expected Result

Test Case ID: TC-AS-FI-DC2-001

1

Description:

- Initiate test case by calling the IUT via the test interface.
- Capture an image with the camera.
- Examine encoding, format and resolution of the returned digitised image data.
- Check the resulting image for visible compression artefacts.
- Check that software-based background elimination and software-based background replacement have not been performed.

Expected Result:

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response contains the correct number of elements based on the XPath expression `//bio:Records/bio:BinaryRecord[@type='bmp' or @type='png' or @type='jpeg' or @type='jpeg2000' and not(@copy-Reasons)]/tradd:BinaryData/tradd:Value/node()`. Exactly 1 element is expected in accordance to the acquisition process.
- The data embedded in the element `//bio:Records/bio:BinaryRecord/tradd:BinaryData/tradd:Value` of the response XML is base64-encoded data that can be successfully decoded resulting in an image file format according to the corresponding @type-attribute.
- The image captured by the camera has a resolution of at least 1244 x 1600.
- The image has a colour space of 24bit RGB.
- If the image is a JPEG or lossily compressed JPEG2000 image: The acquisition device does not support a lossless mode and the image is provided with the least compression possible.
- The image does not contain any visible compression artefacts.
- The image depicts the testperson in the moment of the acquisition.
- The image has the same background as in the moment of the acquisition.

Table 2.50 Test Case ID: TC-AS-FI-DC2-001

2.2.3 Test Cases FM AS-FI-FBS

Test Case ID: TC-AS-FI-FBS-001

<i>Scope</i>	Examination of acquisition software for flat bed scanners that are used for the provisioning of digitised application forms. This test case covers the image format used by the acquisition software of the scanner system.
<i>Preconditions</i>	• The IUT is in operation, required modules are loaded. • A document scanner is connected. • As test resource: a application form preferable for German Identity Document (GID). • As test resource: a colored image to perform scanning in combination with the application form. • As test resource: a greyscale image to perform scanning in combination with the application form.
<i>CTS Mode</i>	• interactive • no provision of pre-defined input data • HTTP method: GET • test case variants:

/TR03122/TC-AS-FI-FBS-001/1

Table 2.53 TC-AS-FI-FBS-001 Test Case Variants

Test Case ID: TC-AS-FI-FBS-001

Description	Step	Description / Expected Result
	1	<i>Description:</i> - Initiate test case by calling the IUT via the test interface. - Scan a colored Image using the application form. - Verify compression of file with a JPEG decoder/viewer. <i>Expected Result:</i> - The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. - The response contains the correct number of elements based on the XPath expression <code>//bio:BinaryRecord[@type="jpeg" and not(@copyReasons="converted")]/tradd:BinaryData/tradd:Value</code>. Exactly 1 element is expected in accordance to the acquisition process. - The image is encoded in JPEG. - The image is colored, i.e. has a colour depth of 24bit RGB. - The image is provided with the least compression possible, meaning that, a quality estimation indicates high quality settings. - The image is provided without any prior or further compression in JPEG, meaning that, there are no artefacts visible. - The image can be displayed and depicts the information that was scanned.
	2	<i>Description:</i> - Initiate test case by calling the IUT via the test interface. - Scan a greyscale Image using the application form. - Verify compression of file with a JPEG decoder/viewer. <i>Expected Result:</i> - The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. - The response contains the correct number of elements based on the XPath expression <code>//bio:BinaryRecord[@type="jpeg" and not(@copyReasons="converted")]/tradd:BinaryData/tradd:Value</code>. Exactly 1 element is expected in accordance to the acquisition process. - The image is encoded in JPEG. - The image is a greyscale image, i.e. has a colour depth of 8bit RGB. - The image is provided with the least compression possible, meaning that, a quality estimation indicates high quality settings. - The image is provided without any prior or further compression in JPEG, meaning that, apart from the compression of JPEG itself there are no artefacts visible. - The image can be displayed and depicts the information that was scanned.

Table 2.52 Test Case ID: TC-AS-FI-FBS-001

2.2.4 Test Cases FM AS-FI-ICS

Test Case ID: TC-AS-FI-ICS-001

Scope	Examination of technical specifications for acquisition software used for integrated camera systems in order to obtain digitised facial images. This test case covers the image formats supported by the acquisition software of the camera system.
Preconditions	Product documentation of the utilised camera system is at hand (e.g. data sheet, manual).
CTS Mode	Not supported

Test Case ID: TC-AS-FI-ICS-001

Description	Step	Description / Expected Result
	1	<i>Description:</i> Consult the product documentation of the camera system and check the supported image formats. <i>Expected Result:</i> The product documentation states that the software provides raw uncompressed image data e.g. YUV422 or sRGB or the software provides a lossless image container format (e.g. BMP, TIFF).

Table 2.54 Test Case ID: TC-AS-FI-ICS-001**Test Case ID: TC-AS-FI-ICS-002**

Scope	Examination of technical specifications for acquisition software used for integrated camera systems in order to obtain digitised facial images. This test case covers the minimum physical resolution of the captured facial image. Note, this requirement is OPTIONAL for scenarios where only a facial verification is performed.	
Preconditions	- The integrated camera system used in the specific application profile is connected and the system is ready for capturing. - Product documentation of the camera unit is at hand (e.g. data sheet, manual).	
CTS Mode	Not supported	
Description	Step	Description / Expected Result
	1	<i>Description:</i> - Verify that the active camera setting (e.g. configurable via camera firmware) used for the specific application provides an image resolution of at least 1200 x 1600 pixels without any up-scaling. - Consult product documentation of the camera and options in the cameras firmware. <i>Expected Result:</i> - The setting provides a native image resolution of at least 1200 x 1600 pixels. - This setting is the active setting for the application profile.

Table 2.55 Test Case ID: TC-AS-FI-ICS-002**Test Case ID: TC-AS-FI-ICS-003**

Scope	Examination of technical specifications for acquisition software used for integrated camera systems in order to obtain digitised facial images. This test case covers the illumination.	
Preconditions	- The integrated camera system used in the specific application profile is connected and the system is ready for capturing. - Product documentation of the integrated camera system is at hand (e.g. data sheet, manual).	
CTS Mode	Not supported	
Description	Step	Description / Expected Result
	1	<i>Description:</i> - Verify that the active setting of the integrated camera system used for the specific application provides diffuse lighting adapting to the environmental light conditions. - Consult product documentation of the integrated camera system and options in the firmware.

Test Case ID: TC-AS-FI-ICS-003*Expected Result:*

- The setting provides diffuse lighting adapting to the environmental light conditions.
- The setting is the active setting for the application profile.

Table 2.56 Test Case ID: TC-AS-FI-ICS-003**Test Case ID: TC-AS-FI-ICS-004**

<i>Scope</i>	Examination of technical specifications for acquisition software used for integrated camera systems in order to obtain digitised facial images. This test case covers the colour space of the captured facial image. Note, this requirement is OPTIONAL for scenarios where only a facial verification is performed.	
<i>Preconditions</i>	• The integrated camera system used in the specific application profile is connected and the system is ready for capturing. • Product documentation of the camera unit is at hand (e.g. data sheet, manual).	
<i>CTS Mode</i>	Not supported	
<i>Description</i>	Step	Description / Expected Result
	1	<i>Description:</i> • Verify that the active setting of the integrated camera system used for the specific application provides the ability to capture images in colour. • Consult product documentation of the integrated camera system and options in the firmware. <i>Expected Result:</i> The setting provides the ability to capture images in colour (24 bit sRGB).

Table 2.57 Test Case ID: TC-AS-FI-ICS-004

2.2.5 Test Cases FM AS-FI-ICS2

Test Case ID: TC-AS-FI-ICS2-001

<i>Scope</i>	Examination of technical specifications for acquisition software used for integrated camera systems in order to obtain digitised facial images. This test case covers the image formats supported by the acquisition software of the camera system.	
<i>Preconditions</i>	Product documentation of the utilised camera system is at hand (e.g. data sheet, manual).	
<i>CTS Mode</i>	Not supported	
<i>Description</i>	Step	Description / Expected Result
	1	<i>Description:</i> Consult the product documentation of the camera system and check the supported image formats. <i>Expected Result:</i> The product documentation states that the software provides raw uncompressed image data e.g. YUV422 or sRGB or the software provides a lossless image container format (e.g. BMP, TIFF).

Table 2.58 Test Case ID: TC-AS-FI-ICS2-001**Test Case ID: TC-AS-FI-ICS2-002**

<i>Scope</i>	Examination of technical specifications for acquisition software used for integrated camera systems in order to obtain digitised facial images. This test case covers the minimum physical resolution of the captured facial image. Note, this requirement is OPTIONAL for scenarios where only a facial verification is performed.
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Test Case ID: TC-AS-FI-ICS2-002

<i>Preconditions</i>	- The integrated camera system used in the specific application profile is connected and the system is ready for capturing. - Product documentation of the camera unit is at hand (e.g. data sheet, manual).				
<i>CTS Mode</i>	Not supported				
<i>Description</i>	<table border="1"> <thead> <tr> <th>Step</th><th>Description / Expected Result</th></tr> </thead> <tbody> <tr> <td>1</td><td> <i>Description:</i> - Verify that the active camera setting (e.g. configurable via camera firmware) used for the specific application provides an image resolution of at least 1244 x 1600 pixels without any up-scaling. - Consult product documentation of the camera and options in the cameras firmware. <i>Expected Result:</i> - The setting provides a native image resolution of at least 1244 x 1600 pixels. - This setting is the active setting for the application profile. </td></tr> </tbody> </table>	Step	Description / Expected Result	1	<i>Description:</i> - Verify that the active camera setting (e.g. configurable via camera firmware) used for the specific application provides an image resolution of at least 1244 x 1600 pixels without any up-scaling. - Consult product documentation of the camera and options in the cameras firmware. <i>Expected Result:</i> - The setting provides a native image resolution of at least 1244 x 1600 pixels. - This setting is the active setting for the application profile.
Step	Description / Expected Result				
1	<i>Description:</i> - Verify that the active camera setting (e.g. configurable via camera firmware) used for the specific application provides an image resolution of at least 1244 x 1600 pixels without any up-scaling. - Consult product documentation of the camera and options in the cameras firmware. <i>Expected Result:</i> - The setting provides a native image resolution of at least 1244 x 1600 pixels. - This setting is the active setting for the application profile.				

Table 2.59 Test Case ID: TC-AS-FI-ICS2-002**Test Case ID: TC-AS-FI-ICS2-003**

<i>Scope</i>	Examination of technical specifications for acquisition software used for integrated camera systems in order to obtain digitised facial images. This test case covers the illumination.				
<i>Preconditions</i>	- The integrated camera system used in the specific application profile is connected and the system is ready for capturing. - Product documentation of the integrated camera system is at hand (e.g. data sheet, manual).				
<i>CTS Mode</i>	Not supported				
<i>Description</i>	<table border="1"> <thead> <tr> <th>Step</th><th>Description / Expected Result</th></tr> </thead> <tbody> <tr> <td>1</td><td> <i>Description:</i> - Verify that the active setting of the integrated camera system used for the specific application provides diffuse lighting adapting to the environmental light conditions. - Consult product documentation of the integrated camera system and options in the firmware. <i>Expected Result:</i> - The setting provides diffuse lighting adapting to the environmental light conditions. - The setting is the active setting for the application profile. </td></tr> </tbody> </table>	Step	Description / Expected Result	1	<i>Description:</i> - Verify that the active setting of the integrated camera system used for the specific application provides diffuse lighting adapting to the environmental light conditions. - Consult product documentation of the integrated camera system and options in the firmware. <i>Expected Result:</i> - The setting provides diffuse lighting adapting to the environmental light conditions. - The setting is the active setting for the application profile.
Step	Description / Expected Result				
1	<i>Description:</i> - Verify that the active setting of the integrated camera system used for the specific application provides diffuse lighting adapting to the environmental light conditions. - Consult product documentation of the integrated camera system and options in the firmware. <i>Expected Result:</i> - The setting provides diffuse lighting adapting to the environmental light conditions. - The setting is the active setting for the application profile.				

Table 2.60 Test Case ID: TC-AS-FI-ICS2-003**Test Case ID: TC-AS-FI-ICS2-004**

<i>Scope</i>	Examination of technical specifications for acquisition software used for integrated camera systems in order to obtain digitised facial images. This test case covers the colour space of the captured facial image.		
<i>Preconditions</i>	- The integrated camera system used in the specific application profile is connected and the system is ready for capturing. - Product documentation of the camera unit is at hand (e.g. data sheet, manual).		
<i>CTS Mode</i>	Not supported		
<i>Description</i>	<table border="1"> <thead> <tr> <th>Step</th><th>Description / Expected Result</th></tr> </thead> <tbody> </tbody> </table>	Step	Description / Expected Result
Step	Description / Expected Result		

Test Case ID: TC-AS-FI-ICS2-004

1

Description:

- Verify that the active setting of the integrated camera system used for the specific application provides the ability to capture images in colour.
- Consult product documentation of the integrated camera system and options in the firmware.

Expected Result:

The setting provides the ability to capture images in colour (24 bit sRGB).

Table 2.61 Test Case ID: TC-AS-FI-ICS2-004**Test Case ID: TC-AS-FI-ICS2-005**

Scope Examination of the software module used for acquisition of digitised facial images

- Preconditions*
- The integrated camera system used in the specific application profile is connected and the system is ready for capturing.
 - Product documentation of the camera unit is at hand (e.g. data sheet, manual).

CTS Mode Not supported

*Description***Step** **Description / Expected Result**

1

Description:

Check proper data encoding, resolution and format of the returned image

- Capture an image with the camera, examine the digitised image data
- Check the resulting image for visible compression artefacts

Expected Result:

- The image is at least of resolution 1244 x 1600 pixels with an inter eye distance of at least 300 pixels and stored in colour (24 bit sRGB).
- There are no visible compression artefacts in the resulting image

Table 2.62 Test Case ID: TC-AS-FI-ICS2-005**Test Case ID: TC-AS-FI-ICS2-006**

Scope Examination of the software module used for acquisition of digitised facial images

- Preconditions*
- The integrated camera system used in the specific application profile is connected and the system is ready for capturing.
 - Product documentation of the camera unit is at hand (e.g. data sheet, manual).

CTS Mode Not supported

*Description***Step** **Description / Expected Result**

1

Description:

Check proper uniform background in case the background meets the requirements of [ISO_FACE] and [ICAO_TR_Portrait_Quality].

Capture an image with the camera, examine the digitised image data.

Expected Result:

- There are no visible background artefacts in the resulting image.
- There is a uniform background according to [ISO_FACE] and [ICAO_TR_Portrait_Quality].

2

Description:

Check proper uniform background in case the background was eliminated and replaced.

Capture an image with the camera, examine the digitised image data.

Test Case ID: TC-AS-FI-ICS2-006

Expected Result:

- There are no visible background artefacts in the resulting image.
- The face, hair and shoulders are not changed (includes cutting off).
- There is a uniform grey background (i.e. R=G=B) and the grey is between #A1A1A1 and #E1E1E1.

Table 2.63 Test Case ID: TC-AS-FI-ICS2-006

Test Case ID: TC-AS-FI-ICS2-007

Scope Examination of the software module used for acquisition of digitised facial images with focus on the active camera settings.

Preconditions

- The IUT is in operation, required modules are loaded.
- An Integrated Camera System (ICS) is connected
- IUT is configured to perform the acquisition of a facial image.
- A testperson is available to test the IUT's behaviour.

CTS Mode

- interactive
- no provision of pre-defined input data
- HTTP method: GET
- test case variants:

/TR03122/TC-AS-FI-ICS2-007/1

Table 2.65 TC-AS-FI-ICS2-007 Test Case Variants

Description

Step	Description / Expected Result
1	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • Capture one image of the testperson with the ICS using the IUT. Examination of the acquired Image. <i>Expected Result:</i> • The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • The response contains the correct number of elements based on the XPath expression <code>./bio:Records/bio:BinaryRecord[@type='bmp']</code>. Exactly 1 element is expected in accordance to the acquisition process. • The data embedded in the element <code>./bio:Records/bio:BinaryRecord[@type='bmp']</code> of the response XML is base64-encoded BMP. • The image captured by the ICS has a resolution of at least 1244 x 1600. • The image has a colour space of 24bit RGB. • The image depicts the testperson in the moment of the acquisition. • The image does not contain compression artefacts.

Table 2.64 Test Case ID: TC-AS-FI-ICS2-007

Test Case ID: TC-AS-FI-ICS2-008

Scope Examination of the software module used for acquisition of digitised facial images with focus on the background elimination.

Applicability This TC only applies to IUT which perform a software-based segmentation.

Test Case ID: TC-AS-FI-ICS2-008*Preconditions*

- The IUT is in operation, required modules are loaded.
- The groundtruth database is at hand.
- The facial images in the database are either 24bit RGB or 8bit grayscale.
- The following metrics shall be used to calculate the performance of the image segmentation: Intersection over Union and Hausdorff Distance.
- The following conditions apply regarding the database:
 - The image data that is sent to the IUT is of the following biometric type: FACE.
 - The groundtruth database contains pairs of images in format BMP.
 - Each pair consists of a facial image with non-uniform background and an image with the segmentation result, i.e. an image of the same dimensions as the facial image with pixels of either red, black or white. A pixel in the segmentation result is black iff the corresponding pixel in the facial image is a background-pixel, otherwise it is red or white. Iff the pixel is part of the critical "core region" of the face (i.e. the area of the face containing mouth, eyes, nose) it is red. Otherwise, the pixels are white.
 - For each image the groundtruth-db contains dedicated thresholds for each of the metrics.
 - The images in the database have dimensions of 1600*1244.
 - The facial images in the database are either 24bit RGB or 8bit grayscale.

CTS Mode

- automatic
- The image data that is sent to the IUT may have the following format:
 - BMP
- HTTP method: POST
- test case variants:

/TR03122/TC-AS-FI-ICS2-008/1

Table 2.67 TC-AS-FI-ICS2-008 Test Case Variants*Description*

Step	Description / Expected Result
1	<i>Description:</i> • The IUT is called via the conformance test interface and an image from the database is provided. The image meets the properties described in the preconditions. • Calculate the Hausdorff distance. • Calculate the IoU-score.

Test Case ID: TC-AS-FI-ICS2-008

Expected Result:

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response contains the correct number of elements based on the XPath expression `./bio:Records/bio:BinaryRecord[@type='bmp']`. Exactly 2 elements are expected in accordance to the acquisition process.
- The response contains the correct number of elements based on the XPath expression `./bio:Records/bio:BinaryRecord[@type='bmp' and @externalReference='facialImage']`. Exactly 1 element is expected in accordance to the acquisition process.
- The response contains the correct number of elements based on the XPath expression `./bio:Records/bio:BinaryRecord[@type='bmp' and @externalReference='segmentationPrediction']`. Exactly 1 element is expected in accordance to the acquisition process.
- The data embedded in the elements `./bio:Records/bio:BinaryRecord[@type='bmp']` of the response XML are BMP.
- The image containing the facial image with eliminated background has a resolution of 1244 x 1600.
- The image containing the segmentation prediction has a resolution of 1244 x 1600.
- The calculated Hausdorff Distance is below the threshold from the ground-truth database.
- The calculated IoU is below the threshold from the groundtruth database.
- The pixels of facial image demarked as "non-background" by the segmentation prediction image are identical to the "non-background" pixels of the corresponding request image.
- The pixels of facial image demarked as "background" by the segmentation prediction image are of gray color in between #A1A1A1 and #E1E1E1.
- None of the pixels classified as critical core region (i.e. the red pixels from the segmentation result) is classified as a background pixel by the IUT (i.e. a black pixel in the segmentation prediction).

Evaluation

Evaluation Step Description / Evaluation Result

1

Description:

- Examine the response returned by the IUT via the test interface.

Expected Result:

- Overall, less than 5 percent of the requested images are not handled as expected.

Table 2.66 Test Case ID: TC-AS-FI-ICS2-008

Test Case ID: TC-AS-FI-ICS2-009

Scope

Examination of the software module used for acquisition of digitised facial images with focus on the background elimination.

Applicability

This TC applies to IUT which perform a segmentation based on additional hardware (for example additional cameras for background information). It shall be executed as an alternative to TC-AS-FI-ICS2-008 when the image-based approach is not feasible.

Test Case ID: TC-AS-FI-ICS2-009*Preconditions*

- The IUT is in operation, required modules are loaded.
- A reference implementation for background replacement is at hand.
- The IUT is configured such that for every captured image two versions of the captured image are released: One of the images is an original presentation of the biometric subject during the capture, the other one is the result of the application of the background elimination feature.
- It is possible to perform this testcase under different conditions including: cluttered Background, uniform background in a distinct color (e.g. blue or green), subject with short hair (ear-length or tied back), subject with long hair (not tied back), subject wearing glasses, subject wearing patterned clothes

CTS Mode

- interactive
- no provision of pre-defined input data
- HTTP method: GET
- test case variants:

/TR03122/TC-AS-FI-ICS2-009/1

Table 2.69 TC-AS-FI-ICS2-009 Test Case Variants*Description*

Step	Description / Expected Result
1	<i>Description:</i> • Capture a facial image of a biometric subject using the IUT in order to test the segmentation. <i>Expected Result:</i> • The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • The response contains the correct number of elements based on the XPath expression <code>./bio:Records/bio:BinaryRecord[@type='jpeg']</code>. Exactly 2 elements are expected in accordance to the acquisition process. • The image in <code>./bio:Records/bio:BinaryRecord[@type='jpeg' and @externalReference='segmentedImage']</code> in the response is the segmented image, i.e. an image of the same dimensions as the facial image with grey pixels. A pixel in the segmented image is grey if the corresponding pixel in the facial image is a background-pixel. • The image in <code>./bio:Records/bio:BinaryRecord[@type='jpeg' and @externalReference='originalImage']</code> in the response is the non-segmented image, i.e. a facial image without any added grey pixels. This is the only difference between the images (i.e. there is no difference regarding cropping, rotation, colorgrading etc.). • The image containing the facial image with background has a resolution of 1244 x 1600. • The image containing the facial image with eliminated background has a resolution of 1244 x 1600. • The pixels of facial image demarked as "background" and replaced by the IUT image are of the same gray color in between #A1A1A1 and #E1E1E1. It is allowed to blur the transition between original and segmented image. • The noise of the image is less than 0.5/FF.
2	<i>Description:</i> • Evaluate the segmented image for changes in the core facial region, i.e. the triangular region including eyes, nose and mouth; glasses are included. • Evaluate the segmented image for changes in the outline. This includes the head, shoulders, clothes and accessoires (i.e. earrings).

Test Case ID: TC-AS-FI-ICS2-009

Expected Result:

- No change is made in the core region. This includes individual pixels.
- No area of at least 3px x 3px in the foreground is removed or blurred. This includes for example the removal of an accessoire, removal of parts of the shoulder or the head. The removal of hair does not alter the shape or hair-style (for example by removing asymmetric parts).

3

Description:

- Use the reference implementation to remove the background in the original image.
- Calculate the IoU and Hausdorff distance between the background replaced images from the IUT and the reference implementation.

Expected Result:

- The IoU in between the IUT response and the reference implementation is less than 10%.
- The Hausdorff distance in between the IUT response and the reference implementation is less than 10.
- No background object bigger than 3px x 3px is visible or reconstructable in the image.
- Note: The difference in between the IUT response and the reference implementation may be greater in case the IUT response performs better than the reference implementation according to a check as per the evaluation step.

Evaluation

Evaluation Step	Description / Evaluation Result
1	<i>Description:</i> • Repeat this testcase 5 times under different conditions including: cluttered Background, uniform background in a distinct color (e.g. blue or green), subject with short hair (earlength or tied back), subject with long hair (not tied back), subject wearing glasses, subject wearing patterned clothes • In case the automatic evaluation of step 3 fails, the IUT response may be evaluated as better than the reference implementation according to the following conditions: <i>Expected Result:</i> • No part of the core facial region is eliminated. • Hair, clothes and accessoires are kept coherently. • No broader (> 3px x 3px) part of the hair, clothes or accessoires is removed or blurred. This includes objects which are located behind the ear or with considerable distance to the facial region (e.g. part of the hairstyle, collar, earrings). • Singular small objects (e.g. individual hairs or fringes) may be altered. • Small gaps (< 3px x 3px) between hairs or between glasses and face do not have to be eliminated. • No object located in the background is kept as a whole. • The transition between foreground and background may be kept (distance 5px). • Blurred regions count as altered/eliminated.

Table 2.68 Test Case ID: TC-AS-FI-ICS2-009

2.2.6 Test Cases FM AS-FI-ICS3

Test Case ID: TC-AS-FI-ICS3-001

<i>Scope</i>	Examination of correctness of the software module used for integrated camera systems in order to obtain digitised facial images for detection of multiple faces.
<i>Preconditions</i>	• The IUT is in operation, required modules are loaded. • The Integrated Camera System (ICS) used in the specific application scenario is connected. • The IUT implements an interface for conformance testing where a single image can be captured. • The IUT is able to return the results for every captured image via the test interface. • At least two test subjects are available to test the IUT's behaviour..
<i>CTS Mode</i>	• interactive • no provision of pre-defined input data • HTTP method: GET • test case variants:

/TR03122/TC-AS-FI-ICS3-001/1

Table 2.71 TC-AS-FI-ICS3-001 Test Case Variants

<i>Description</i>	Step	Description / Expected Result
	1	<i>Description:</i> • A single biometric subject is present in the IUT's capture area. • Initiate test case by calling the IUT via the test interface. • Perform a compliant face presentation according to FM AS-FI-ICS3. <i>Expected Result:</i> • The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • The response contains the correct number of elements based on the XPath expression //bio:FaceAcquisition. Exactly 1 element is expected in accordance to the acquisition process. • The acceptance conditions according are satisfied and achieved (i.e. no multiple faces detected). • The response contains the encoded facial image. • The acquired image contained in the response depicts the compliant facial image presentation according to FM AS-FI-ICS3.
	2	<i>Description:</i> • More than one biometric subjects are present in the IUT's capture area. • Initiate test case by calling the IUT via the test interface. • Perform a face presentation containing multiple faces, i.e. the face presentation is compliant apart from the fact that multiple faces are visible.

Test Case ID: TC-AS-FI-ICS3-001

Expected Result:

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- Facial image acquisition is omitted by the IUT.
- The response contains the correct number of elements based on the XPath expression `//bio:Records/bio:BinaryRecord/tradd:BinaryData/tradd:Value/node()`. No elements are expected to appear in accordance to the acquisition process.
- In the response, the acceptance conditions are not met, i.e. the total result of the acquisition is negative.

Table 2.70 Test Case ID: TC-AS-FI-ICS3-001

Test Case ID: TC-AS-FI-ICS3-002

Scope	Examination of correctness of the software module used for integrated camera systems in order to obtain digitised facial images for detection of multiple faces.
Preconditions	• The IUT is in operation, required modules are loaded. • The Integrated Camera System (ICS) used in the specific application scenario is connected • The IUT implements an interface for conformance testing where a single image can be captured. • The IUT is able to return the results for every captured image via the test interface. • A non-zero timeout is set in the IUT for terminating the process if the face of the biometric subject is leaving the capture area. • A testperson is available to test the IUT's behaviour..
CTS Mode	• interactive • no provision of pre-defined input data • HTTP method: GET • test case variants:

/TR03122/TC-AS-FI-ICS3-002/1

Table 2.73 TC-AS-FI-ICS3-002 Test Case Variants

Description	Step	Description / Expected Result
	1	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • By leaving the capture area the test subject does not perform a face presentation, i.e. the test subject moves such that the face is not further recognisable, e.g. through moving sideways according to FM AS-FI-ICS3. <i>Expected Result:</i> • The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • Facial image acquisition by the IUT is omitted. • The process terminates after the timer which starts at the point of leaving the capture area reaches the preset timeout. • The response contains the correct number of elements based on the XPath expression <code>//bio:Records/bio:BinaryRecord/tradd:BinaryData/tradd:Value/node()</code>. No elements are expected to appear in accordance to the acquisition process. • The response contains information about the configured timeout and that it was reached.

Table 2.72 Test Case ID: TC-AS-FI-ICS3-002

2.2.7 Test Cases FM AS-FP-MF

Test Case ID: TC-AS-FP-MF-001

Scope	Examination of correct functionality of the fingerprint scanners pre-qualification regarding a medium/default setting and test configurability of pre-qualification thresholds, regarding relaxed and strict settings (if supported by the IUT).
Applicability	The TC is only applicable if the IUT allows multiple thresholds for pre-qualification.
Preconditions	• The IUT is in operation, required modules are loaded. • The multi fingerprint scanner used in the specific application scenario is connected • The IUT implements an interface for conformance testing where a single fingerprint image can be captured. This might be done by de-selecting not required fingers. • The IUT is able to return the results for every captured image via the test interface. • A medium pre-qualification threshold (e.g. default value, via software/firmware, consult product documentation) is configured in the IUT. • A testperson is available to test the IUT's behaviour.
CTS Mode	• interactive • no provision of pre-defined input data • HTTP method: GET • test case variants:

/TR03122/TC-AS-FP-MF-001/1

Table 2.75 TC-AS-FP-MF-001 Test Case Variants

Description	Step	Description / Expected Result
	1	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • Place fingers on the scanner, at first with slightest pressure. • Successively increase pressure and improve finger position on the scanner, until auto capture is performed. <i>Expected Result:</i> • The automatic activation of the acquisition does not occur immediately. • With improving fingerprint quality over time, the scanner performs the automatic acquisition. • A message (i.e. the IUT informs the user) is given that the required fingerprint was captured successfully. • The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • The response contains the encoded fingerprint images in Windows Bitmap Version 3 (BMP) format. • The image depicts the testperson (i.e. their fingers or face) in the moment of the acquisition.
	2	<i>Description:</i> • Configure pre-qualification threshold as relaxed as possible. <i>Expected Result:</i> • The configuration is successful.

Test Case ID: TC-AS-FP-MF-001

3	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • Place fingers on the scanner, at first with slightest pressure. • Successively increase pressure and improve finger position on the scanner <i>Expected Result:</i> • The automatic activation of the scanner triggers early (in comparison to step 1.). • A message (i.e. the IUT informs the user) is given that the required fingerprint was captured successfully. • The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • The response contains the encoded fingerprint images in Windows Bitmap Version 3 (BMP) format. • The image depicts the testperson (i.e. their fingers or face) in the moment of the acquisition.
4	<i>Description:</i> • Configure pre-qualification threshold as strict as possible. <i>Expected Result:</i> • The configuration is successful.
5	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • Place fingers on the scanner, at first with slightest pressure. • Successively increase pressure and improve finger position on the scanner. <i>Expected Result:</i> • The automatic activation of the scanner triggers very late (in comparison to step 1.) or not at all. • A message (i.e. the IUT informs the user) is given that the required fingerprint was captured successfully respectively no message is given at all. • The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • If images were acquired: The response contains the encoded fingerprint image in Windows Bitmap Version 3 (BMP) format.

Table 2.74 Test Case ID: TC-AS-FP-MF-001

Test Case ID: TC-AS-FP-MF-002

<i>Scope</i>	Examination of the fingerprint scanners resulting image resolution.	
<i>Preconditions</i>	Product documentation of the fingerprint scanner model is at hand (e.g. data sheet, manual, manufacturer declaration).	
<i>CTS Mode</i>	Not supported	
<i>Description</i>	Step	Description / Expected Result
	1	<i>Description:</i> • Verify that the resolution of the fingerprint scanner fulfils the requirements for acquisition setting levels 31 or 41 [ISO_FINGER] • Consult product documentation

Test Case ID: TC-AS-FP-MF-002*Expected Result:*

The documentation attests that the requirements are fulfilled.

Table 2.76 Test Case ID: TC-AS-FP-MF-002**Test Case ID: TC-AS-FP-MF-003**

<i>Scope</i>	Examination of the fingerprint scanners pre-qualification regarding their configuration.	
<i>Preconditions</i>	Product documentation of the fingerprint scanner model is at hand (e.g. data sheet, manual, manufacturer declaration).	
<i>CTS Mode</i>	Not supported	
<i>Description</i>	Step	Description / Expected Result
	1	<i>Description:</i> • Verify that the pre-qualification is configurable by a system administrator • Consult product documentation <i>Expected Result:</i> The documentation attests that the requirements are fulfilled.

Table 2.77 Test Case ID: TC-AS-FP-MF-003

2.2.8 Test Cases FM AS-FP-ROLL

Test Case ID: TC-AS-FP-ROLL-001

Scope	Examination of the software module used for rolled fingerprint acquisition
Preconditions	• The IUT is in operation, required modules are loaded. • The fingerprint scanner providing rolled fingerprints used in the specific application scenario is connected • The IUT implements an interface for conformance testing where a rolled fingerprint image can be captured. • The IUT is able to return the results for every captured image via the test interface. • The following test resources are necessary: • 5 test subjects are available to perform the acquisition with. • A trained operator is available to perform the capture process with the test persons.
CTS Mode	• interactive • no provision of pre-defined input data • HTTP method: GET • test case variants:
/TR03122/TC-AS-FP-ROLL-001/1	

Test Case ID: TC-AS-FP-ROLL-001

1

Description:

- Initiate test case by calling the IUT via the test interface.
- Perform the flat acquisition process of all fingers in order to obtain the necessary reference data for the control verification. Ensure sufficient quality and correct execution throughout the process.
- For each finger of the test person: Perform a rolled fingerprint acquisition, rolling the finger uniformly from nail to nail. It is not required to acquire the fingerprints of all test persons directly in line.
- Examine the response returned by the IUT via the test interface.

Expected Result:

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response contains the encoded "flat" fingerprint images.
- The response contains the correct number of elements based on the XPath expression `//bio:FingerAcquisition[bio:FingerCaptureMode = "roll"]/bio:Records/bio:BinaryRecord/tradd:BinaryData/tradd:Value/node()`. Exactly 10 elements are expected in accordance to the acquisition process.
- The resulting image shows the full width of the rolled fingerprint. Typically, the image will roughly be of rectangular shape.
- The captured fingerprint image does not depict visible distortion or interruptions. The captured fingerprint image depicts the fingerprint from nail to nail.
- The captured fingerprint image depicts a faithful reproduction of the fingerprint, especially in the areas where the rolled fingerprint overlaps with the corresponding plain print.
- The captured fingerprint image does not depict puzzle effects such that parts of the fingerprint image are displaced from their actual position
- The captured fingerprint image clearly depicts friction ridges.
- The captured fingerprint image does not depict blurring and smearing.
- The captured fingerprint image clearly depicts ridge patterns.
- If features exists for the given fingerprint: The captured fingerprint image clearly depicts features.
- If loop features exists for the given fingerprint: The captured fingerprint image clearly depicts loop features (core and delta).
- The captured fingerprint image clearly depicts existing features at the border zone of the image.
- The captured fingerprint image depicts the fingerprint's upper part.
- The captured fingerprint image depicts the fingerprint's core area with ridge lines
- If delta features exists for the given fingerprint: The captured fingerprint image depicts the fingerprints delta features.
- The captured fingerprint image depicts the fingerprint's baseline (bottom area).
- The vertical axis of the fingerprint depicted in the captured image is in parallel with the fingerprint image's vertical axis.

Evaluation

Evaluation Step	Description / Evaluation Result
-----------------	---------------------------------

Test Case ID: TC-AS-FP-ROLL-001

1

Description:

- The TC shall be repeated for each test person. It is required to acquire all ten fingerprints of different test persons and not from one test person several times to achieve five capture processes.
- For each capture process: Check the Expected Results. In case of doubt whether one or several of the expected results hold true for the given fingerprint, consult a dactyloscopy expert for clarification.

Expected Result:

- The fingerprints resulting from each of the five capture process meet the expectations formulated above.

Table 2.78 Test Case ID: TC-AS-FP-ROLL-001**Test Case ID: TC-AS-FP-ROLL-002**

Scope Examination of the fingerprint scanners resulting image resolution

Preconditions Product documentation of the fingerprint scanner model is at hand (e.g. data sheet, manual, manufacturer declaration)

CTS Mode Not supported

Description

Step Description / Expected Result

1

Description:

Verify that the resolution of the fingerprint scanner fulfils the requirements for acquisition setting levels 31 or 41 [ISO_FINGER]

Consult product documentation

Expected Result:

The documentation attests that the requirements are fulfilled

Table 2.80 Test Case ID: TC-AS-FP-ROLL-002**Test Case ID: TC-AS-FP-ROLL-003**

Scope Examination of the software module used for rolled fingerprint acquisition

- Preconditions*
- The IUT is in operation, required modules are loaded.
 - The fingerprint scanner providing rolled fingerprints used in the specific application scenario is connected
 - The following test resources are necessary:
 - One test subject is available to perform the acquisition with.
 - A trained operator is available to perform the capture process with the test person. The operator is required to be trained in capturing fingerprints of persons to ensure a realistic finger roll process.

- CTS Mode*
- interactive
 - no provision of pre-defined input data
 - HTTP method: GET
 - test case variants:

/TR03122/TC-AS-FP-ROLL-003/1

Table 2.82 TC-AS-FP-ROLL-003 Test Case Variants

Description

Step Description / Expected Result

Test Case ID: TC-AS-FP-ROLL-003

1

Description:

- Configure the IUT for the rolled acquisition of fingerprints.
- Capture rolled fingerprint images of the test person.
- Roll finger from nail to nail, causing blurring and smearing each time.
- Examine the images resulting from the acquisition.

Expected Result:

- The captured fingerprint images both depict blurring and smearing.
- In both cases the acquisition software detects the occurrence of blurring or smearing.
- In both cases a warning is shown to the operator.
- In both cases the acquisition software does not perform any correction concerning the blurring and smearing.

Table 2.81 Test Case ID: TC-AS-FP-ROLL-003

2.2.9 Test Cases FM AS-FP-SF

Test Case ID: TC-AS-FP-SF-001

Scope Examination of correct functionality of the fingerprint scanners pre-qualification regarding a medium/default setting and test configurability of pre-qualification thresholds, regarding relaxed and strict settings (if supported by the IUT).

Applicability The TC is only applicable if the IUT allows multiple thresholds for pre-qualification.

Preconditions

- The IUT is in operation, required modules are loaded.
- The single fingerprint scanner used in the specific application scenario is connected
- The IUT implements an interface for conformance testing where a single fingerprint image can be captured. This might be done by de-selecting not required fingers.
- The IUT is able to return the results for every captured image via the test interface.
- A medium pre-qualification threshold (e.g. default value, via software/firmware, consult product documentation) is configured in the IUT.
- A testperson is available to test the IUT's behaviour..

CTS Mode

- interactive
- no provision of pre-defined input data
- HTTP method: GET
- test case variants:

/TR03122/TC-AS-FP-SF-001/1

Table 2.84 TC-AS-FP-SF-001 Test Case Variants

Description

Step Description / Expected Result

1

Description:

- Initiate test case by calling the IUT via the test interface.
- Place single finger on the scanner, at first with slightest pressure.
- Successively increase pressure and improve finger position on the scanner, until auto capture is performed.

Test Case ID: TC-AS-FP-SF-001

Expected Result:

- The automatic activation of the acquisition does not occur immediately.
- With improving fingerprint quality over time, the scanner performs the automatic acquisition.
- A message (i.e. the IUT informs the user) is given that the required fingerprint was captured successfully.
- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response contains the correct number of elements based on the XPath expression `//bio:Records/bio:BinaryRecord/tradd:BinaryData/tradd:Value/node()`. Exactly 1 element is expected in accordance to the acquisition process.

2

Description:

- Configure pre-qualification threshold as relaxed as possible.

Expected Result:

- The configuration is successful.

3

Description:

- Initiate test case by calling the IUT via the test interface.
- Place single finger on the scanner, at first with slightest pressure.
- Successively increase pressure and improve finger position on the scanner, until auto capture is performed.

Expected Result:

- With improving fingerprint quality over time, the scanner performs the automatic acquisition.
- The automatic activation of the scanner triggers early (in comparison to step 1.).
- A message (i.e. the IUT informs the user) is given that the required fingerprint was captured successfully.
- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response contains the correct number of elements based on the XPath expression `//bio:Records/bio:BinaryRecord/tradd:BinaryData/tradd:Value/node()`. Exactly 1 element is expected in accordance to the acquisition process.

4

Description:

- Configure pre-qualification threshold as strict as possible.

Expected Result:

- The configuration is successful.

5

Description:

- Initiate test case by calling the IUT via the test interface.
- Place single finger on the scanner, at first with slightest pressure.
- Successively increase pressure and improve finger position on the scanner, until auto capture is performed.

Test Case ID: TC-AS-FP-SF-001*Expected Result:*

- With improving fingerprint quality over time, the scanner performs the automatic acquisition or a timeout is reached.
- The automatic activation of the scanner triggers very late (in comparison to step 1.) or not at all.
- A message (i.e. the IUT informs the user) is given that the required fingerprint was captured successfully respectively no message is given at all.
- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- If an image was acquired: The response contains the encoded fingerprint image in Windows Bitmap Version 3 (BMP) format.

Table 2.83 Test Case ID: TC-AS-FP-SF-001**Test Case ID: TC-AS-FP-SF-002**

<i>Scope</i>	Examination of the fingerprint scanners pre-qualification regarding their configuration.	
<i>Preconditions</i>	Product documentation of the fingerprint scanner model is at hand (e.g. data sheet, manual, manufacturer declaration).	
<i>CTS Mode</i>	Not supported	
<i>Description</i>	Step	Description / Expected Result
	1	<i>Description:</i> • Verify that the pre-qualification is configurable by a system administrator • Consult product documentation <i>Expected Result:</i> The documentation attests that the requirements are fulfilled.

Table 2.85 Test Case ID: TC-AS-FP-SF-002

2.2.10 Test Cases FM AS-FP-SLP

Test Case ID: TC-AS-FP-SLP-001

Scope	Examination of correctness of automatic classification of left and right hand slap.	
Preconditions	• The IUT is in operation, required modules are loaded. • The multi fingerprint scanner used in the specific application scenario is connected • The IUT implements an interface for conformance testing where a single slap image can be captured. • The IUT is able to return the results for every captured image via the test interface. • The following test resources are necessary: • One test subject is available to perform the acquisition with.	
CTS Mode	• interactive • no provision of pre-defined input data • HTTP method: GET • test case variants:	
	<table><tr><td>/TR03122/TC-AS-FP-SLP-001/1</td></tr></table>	/TR03122/TC-AS-FP-SLP-001/1
/TR03122/TC-AS-FP-SLP-001/1		

Table 2.87 TC-AS-FP-SLP-001 Test Case Variants

<i>Description</i>	Step	Description / Expected Result
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Test Case ID: TC-AS-FP-SLP-001

1

Description:

- Initiate test case by calling the IUT via the test interface.
- Place left hand slap properly on the scanner.
- Perform a capture of the left hand slap, e.g. according to PAP Task ACQ-FPS-SV-2: Capture Slap Supervised.
- Examine the response returned by the IUT via the test interface.

Expected Result:

- The activation of the acquisition occurs automatically.
- A message is given that the required fingerprints were captured successfully.
- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response contains correct information about the slap classification.
- The image depicts the testperson (i.e. their fingers or face) in the moment of the acquisition.

2

Description:

- Initiate test case by calling the IUT via the test interface.
- Place right hand slap properly on the scanner.
- Perform a capture of the right hand slap, e.g. according to PAP Task ACQ-FPS-SV-2: Capture Slap Supervised.
- Examine the response returned by the IUT via the test interface.

Expected Result:

- The activation of the acquisition occurs automatically.
- A message is given that the required fingerprints were captured successfully.
- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response contains correct information about the slap classification.
- The image depicts the testperson (i.e. their fingers or face) in the moment of the acquisition.

Table 2.86 Test Case ID: TC-AS-FP-SLP-001

Test Case ID: TC-AS-FP-SLP-002

Scope Examination of the performance and configurability of the automated hand slap classification.

Preconditions The vendor of the algorithm provides a documentation which gives information about the accuracy of the classification algorithm.

CTS Mode Not supported

*Description***Step****Description / Expected Result**

1

Description:

Examination of the documentation provided by the vendor of the classification algorithm.

Test Case ID: TC-AS-FP-SLP-002

Expected Result:

- The documentation gives information of the performance of the algorithm based on a reference database.
- The documentation gives evidence that the classification algorithm ensures an accuracy of at least 99%, i.e. 99% of left and right hand slaps are correctly classified.
- The documentation gives evidence that the classification algorithm can be calibrated for accuracy levels using configurable thresholds in case the classification algorithm returns more than two possible results, e.g. "left", "right" and "unknown".

Table 2.88 Test Case ID: TC-AS-FP-SLP-002

Test Case ID: TC-AS-FP-SLP-003

Scope Examination of correctness of automatic classification of left and right hand slap.

- Preconditions*
- The IUT is in operation, required modules are loaded.
 - The multi fingerprint scanner used in the specific application scenario is connected
 - The IUT implements an interface for conformance testing where a single slap image can be captured.
 - The IUT is able to return the results for every captured image via the test interface.
 - The following test resources are necessary:
 - One test subject is available to perform the acquisition with.

- CTS Mode*
- interactive
 - no provision of pre-defined input data
 - HTTP method: GET
 - test case variants:

/TR03122/TC-AS-FP-SLP-003/1

Table 2.90 TC-AS-FP-SLP-003 Test Case Variants

Description

Step	Description / Expected Result
1	<i>Description:</i> • Activate automated hand slap classification (e.g. via software/firmware, consult product documentation). <i>Expected Result:</i> • The configuration is successful.
2	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • Place hand slap properly on the scanner. • Perform a capture of the hand slap, e.g. according to PAP Task ACQ-FPS-SV-2: Capture Slap Supervised. • Examine the response returned by the IUT via the test interface.

Test Case ID: TC-AS-FP-SLP-003

Expected Result:

- The activation of the acquisition occurs automatically.
- A message is given that the required fingerprints were captured successfully.
- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- If a left slap capture was performed: The response contains correct information about the slap classification.
- If a right slap capture was performed: The response contains correct information about the slap classification.
- The image depicts the testperson (i.e. their fingers or face) in the moment of the acquisition.

3

Description:

- Deactivate automated hand slap classification (e.g. via software/firmware, consult product documentation).

Expected Result:

- The configuration is successful.

4

Description:

- Initiate test case by calling the IUT via the test interface.
- Place hand slap properly on the scanner.
- Perform a capture of the hand slap, e.g. according to PAP Task ACQ-FPS-SV-2: Capture Slap Supervised.
- Examine the response returned by the IUT via the test interface.

Expected Result:

- The activation of the acquisition occurs automatically.
- A message is given that the required fingerprints were captured successfully.
- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response does not contain information about the slap classification.
- The image depicts the testperson (i.e. their fingers or face) in the moment of the acquisition.

Table 2.89 Test Case ID: TC-AS-FP-SLP-003

2.3 Test Cases Biometric Image Processing

2.3.1 Test Cases FM BIP-FI-APP

Test Case ID: TC-BIP-FI-APP-001

Scope

Verify correct facial image processing with respect to the output of integrated camera systems to obtain a facial image. The metrics applied for assessment are based on [ISO_FACE] .

Test Case ID: TC-BIP-FI-APP-001*Preconditions*

- The IUT is in operation, required modules are loaded.
- The IUT implements an interface for conformance testing where a single image can be provided and the according biometric container including the image data is returned.
- The following definitions according to [ISO_FACE] are relevant for verifying the geometric requirements: The intersection of the vertical line passing through the approximate horizontal midpoints of the mouth and the bridge of the nose, and the horizontal line connecting the centres of the two eyes, is called the centre of the face. The width of the head is defined as the distance between the two imaginary lines parallel to the vertical line mentioned above. Each line is drawn between the upper and lower lobes of each ear and positioned where the external ear connects the head. The length of the head is defined as the distance between the base of the chin and the crown (i.e. the top of the head ignoring any hair) measured on the vertical line mentioned above.
- The following conditions apply regarding the database:
 - The image data that is sent to the IUT is of the following biometric type: FACE.
 - The groundtruth database contains noncompliant images w.r.t. the required quality values.
 - The database contains images, that are not cropped to the face.

CTS Mode

- automatic
- The image data that is sent to the IUT may have the following format:
 - JPEG
- HTTP method: POST
- test case variants:

/TR03122/TC-BIP-FI-APP-001/1

Table 2.92 TC-BIP-FI-APP-001 Test Case Variants*Description*

Step	Description / Expected Result
1	<i>Description:</i> • The IUT is called via the conformance test interface and an image from the database is provided. The image meets the properties described in the preconditions. • Check correct processing of the output of integrated camera systems. • Determine the width of the processed image and the head width. • Determine the height of the processed image and the head length. • Determine the X and Y coordinates of the centre of the face, assuming that the origin of the coordinate system is in the upper left corner of the image.

Test Case ID: TC-BIP-FI-APP-001*Expected Result:*

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response contains a single facial image.
- If the facial image is used in a different scenario than verification: The colour space is sRGB.
- If the facial image is used in a different scenario than verification: The colour depth is 24 bit.
- If the facial image is used in a verification scenario: The colour space is sRGB or grey scale is used.
- If the facial image is used in a verification scenario: The colour depth is 24 bit or a colour depth of 8 bit with grey scale used.
- The face is fully visible in foreground.
- The minimum distance between both eyes is 120 pixels.
- The processed image is cropped and de-rotated.
- The position of the face within the processed image meets the geometric requirements of [ISO_FACE] , chapter 8.3, i.e.: The X coordinate of the centre of the face is between 45 % and 55 % of the image width.
- The position of the face within the processed image meets the geometric requirements of [ISO_FACE] , chapter 8.3, i.e.: The Y coordinate of the centre of the face is between 30 % and 50 % (for children under the age of 11 between 30 % and 60 %) of the image height.
- The size of the face within the processed image meets the geometric requirements of [ISO_FACE] , chapter 8.3, i.e.: The head width is between 50 % and 75 % of the image width.
- The size of the face within the processed image meets the geometric requirements of [ISO_FACE] , chapter 8.3, i.e.: The head length is between 60 % and 90 % (for children under the age of 11 between 50 % and 90 %) of the vertical length of the image.

Evaluation

Evaluation Step	Description / Evaluation Result
1	<i>Description:</i> • Examine the response returned by the IUT via the test interface. <i>Expected Result:</i> • Overall, less than 0 percent of the requested images are not handled as expected.

Table 2.91 Test Case ID: TC-BIP-FI-APP-001

2.3.2 Test Cases FM BIP-FI-DC-HQ

Test Case ID: TC-BIP-FI-DC-HQ-001*Scope*

Verify correct facial image processing.

Test Case ID: TC-BIP-FI-DC-HQ-001*Preconditions*

- The IUT is in operation, required modules are loaded.
- The IUT implements an interface for conformance testing where a single image can be provided and the according biometric container including the image data is returned. If producing GSAT3 data is in scope of the considered Application Profile the IUT SHALL provide both output types in parallel (generic facial image, facial image for German Standard for AFIS Transactions Version 3 (GSAT3) transaction).
- The following conditions apply regarding the database:
 - The image data that is sent to the IUT is of the following biometric type: FACE.
 - The groundtruth database contains noncompliant images w.r.t. the required quality values.
 - The database contains images, that are not cropped to the face and are greater than 1600*1200 pixels (in both height and width).

CTS Mode

- automatic
- The image data that is sent to the IUT may have the following format:
 - JPEG
- HTTP method: POST
- test case variants:

/TR03122/TC-BIP-FI-DC-HQ-001/1

Table 2.94 TC-BIP-FI-DC-HQ-001 Test Case Variants*Description*

Step	Description / Expected Result
1	<i>Description:</i> • The IUT is called via the conformance test interface and an image from the database is provided. The image meets the properties described in the pre-conditions. • Check correct processing of generic facial images. • Examine the processed image(s). <i>Expected Result:</i> • The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • The response contains a single facial image (outside a possibly existing GSAT3 XML). • The processed image (outside a possibly existing GSAT3 XML) has dimensions of exactly 1600*1200 pixels (in regard to height and width). • If producing GSAT3 data is in scope of the considered Application Profile: The response contains a single facial image inside the GSAT3 XML. • If producing GSAT3 data is in scope of the considered Application Profile: The processed image inside the GSAT3 XML has dimensions of exactly 800*600 pixels (in regard to height and width). • The minimum distance between both eyes is at least 120 pixels for all facial images. • Padding has not been used during image processing.

Evaluation

Evaluation Step	Description / Evaluation Result
1	<i>Description:</i> • Examine the response returned by the IUT via the test interface. <i>Expected Result:</i> • Overall, less than 0 percent of the requested images are not handled as expected.

Table 2.93 Test Case ID: TC-BIP-FI-DC-HQ-001

2.3.3 Test Cases FM BIP-FI-FBS

Test Case ID: TC-BIP-FI-FBS-001

- Scope** Verify correct image processing of application forms and the cropping of the facial image on the application form using a flat bed scanner.
- Preconditions**
- The IUT is in operation, required modules are loaded.
 - The flat bed scanner used in the specific application scenario is connected
 - The IUT is able to return the results for every captured image via the test interface.
 - The following test resources are necessary:
 - As test resource: an application form with a facial image that fulfills the requirements of "full frontal" specified in [ISO_FACE].
 - No provision of pre-defined input data.

- CTS Mode**
- interactive
 - no provision of pre-defined input data
 - HTTP method: GET
 - test case variants:

/TR03122/TC-BIP-FI-FBS-001/1

Table 2.96 TC-BIP-FI-FBS-001 Test Case Variants

Description

Step	Description / Expected Result
1	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • Position the application form and the image properly on the FBS. • Perform the scanning process. • Examine the response returned by the IUT via the test interface. • Examine the image contained in the response regarding crop, width, height and colour depth and colour space. <i>Expected Result:</i> • The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • The response contains the correct number of elements based on the XPath expression <code>//bio:FaceDelivery/bio:Records/bio:BinaryRecord[contains(@copyReasons, "cropped_automatically") or contains(@copyReasons, "rotated_automatically") or not(@copyReasons)]/tradd:BinaryData/tradd:Value/node()</code>. Exactly 1 element is expected in accordance to the acquisition process. • The image image contained in the response has a resolution of 622 x 800. • The image is a coloured image with a colour space of 8 bit (greyscale) or 24 bit sRGB. • The processed image is cropped and de-rotated. • The face is centered and fully visible in the foreground. • The image depicts the testperson (i.e. their fingers or face) in the moment of the acquisition.

Table 2.95 Test Case ID: TC-BIP-FI-FBS-001

2.3.4 Test Cases FM BIP-FI-GID

Test Case ID: TC-BIP-FI-GID-001

<i>Scope</i>	Verify correct image processing and the cropping of the facial image using a digital camera or integrated camera system (ICS).
<i>Preconditions</i>	• The IUT is in operation, required modules are loaded. • A digital camera or ICS is connected • The IUT implements an interface for conformance testing where biometric data can be acquired. • The following test resources are necessary: • A test person to perform the biometric acquisition with. • No provision of pre-defined input data.
<i>CTS Mode</i>	• interactive • no provision of pre-defined input data • HTTP method: GET • test case variants:

/TR03122/TC-BIP-FI-GID-001/1

Table 2.98 TC-BIP-FI-GID-001 Test Case Variants

<i>Description</i>	Step	Description / Expected Result
	1	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • Position the biometric test subject properly in front of the digital camera or ICS. • Perform a facial acquisition process. • Examine the response returned by the IUT via the test interface. • Examine the image contained in the response regarding crop, width, height and colour depth and colour space. <i>Expected Result:</i> • The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • The response contains the correct number of elements based on the XPath expression <code>//bio:FaceAcquisition/bio:Records/bio:BinaryRecord[contains(@copyReasons, "cropped_automatically") or contains(@copyReasons, "rotated_automatically")]/tradd:BinaryData/tradd:Value/node()</code>. Exactly 1 element is expected in accordance to the acquisition process. • The image has a resolution of at least 1244 x 1600. • The image is a coloured image with a colour space of 24 bit sRGB. • The processed image is cropped and de-rotated. • The face is centered and fully visible in the foreground. • The inter eye distance is at least 300 pixels. • Padding has not been used during image processing. • The image depicts the testperson (i.e. their fingers or face) in the moment of the acquisition.

Table 2.97 Test Case ID: TC-BIP-FI-GID-001

2.3.5 Test Cases FM BIP-FP-APP

The acquisition of fingerprints can be performed with the help of a single fingerprint sensor or a multi fingerprint sensor. In the first case TC-BIP-FP-APP-001 has to be applied, in the second case TC-BIP-FP-APP-002. Furthermore, in the case of four finger sensors TC-BIP-FP-APP-003 has to be applied additionally.

Test Case ID: TC-BIP-FP-APP-001

Scope	Examination of segmentation in connection with single finger scanners.
Applicability	This TC only applies when single finger sensors are being used.
Preconditions	• The IUT is in operation, required modules are loaded. • The single fingerprint scanner used in the specific application scenario is connected • The IUT implements an interface for conformance testing where biometric data can be acquired. • The following test resources are necessary: • A test person to perform the biometric acquisition with. • No provision of pre-defined input data.
CTS Mode	• interactive • no provision of pre-defined input data • HTTP method: GET • test case variants:

/TR03122/TC-BIP-FP-APP-001/1

Table 2.100 TC-BIP-FP-APP-001 Test Case Variants

Description	Step	Description / Expected Result
	1	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • Perform a fingerprint acquisition process of one fingerprint by correct placement of this finger on the sensor (de-select the remaining fingers for the acquisition). • If applicable: Examine the image contained in the response regarding segmentation and signs of upscaling. <i>Expected Result:</i> • The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • The result contains a single element "//bio:FingerAcquisition" containing the base64 image of the fingerprint beneath ("bio:BinaryRecord/tradd:BinaryData/tradd:Value"). • A correct segmentation has been performed. In case of uncertainty, the test SHALL be repeated. Note: segmentation for single finger scanners is optional. • No upscaling has been applied to the fingerprint image. The fingerprint image may have been evenly surrounded with white pixels to reach the desired size if the targeted system or database requires fingerprint images of higher size than the captured fingerprint image.

Test Case ID: TC-BIP-FP-APP-001

2

Description:

- Initiate test case by calling the IUT via the test interface.
- Perform a fingerprint acquisition process of one fingerprint by positioning this finger rotated up to 45 degree on the sensor (de-select the remaining fingers for the acquisition). Note: In case the IUT rejects strongly rotated fingerprints such that the capture process does not yield a biometric sample, the capture process can be initiated manually.
- If applicable: Examine the image contained in the response regarding segmentation and signs of upscaling.

Expected Result:

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The result contains a single element "//bio:FingerAcquisition" containing the base64 image of the fingerprint beneath ("bio:BinaryRecord/tradd:BinaryData/tradd:Value").
- The fingerprint is corrected to be vertical in the image which is manually verified by the examiner.
- A correct segmentation has been performed. In case of uncertainty, the test SHALL be repeated. Note: segmentation for single finger scanners is optional.
- No upscaling has been applied to the fingerprint image. The fingerprint image may have been evenly surrounded with white pixels to reach the desired size if the targeted system or database requires fingerprint images of higher size than the captured fingerprint image.

3

Description:

- Initiate test case by calling the IUT via the test interface.
- Perform a fingerprint acquisition process of one fingerprint by positioning this finger such that all phalanges are captured (de-select the remaining fingers for the acquisition).
- If applicable: Examine the image contained in the response regarding segmentation and signs of upscaling.

Expected Result:

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The result contains a single element "//bio:FingerAcquisition" containing the base64 image of the fingerprint beneath ("bio:BinaryRecord/tradd:BinaryData/tradd:Value").
- The image contains only the first part of the finger.
- A correct segmentation has been performed. In case of uncertainty, the test SHALL be repeated. Note: segmentation for single finger scanners is optional.
- No upscaling has been applied to the fingerprint image. The fingerprint image may have been evenly surrounded with white pixels to reach the desired size if the targeted system or database requires fingerprint images of higher size than the captured fingerprint image.

*Evaluation***Evaluation Step****Description / Evaluation Result**

1

Description:

- The test case is repeated 10 times for every test step in order to be able to estimate the quality of the segmentation process. The fingers have to be lifted after every acquisition.

Test Case ID: TC-BIP-FP-APP-001

Expected Result:

- The results obtained during the repetition conform to the expected results. If in any repetitions the actual results diverge from the expected results then this behaviour can be related to some external influence that is not in scope of the FM, e.g. overrotating of the fingerprint, acquisition with too little pressure etc.

Table 2.99 Test Case ID: TC-BIP-FP-APP-001

Test Case ID: TC-BIP-FP-APP-002

Scope	Examination of segmentation in connection with multi finger scanners.
Applicability	This TC only applies when multi finger scanners are being used.
Preconditions	• The IUT is in operation, required modules are loaded. • The multi fingerprint scanner used in the specific application scenario is connected • The IUT implements an interface for conformance testing where biometric data can be acquired. • The following test resources are necessary: • A test person to perform the biometric acquisition with. • No provision of pre-defined input data.
CTS Mode	• interactive • no provision of pre-defined input data • HTTP method: GET • test case variants:

/TR03122/TC-BIP-FP-APP-002/1

Table 2.102 TC-BIP-FP-APP-002 Test Case Variants

Description	Step	Description / Expected Result
	1	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • Perform a fingerprint acquisition process of two fingerprints by correct placement of these fingers on the sensor (de-select the remaining fingers for the acquisition). • If applicable: Examine the images contained in the response regarding segmentation and signs of upscaling. <i>Expected Result:</i> • The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • The result contains a single element "//bio:FingerAcquisition" containing the base64 images of both fingerprints (one for each) beneath ("bio:Binary-Record/tradd:BinaryData/tradd:Value"). • A correct segmentation has been performed. In case of uncertainty, the test SHALL be repeated. • No upscaling has been applied to the fingerprint images. The fingerprint images may have been evenly surrounded with white pixels to reach the desired size if the targeted system or database requires fingerprint images of higher size than the captured fingerprint images.

Test Case ID: TC-BIP-FP-APP-002

2

Description:

- Initiate test case by calling the IUT via the test interface.
- Perform a fingerprint acquisition process of one fingerprint by correct placement of this finger on the sensor (de-select the remaining fingers for the acquisition).
- If applicable: Examine the image contained in the response regarding segmentation and signs of upscaling.

Expected Result:

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The result contains a single element "//bio:FingerAcquisition" containing the base64 image of the fingerprint beneath ("bio:BinaryRecord/tradd:BinaryData/tradd:Value").
- A correct segmentation has been performed. In case of uncertainty, the test SHALL be repeated.
- No upscaling has been applied to the fingerprint image. The fingerprint image may have been evenly surrounded with white pixels to reach the desired size if the targeted system or database requires fingerprint images of higher size than captured the fingerprint image.

3

Description:

- Initiate test case by calling the IUT via the test interface.
- Perform a fingerprint acquisition process of two fingerprints by positioning this finger rotated up to 45 degree (both fingers in the same direction) on the sensor (de-select the remaining fingers for the acquisition). Note: In case the IUT rejects strongly rotated fingerprints such that the capture process does not yield a biometric sample, the capture process can be initiated manually.
- If applicable: Examine the images contained in the response regarding segmentation and signs of upscaling.

Expected Result:

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The result contains a single element "//bio:FingerAcquisition" containing the base64 images of both fingerprints (one for each) beneath ("bio:BinaryRecord/tradd:BinaryData/tradd:Value").
- The fingerprints are corrected to be vertical in the images which is manually verified by the examiner.
- A correct segmentation has been performed. In case of uncertainty, the test SHALL be repeated.
- No upscaling has been applied to the fingerprint images. The fingerprint images may have been evenly surrounded with white pixels to reach the desired size if the targeted system or database requires fingerprint images of higher size than the captured fingerprint images.

4

Description:

- Initiate test case by calling the IUT via the test interface.
- Perform a fingerprint acquisition process of one fingerprint by positioning this finger rotated up to 45 degree on the sensor (de-select the remaining fingers for the acquisition). Note: In case the IUT rejects strongly rotated fingerprints such that the capture process does not yield a biometric sample, the capture process can be initiated manually.
- If applicable: Examine the image contained in the response regarding segmentation and signs of upscaling.

Test Case ID: TC-BIP-FP-APP-002

Expected Result:

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The result contains a single element "//bio:FingerAcquisition" containing the base64 image of the fingerprint beneath ("bio:BinaryRecord/tradd:BinaryData/tradd:Value").
- The fingerprint is corrected to be vertical in the image which is manually verified by the examiner.
- A correct segmentation has been performed. In case of uncertainty, the test SHALL be repeated.
- No upscaling has been applied to the fingerprint image. The fingerprint image may have been evenly surrounded with white pixels to reach the desired size if the targeted system or database requires fingerprint images of higher size than captured the fingerprint image.

5

Description:

- Initiate test case by calling the IUT via the test interface.
- Perform a fingerprint acquisition process of two fingerprint by positioning these fingers such that all phalanges are captured (de-select the remaining fingers for the acquisition).
- If applicable: Examine the images contained in the response regarding segmentation and signs of upscaling.

Expected Result:

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The result contains a single element "//bio:FingerAcquisition" containing both base64 images of the fingerprints (one for each) beneath ("bio:BinaryRecord/tradd:BinaryData/tradd:Value").
- The images contain only the first part of the fingers.
- A correct segmentation has been performed. In case of uncertainty, the test SHALL be repeated.
- No upscaling has been applied to the fingerprint images. The fingerprint images may have been evenly surrounded with white pixels to reach the desired size if the targeted system or database requires fingerprint images of higher size than the captured fingerprint images.

6

Description:

- Initiate test case by calling the IUT via the test interface.
- Perform a fingerprint acquisition process of one fingerprint by positioning this finger such that all phalanges are captured (de-select the remaining fingers for the acquisition).
- If applicable: Examine the image contained in the response regarding segmentation and signs of upscaling.

Test Case ID: TC-BIP-FP-APP-002*Expected Result:*

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The result contains a single element "//bio:FingerAcquisition" containing the base64 image of the fingerprint beneath ("bio:BinaryRecord/tradd:BinaryData/tradd:Value").
- The image contains only the first part of the finger.
- A correct segmentation has been performed. In case of uncertainty, the test SHALL be repeated.
- No upscaling has been applied to the fingerprint image. The fingerprint image may have been evenly surrounded with white pixels to reach the desired size if the targeted system or database requires fingerprint images of higher size than captured the fingerprint image.

Evaluation

Evaluation Step	Description / Evaluation Result
1	<i>Description:</i> • The test case is repeated 10 times for every test step in order to be able to estimate the quality of the segmentation process. The fingers have to be lifted after every acquisition. <i>Expected Result:</i> • The results obtained during the repetition conform to the expected results. If in any repetitions the actual results diverge from the expected results then this behaviour can be related to some external influence that is not in scope of the FM, e.g. overrotating of the fingerprint, acquisition with too little pressure etc.

Table 2.101 Test Case ID: TC-BIP-FP-APP-002**Test Case ID: TC-BIP-FP-APP-003**

<i>Scope</i>	Examination of segmentation in connection with four finger scanners.
<i>Applicability</i>	This TC only applies when multi finger scanners are being used.
<i>Preconditions</i>	• The IUT is in operation, required modules are loaded. • The multi fingerprint scanner used in the specific application scenario is connected • The IUT implements an interface for conformance testing where biometric data can be acquired. • The following test resources are necessary: • A test person to perform the biometric acquisition with. • No provision of pre-defined input data.
<i>CTS Mode</i>	• interactive • no provision of pre-defined input data • HTTP method: GET • test case variants:

/TR03122/TC-BIP-FP-APP-003/1

Table 2.104 TC-BIP-FP-APP-003 Test Case Variants*Description*

Step	Description / Expected Result
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Test Case ID: TC-BIP-FP-APP-003

1

Description:

- Initiate test case by calling the IUT via the test interface.
- Perform a fingerprint acquisition process of three fingerprints by correct placement of these fingers on the sensor (de-select the remaining fingers for the acquisition).
- If applicable: Examine the images contained in the response regarding segmentation and signs of upscaling.

Expected Result:

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The result contains a single element "//bio:FingerAcquisition" containing the base64 images of three fingerprints beneath (one for each) ("bio:BinaryRecord/tradd:BinaryData/tradd:Value").
- A correct segmentation has been performed. In case of uncertainty, the test SHALL be repeated. Note: segmentation for single finger scanners is optional.
- No upscaling has been applied to the fingerprint image. The fingerprint image may have been evenly surrounded with white pixels to reach the desired size if the targeted system or database requires fingerprint images of higher size than captured the fingerprint image.

2

Description:

- Initiate test case by calling the IUT via the test interface.
- Perform a fingerprint acquisition process of four fingerprints by correct placement of these fingers on the sensor (de-select the remaining fingers for the acquisition).
- If applicable: Examine the images contained in the response regarding segmentation and signs of upscaling.

Expected Result:

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The result contains a single element "//bio:FingerAcquisition" containing the base64 images of four fingerprints beneath (one for each) ("bio:BinaryRecord/tradd:BinaryData/tradd:Value").
- A correct segmentation has been performed. In case of uncertainty, the test SHALL be repeated.
- No upscaling has been applied to the fingerprint image. The fingerprint image may have been evenly surrounded with white pixels to reach the desired size if the targeted system or database requires fingerprint images of higher size than captured the fingerprint image.

3

Description:

- Initiate test case by calling the IUT via the test interface.
- Perform a fingerprint acquisition process of three fingerprints by positioning these fingers rotated up to 45 degree (both fingers in the same direction) on the sensor (de-select the remaining fingers for the acquisition). Note: In case the IUT rejects strongly rotated fingerprints such that the capture process does not yield a biometric sample, the capture process can be initiated manually.
- If applicable: Examine the image contained in the response regarding segmentation and signs of upscaling.

Test Case ID: TC-BIP-FP-APP-003

Expected Result:

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The result contains a single element "//bio:FingerAcquisition" containing the base64 images of both fingerprints (one for each) beneath ("bio:BinaryRecord/tradd:BinaryData/tradd:Value").
- The fingerprints are corrected to be vertical in the images which is manually verified by the examiner.
- A correct segmentation has been performed. In case of uncertainty, the test SHALL be repeated.
- No upscaling has been applied to the fingerprint image. The fingerprint image may have been evenly surrounded with white pixels to reach the desired size if the targeted system or database requires fingerprint images of higher size than captured the fingerprint image.

4

Description:

- Initiate test case by calling the IUT via the test interface.
- Perform a fingerprint acquisition process of four fingerprints by positioning these fingers rotated up to 45 degree (all fingers in the same direction on the sensor (de-select the remaining fingers for the acquisition). Note: In case the IUT rejects strongly rotated fingerprints such that the capture process does not yield a biometric sample, the capture process can be initiated manually.
- If applicable: Examine the images contained in the response regarding segmentation and signs of upscaling.

Expected Result:

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The result contains a single element "//bio:FingerAcquisition" containing the base64 images of the four fingerprints (one for each) beneath ("bio:BinaryRecord/tradd:BinaryData/tradd:Value").
- The fingerprints are corrected to be vertical in the images which is manually verified by the examiner.
- A correct segmentation has been performed. In case of uncertainty, the test SHALL be repeated.
- No upscaling has been applied to the fingerprint images. The fingerprint images may have been evenly surrounded with white pixels to reach the desired size if the targeted system or database requires fingerprint images of higher size than the captured fingerprint images.

5

Description:

- Initiate test case by calling the IUT via the test interface.
- Perform a fingerprint acquisition process of three fingerprint by positioning these fingers such that all phalanges are captured (de-select the remaining fingers for the acquisition).
- If applicable: Examine the images contained in the response regarding segmentation and signs of upscaling.

Test Case ID: TC-BIP-FP-APP-003

Expected Result:

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The result contains a single element "//bio:FingerAcquisition" containing both base64 images of the three fingerprints (one for each) beneath ("bio:BinaryRecord/tradd:BinaryData/tradd:Value").
- The images contain only the first part of the fingers.
- A correct segmentation has been performed. In case of uncertainty, the test SHALL be repeated. Note: segmentation for single finger scanners is optional.
- No upscaling has been applied to the fingerprint images. The fingerprint images may have been evenly surrounded with white pixels to reach the desired size if the targeted system or database requires fingerprint images of higher size than the captured fingerprint images.

6

Description:

- Initiate test case by calling the IUT via the test interface.
- Perform a fingerprint acquisition process of four fingerprints by positioning this finger such that all phalanges are captured (de-select the remaining fingers for the acquisition).
- If applicable: Examine the images contained in the response regarding segmentation and signs of upscaling.

Expected Result:

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The result contains a single element "//bio:FingerAcquisition" containing the base64 images of the four fingerprints (one for each) beneath ("bio:BinaryRecord/tradd:BinaryData/tradd:Value").
- The image contains only the first part of the fingers.
- A correct segmentation has been performed. In case of uncertainty, the test SHALL be repeated.
- No upscaling has been applied to the fingerprint image. The fingerprint images may have been evenly surrounded with white pixels to reach the desired size if the targeted system or database requires fingerprint images of higher size than the captured fingerprint images.

Evaluation

Evaluation Step	Description / Evaluation Result
1	<i>Description:</i> • The test case is repeated 10 times for every test step in order to be able to estimate the quality of the segmentation process. The fingers have to be lifted after every acquisition. <i>Expected Result:</i> • The results obtained during the repetition conform to the expected results. If in any repetitions the actual results diverge from the expected results then this behaviour can be related to some external influence that is not in scope of the FM, e.g. overrotating of the fingerprint, acquisition with too little pressure etc.

Table 2.103 Test Case ID: TC-BIP-FP-APP-003

2.4 Test Cases Quality Assessment

2.4.1 Test Cases FM QA-FI-ARE

Test Case ID: TC-QA-FI-ARE-001

Scope Examination of the software-based quality assessment for facial images within the context of alien register enrolment. The test object consists of the quality assessment software and a fixed parameter set describing the application profile specific tolerances threshold regarding several assessment components.

Preconditions

- The IUT is in operation, required modules are loaded.
- An Integrated Camera System (ICS) is connected.
- The IUT implements an interface for conformance testing where a single image can be provided and the computed quality values and configuration data are returned.
- The following conditions apply regarding the database:
 - The image data that is sent to the IUT is of the following biometric type: FACE.
 - The groundtruth database contains compliant and noncompliant images w.r.t. the required quality values.

CTS Mode

- automatic
- The image data that is sent to the IUT may have the following formats:
 - JPEG
 - JPEG2000
- HTTP method: POST
- test case variants:

/TR03122/TC-QA-FI-ARE-001/1

Table 2.106 TC-QA-FI-ARE-001 Test Case Variants

Description

Step	Description / Expected Result
1	<i>Description:</i> • The IUT is called via the conformance test interface and an image from the database is provided. The image meets the properties described in the preconditions. • The quality values contained in the IUT-response are checked against the groundtruth.

Test Case ID: TC-QA-FI-ARE-001

Expected Result:

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response contains the correct number of elements based on the XPath expression `//bio:FaceImageQuality`. Exactly 1 element is expected in accordance to the acquisition process.
- The IUT provides the quality values for all components marked with "M" in the respective table for quality threshold requirements of Function Module QA-FI-ARE
- Mandatory components: The minimum, maximum and threshold values are correctly set.
- Optional components: The minimum, maximum and threshold values are correctly set.
- For all components: The relation $min \leq tmin \leq tmax \leq max$ holds.
- If the image is a compliant presentation (w.r.t. all possible components) the IUT's assessment is positive.
- If the image is a noncompliant presentation (w.r.t. the mandatory components) the IUT's overall assessment is negative.
- Mandatory components: If the image is a compliant presentation the IUT assessment is positive for all components.
- Mandatory components: If the image is a noncompliant presentation the IUT assessment is negative for the respective component.

Evaluation

Evaluation Step	Description / Evaluation Result
1	<i>Description:</i> • Repeat the TC for all applicable images from the database. • Examine the response returned by the IUT via the test interface. <i>Expected Result:</i> • Overall, less than 5 percent of the requested images are not handled as expected.

Table 2.105 Test Case ID: TC-QA-FI-ARE-001

2.4.2 Test Cases FM QA-FI-BCL

Test Case ID: TC-QA-FI-BCL-001

<i>Scope</i>	Examination of the software-based quality assessment for facial images within the context of border control. The test object consists of the quality assessment software and a fixed parameter set describing the application profile specific tolerances threshold regarding several assessment components.
<i>Preconditions</i>	• The IUT is in operation, required modules are loaded. • The IUT implements an interface for conformance testing where a single image can be provided and the computed quality values and configuration data are returned. • The following conditions apply regarding the database: • The image data that is sent to the IUT is of the following biometric type: FACE. • The groundtruth database contains compliant and noncompliant images w.r.t. the required quality values.

Test Case ID: TC-QA-FI-BCL-001
CTS Mode

- automatic
- The image data that is sent to the IUT may have the following formats:
 - JPEG
 - JPEG2000
- HTTP method: POST
- test case variants:

/ TR03122 / TC-QA-FI-BCL-001 / 1

Table 2.108 TC-QA-FI-BCL-001 Test Case Variants

Description

Step	Description / Expected Result
1	<i>Description:</i> • The IUT is called via the conformance test interface and an image from the database is provided. The image meets the properties described in the pre-conditions. • The quality values contained in the IUT-response are checked against the groundtruth. <i>Expected Result:</i> • The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • The response contains the correct number of elements based on the XPath expression //bio:FaceImageQuality. Exactly 1 element is expected in accordance to the acquisition process. • The IUT provides the quality values for all components marked with "M" in the respective table for quality threshold requirements of Function Module QA-FI-ARE • Mandatory components: The minimum, maximum and threshold values are correctly set. • Optional components: The minimum, maximum and threshold values are correctly set. • For all components: The relation $min \leq tmin \leq tmax \leq max$ holds. • If the image is a compliant presentation (w.r.t. all possible components) the IUTs assessment is positive. • If the image is a noncompliant presentation (w.r.t. the mandatory components) the IUTs overall assessment is negative. • Mandatory components: If the image is a compliant presentation the IUT assessment is positive for all components. • Mandatory components: If the image is a noncompliant presentation the IUT assessment is negative for the respective component.

Evaluation

Evaluation Step	Description / Evaluation Result
1	<i>Description:</i> • Repeat the TC for all applicable images from the database. • Examine the response returned by the IUT via the test interface. <i>Expected Result:</i> • Overall, less than 5 percent of the requested images are not handled as expected.

Table 2.107 Test Case ID: TC-QA-FI-BCL-001

2.4.3 Test Cases FM QA-FI-GENERIC

Test Case ID: TC-QA-FI-GENERIC-001

Scope Examination of the software-based quality assessment for facial images regarding the generic quality evaluation components. Note that the examination for fixed parameters and thresholds are not part of this test case, as they are specified by the volume specific Function Module. Therefore, this test case is optional if a volume-specific QA-TC applies.

Preconditions

- The IUT is in operation, required modules are loaded.
- The IUT implements an interface for conformance testing where a single image can be provided and the computed quality values and configuration data are returned.
- The following conditions apply regarding the database:
 - The image data that is sent to the IUT is of the following biometric type: FACE.
 - The groundtruth database contains compliant and noncompliant images w.r.t. the required quality values.

CTS Mode

- automatic
- The image data that is sent to the IUT may have the following formats:
 - JPEG
 - JPEG2000
- HTTP method: POST
- test case variants:

/TR03122/TC-QA-FI-GENERIC-001/1

Table 2.110 TC-QA-FI-GENERIC-001 Test Case Variants

Description

Step	Description / Expected Result
1	<i>Description:</i> • The IUT is called via the conformance test interface and an image from the database is provided. The image meets the properties described in the preconditions. • The quality values contained in the IUT-response are checked against the groundtruth. <i>Expected Result:</i> • The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • The response contains the correct number of elements based on the XPath expression //bio:FaceImageQuality. Exactly 1 element is expected in accordance to the acquisition process. • The IUT provides the quality values for all components marked with "M" in the respective table for quality threshold requirements in the respective Function Module. • Mandatory components: The minimum, maximum and threshold values are correctly set and result is of the correct type (if it exists). • For all components: The relation $min \leq tmin \leq tmax \leq max$ holds. • For all components: The relation $min \leq result \leq max$ holds.

Evaluation

Evaluation Step	Description / Evaluation Result
1	<i>Description:</i> • Repeat the TC for all applicable images from the database. • Examine the response returned by the IUT via the test interface.

Test Case ID: TC-QA-FI-GENERIC-001*Expected Result:*

- Overall, less than 5 percent of the requested images are not handled as expected.

Table 2.109 Test Case ID: TC-QA-FI-GENERIC-001**Test Case ID: TC-QA-FI-GENERIC-002**

Scope Examination of correctness of the Quality Assessment module for identification of the best capture. The test object consists of the quality assessment software and a fixed parameter set describing the application profile specific tolerances threshold regarding several assessment components

- Preconditions*
- The IUT is in operation, required modules are loaded.
 - The IUT implements an interface for conformance testing where a sequence of images can be provided and the best capture as well as the computed quality values and configuration data are returned.
 - Each sequence contains precisely one image, that ranks higher than the other provided images w.r.t. the order described in FM QA-FI-GENERIC.
 - Timestamps may be provided as part of the exif-metadata (for jpeg).
 - To match the returned image it is expected that the it is bitwise identical to the respective image of the sequence.
 - The evaluator specifies the which application profile specific tolerances thresholds should be used (i.e. QA-FI-ARE/BCL/GID/IMA).
 - The following conditions apply regarding the database:
 - The image data that is sent to the IUT is of the following biometric type: FACE.
 - The groundtruth database contains compliant and noncompliant images w.r.t. the required quality values.

- CTS Mode*
- automatic
 - The image data that is sent to the IUT may have the following formats:
 - JPEG
 - JPEG2000
 - HTTP method: POST
 - test case variants:

/TR03122/TC-QA-FI-GENERIC-002/1

Table 2.112 TC-QA-FI-GENERIC-002 Test Case Variants*Description*

Step	Description / Expected Result
1	<i>Description:</i> • The IUT is called via the conformance test interface and an image from the database is provided. The image meets the properties described in the preconditions. • A sequence of images from the database is provided. The sequences and images meet the properties described in the preconditions. • Compare the image that is contained in the response with the best ranked image from the request sequence. <i>Expected Result:</i> • The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • An element //bio:FaceImageQuality exists for the selected best capture. • The response contains a single image that is bitwise identical to the best ranked image from the request sequence.

Test Case ID: TC-QA-FI-GENERIC-002

<i>Evaluation</i>	Evaluation Step	Description / Evaluation Result
	1	<i>Description:</i> - Repeat the TC for all applicable test sequences derived from the database. - Examine the response returned by the IUT via the test interface. <i>Expected Result:</i> - Overall, less than 5 percent of the requested images are not handled as expected.

Table 2.111 Test Case ID: TC-QA-FI-GENERIC-002

2.4.4 Test Cases FM QA-FI-GID

Test Case ID: TC-QA-FI-GID-001

Scope	Examination of the software-based quality assessment for facial images within the context of german identity documents. The test object consists of the quality assessment software and a fixed parameter set describing the application profile specific tolerances threshold regarding several assessment components. Note: This testcase uses the QA-thresholds for live/cloud images.
Preconditions	• The IUT is in operation, required modules are loaded. • The IUT implements an interface for conformance testing where a single image can be provided and the computed quality values and configuration data are returned. • The following conditions apply regarding the database: • The image data that is sent to the IUT is of the following biometric type: FACE. • The groundtruth database contains compliant and noncompliant images w.r.t. the required quality values.
CTS Mode	• automatic • The image data that is sent to the IUT may have the following formats: • JPEG • JPEG2000 • HTTP method: POST • test case variants:
/TR03122/TC-QA-FI-GID-001/1	

Table 2.114 TC-QA-FI-GID-001 Test Case Variants

<i>Description</i>	Step	Description / Expected Result
	1	<i>Description:</i> - The IUT is called via the conformance test interface and an image from the database is provided. The image meets the properties described in the preconditions. - The quality values contained in the IUT-response are checked against the groundtruth.

Test Case ID: TC-QA-FI-GID-001

Expected Result:

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response contains the correct number of elements based on the XPath expression `//bio:FaceImageQuality`. Exactly 1 element is expected in accordance to the acquisition process.
- The IUT provides the quality values for all components marked with "M" in the respective table for quality threshold requirements of Function Module QA-FI-GID
- Mandatory components: The minimum, maximum and threshold values are correctly set.
- Optional components: The minimum, maximum and threshold values are correctly set.
- For all components: The relation $min \leq tmin \leq tmax \leq max$ holds.
- If the image is a compliant presentation (w.r.t. all possible components) the IUT's assessment is positive.
- If the image is a noncompliant presentation (w.r.t. the mandatory components) the IUT's overall assessment is negative.
- Mandatory components: If the image is a compliant presentation the IUT assessment is positive for all components.
- Mandatory components: If the image is a noncompliant presentation the IUT assessment is negative for the respective components.

Evaluation

Evaluation Step	Description / Evaluation Result
1	<i>Description:</i> • Repeat the TC for all applicable images from the database. • Examine the response returned by the IUT via the test interface. <i>Expected Result:</i> • Overall, less than 5 percent of the requested images are not handled as expected.

Table 2.113 Test Case ID: TC-QA-FI-GID-001

Test Case ID: TC-QA-FI-GID-002

Scope

Examination of the software-based quality assessment for facial images within the context of german identity documents. The test object consists of the quality assessment software and a fixed parameter set describing the application profile specific tolerances threshold regarding several assessment components. Note: This testcase uses the QA-thresholds for scanned images.

Preconditions

- The IUT is in operation, required modules are loaded.
- The IUT implements an interface for conformance testing where a single image can be provided and the computed quality values and configuration data are returned.
- The following conditions apply regarding the database:
 - The image data that is sent to the IUT is of the following biometric type: FACE.
 - The groundtruth database contains compliant and noncompliant images w.r.t. the required quality values.

Test Case ID: TC-QA-FI-GID-002

CTS Mode

- automatic
- The image data that is sent to the IUT may have the following formats:
 - JPEG
 - JPEG2000
- HTTP method: POST
- test case variants:

/TR03122/TC-QA-FI-GID-002/1

Table 2.116 TC-QA-FI-GID-002 Test Case Variants

Description

Step	Description / Expected Result
1	<i>Description:</i> • The IUT is called via the conformance test interface and an image from the database is provided. The image meets the properties described in the pre-conditions. • The quality values contained in the IUT-response are checked against the groundtruth. <i>Expected Result:</i> • The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • The response contains the correct number of elements based on the XPath expression //bio:FaceImageQuality. Exactly 1 element is expected in accordance to the acquisition process. • The IUT provides the quality values for all components marked with "M" in the respective table for quality threshold requirements of Function Module QA-FI-GID • Mandatory components: The minimum, maximum and threshold values are correctly set. • Optional components: The minimum, maximum and threshold values are correctly set. • For all components: The relation $min \leq tmin \leq tmax \leq max$ holds. • If the image is a compliant presentation (w.r.t. all possible components) the IUTs assessment is positive. • If the image is a noncompliant presentation (w.r.t. the mandatory components) the IUTs overall assessment is negative. • Mandatory components: If the image is a compliant presentation the IUT assessment is positive for all components. • Mandatory components: If the image is a noncompliant presentation the IUT assessment is negative for the respective component.

Evaluation

Evaluation Step	Description / Evaluation Result
1	<i>Description:</i> • Repeat the TC for all applicable images from the database. • Examine the response returned by the IUT via the test interface. <i>Expected Result:</i> • Overall, less than 5 percent of the requested images are not handled as expected.

Table 2.115 Test Case ID: TC-QA-FI-GID-002

2.4.5 Test Cases FM QA-FI-IMA

Test Case ID: TC-QA-FI-IMA-001

<i>Scope</i>	Examination of the software-based quality assessment for facial images within the context of immigration authorities. The test object consists of the quality assessment software and a fixed parameter set describing the application profile specific tolerances threshold regarding several assessment components.
<i>Preconditions</i>	• The IUT is in operation, required modules are loaded. • The IUT implements an interface for conformance testing where a single image can be provided and the computed quality values and configuration data are returned. • The following conditions apply regarding the database: • The image data that is sent to the IUT is of the following biometric type: FACE. • The groundtruth database contains compliant and noncompliant images w.r.t. the required quality values.
<i>CTS Mode</i>	• automatic • The image data that is sent to the IUT may have the following formats: • JPEG • JPEG2000 • HTTP method: POST • test case variants:

/TR03122/TC-QA-FI-IMA-001/1

Table 2.118 TC-QA-FI-IMA-001 Test Case Variants

<i>Description</i>	Step	Description / Expected Result
	1	<i>Description:</i> • The IUT is called via the conformance test interface and an image from the database is provided. The image meets the properties described in the preconditions. • The quality values contained in the IUT-response are checked against the groundtruth.

Test Case ID: TC-QA-FI-IMA-001

Expected Result:

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response contains the correct number of elements based on the XPath expression `//bio:FaceImageQuality`. Exactly 1 element is expected in accordance to the acquisition process.
- The IUT provides the quality values for all components marked with "M" in the respective table for quality threshold requirements of Function Module QA-FI-ARE
- Mandatory components: The minimum, maximum and threshold values are correctly set.
- Optional components: The minimum, maximum and threshold values are correctly set.
- For all components: The relation $min \leq tmin \leq tmax \leq max$ holds.
- If the image is a compliant presentation (w.r.t. all possible components) the IUT's assessment is positive.
- If the image is a noncompliant presentation (w.r.t. the mandatory components) the IUT's overall assessment is negative.
- Mandatory components: If the image is a compliant presentation the IUT assessment is positive for all components.
- Mandatory components: If the image is a noncompliant presentation the IUT assessment is negative for the respective components.

Evaluation

Evaluation Step	Description / Evaluation Result
1	<i>Description:</i> • Repeat the TC for all applicable images from the database. • Examine the response returned by the IUT via the test interface. <i>Expected Result:</i> • Overall, less than 5 percent of the requested images are not handled as expected.

Table 2.117 Test Case ID: TC-QA-FI-IMA-001

2.4.6 Test Cases FM QA-FP-APP

Test Case ID: TC-QA-FP-APP-001

<i>Scope</i>	Examination of correctness of the Quality Assessment module for fingerprint acquisition sequence 4-4-2.
<i>Preconditions</i>	• The IUT is in operation, required modules are loaded. • The IUT implements an interface for conformance testing where a single fingerprint image can be provided and the computed quality values and configuration data are returned. • The following conditions apply regarding the database: • The image data that is sent to the IUT is of the following biometric type: FINGER. • The database contains segmented fingerprint images including corresponding finger codes. • The database may contain Fingerprint images with 500 ppi. The resolution may be provided via the DIB-Header of each image. By default (i.e. if not specified): the resolution shall be assumed to be 500ppi. • The database does not contain images expected to return an error code during quality assessment (e.g. corrupt BMP). • The database contains corresponding NFIQ2 [NFIQ2] quality values as Ground Truth for images, that are not expected to return an error code.

Test Case ID: TC-QA-FP-APP-001*CTS Mode*

- automatic
- The image data that is sent to the IUT may have the following format:
 - BMP
- HTTP method: POST
- test case variants:

/TR03122/TC-QA-FP-APP-001/1

Table 2.120 TC-QA-FP-APP-001 Test Case Variants*Description*

Step	Description / Expected Result
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- | | |
|---|--|
| 1 | <p><i>Description:</i></p> <ul style="list-style-type: none"> • The IUT is called via the conformance test interface and an image from the database is provided. The image meets the properties described in the pre-conditions. • The images provided via the conformance test interface are sequences of fingerprint images including the corresponding finger codes from the database simulating 4-4-2 slaps. The respective finger code is provided in the attribute '@subtype' as part of the request. The same fingercode (i.e. '@subtype') occurs three times to simulate repeated captures. Thereby, the i-th occurrence of /biocts:biocts-testsetup/biocts:Parameter[@subtype="n"] is defined as the capture of the finger "n" from the i-th repetition of the slap capture. Among all these synthesized repetitions, there is exactly one slap capture that is the best according to the metric defined in TR-03121. The images meet the properties described in the preconditions. • Examination of correct identification and comparison of the IUTs selection of the best capture out of multiple captures with the Ground Truth database. <p><i>Expected Result:</i></p> <ul style="list-style-type: none"> • The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • The response contains capture-information corresponding to the repeated acquisition simulated in the request. • For each slap: The information contained in the response correctly describes the repeated capture attempts and identifies the best capture attempt in the request (according to the metric defined in FM QA-FP-APP). I.e. the base64 encoded images (bio:BinaryRecord/tradd:BinaryData/tradd:Value) of segmented fingerprints that are identified as the best attempt for each slap are bitwise identical to the corresponding input images. • The response contains qa-information for every fingerprint image from the best capture for the attempt from the request sequence (according to the metric defined in FM QA-FP-APP). I.e. for each reference to a record block x in /bio:FingerCapture/@selected of each capture the response contain elements bio:FingerPrintQuality where bio:FingerPrintQuality/@ref matches x for every fingercode relevant @fc. • The qa-information contained in the response for every finger from the best capture matches the groundtruth. I.e. for each reference to a record block x in /bio:FingerCapture/@selected of each capture the value of bio:FingerPrintQuality[@ref=x][@fc=y]/@result is equal to the groundtruth value for the image corresponding to the capture of finger y. • The qa-thresholds for every finger contained in the response match the values required for the specific scenario (enrolment, plain control verification/identification or rolled acquisition). I.e. each value //bio:FingerQuality/bio:FingerPrintQuality[@fc="x"]/@threshold shall match the required value for the finger x. |
|---|--|

Evaluation

Evaluation Step	Description / Evaluation Result
-----------------	---------------------------------

Test Case ID: TC-QA-FP-APP-001

1

Description:

- Repeat the TC for all applicable images from the database.
- Examine the response returned by the IUT via the test interface.

Expected Result:

- Overall, less than 5 percent of the requested images are not handled as expected.

Table 2.119 Test Case ID: TC-QA-FP-APP-001**Test Case ID: TC-QA-FP-APP-002**

Scope Examination of correctness of the Quality Assessment module for fingerprint acquisition sequence 4-1-4-1.

Preconditions

- The IUT is in operation, required modules are loaded.
- The IUT implements an interface for conformance testing where a single fingerprint image can be provided and the computed quality values and configuration data are returned.
- The following conditions apply regarding the database:
 - The image data that is sent to the IUT is of the following biometric type: FINGER.
 - The database contains segmented fingerprint images including corresponding finger codes.
 - The database may contain Fingerprint images with 500 ppi. The resolution may be provided via the DIB-Header of each image. By default (i.e. if not specified): the resolution shall be assumed to be 500ppi.
 - The database does not contain images expected to return an error code during quality assessment (e.g. corrupt BMP).
 - The database contains corresponding NFIQ2 [NFIQ2] quality values as Ground Truth for images, that are not expected to return an error code.

CTS Mode

- automatic
- The image data that is sent to the IUT may have the following format:
 - BMP
- HTTP method: POST
- test case variants:

/TR03122/TC-QA-FP-APP-002/1

Table 2.122 TC-QA-FP-APP-002 Test Case Variants*Description*

Step	Description / Expected Result
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Test Case ID: TC-QA-FP-APP-002

1

Description:

- The IUT is called via the conformance test interface and an image from the database is provided. The image meets the properties described in the preconditions.
- The images provided via the conformance test interface are sequences of fingerprint images including the corresponding finger codes from the database simulating 4-1-4-1 slaps. The respective finger code is provided in the attribute @subtype as part of the request. The same fingeCode (i.e. @subtype) occurs three times to simulate repeated captures. Thereby, the i-th occurrence of /biocts:biocts-testsetup/biocts:Parameter[@subtype="n"] is defined as the capture of the finger "n" from the i-th repetition of the slap capture. Among all these synthesized repetitions, there is exactly one slap capture that is the best according to the metric defined in TR-03121. The images meet the properties described in the preconditions.
- Examination of correct identification and comparison of the IUTs selection of the best capture out of multiple captures with the Ground Truth database.

Expected Result:

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response contains capture-information corresponding to the repeated acquisition simulated in the request.
- For each slap: The information contained in the response correctly describes the repeated capture attempts and identifies the best capture attempt in the request (according to the metric defined in FM QA-FP-APP). I.e. the base64 encoded images (bio:BinaryRecord/tradd:BinaryData/tradd:Value) of segmented fingerprints that are identified as the best attempt for each slap are bitwise identical to the corresponding input images.
- The response contains qa-information for every fingerprint image from the best capture for the attempt from the request sequence (according to the metric defined in FM QA-FP-APP). I.e. for each reference to a record block x in //bio:FingerCapture/@selected of each capture the response contains elements bio:FingerPrintQuality where bio:FingerPrintQuality/@ref matches x for every fingeCode relevant @fc.
- The qa-information contained in the response for every finger from the best capture matches the groundtruth. I.e. for each reference to a record block x in /bio:FingerCapture/@selected of each capture the value of bio:FingerPrintQuality[@ref=x][@fc=y]/@result is equal to the groundtruth value for the image corresponding to the capture of finger y.
- The qa-thresholds for every finger contained in the response match the values required for the specific scenario (enrolment, plain control verification/identification or rolled acquisition). I.e. each value //bio:FingerQuality/bio:FingerPrintQuality[@fc="x"]/@threshold shall match the required value for the finger x.

Evaluation

Evaluation Step

Description / Evaluation Result

1

Description:

- Repeat the TC for all applicable images from the database.
- Examine the response returned by the IUT via the test interface.

Expected Result:

- Overall, less than 5 percent of the requested images are not handled as expected.

Table 2.121 Test Case ID: TC-QA-FP-APP-002

Test Case ID: TC-QA-FP-APP-003

Scope Examination of correctness of NFIQ2 algorithm implementation. Note, that this testcase is applicable for IUTs that are able to process fingerprints in both 1000ppi and 500ppi).

Preconditions

- The IUT is in operation, required modules are loaded.
- The IUT implements an interface for conformance testing where a single fingerprint image can be provided and the computed quality values and configuration data are returned.
- The following conditions apply regarding the database:
 - The image data that is sent to the IUT is of the following biometric type: FINGER.
 - The database contains Fingerprint images including corresponding finger codes. The respective finger code is provided in the attribute '@subtype' as part of the request.
 - The database contains Fingerprint images with 500 ppi and 1000 ppi. The resolution is provided via the DIB-Header of each image. By default (i.e. if not specified); the resolution shall be assumed to be 500ppi.
 - The database contains images expected to return an error code during quality assessment (e.g. corrupt BMP).
 - The database contains corresponding NFIQ2 [NFIQ2] quality values as Ground Truth for images, that are not expected to return an error code.

CTS Mode

- automatic
- The image data that is sent to the IUT may have the following format:
 - BMP
- HTTP method: POST
- test case variants:

/TR03122/TC-QA-FP-APP-003/1

Table 2.124 TC-QA-FP-APP-003 Test Case Variants

Description

Step	Description / Expected Result
1	<i>Description:</i> • The IUT is called via the conformance test interface and an image from the database is provided. The image meets the properties described in the preconditions. • The quality values contained in the IUT-response are checked against the groundtruth. • If the image sent to the IUT is expected to return an error code: Check whether an error code is returned by the IUT. • If the image sent to the IUT is not expected to return an error code: Check the resolution of the image returned by the IUT.

Test Case ID: TC-QA-FP-APP-003*Expected Result:*

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- If the image is not expected to return an error code: The response contains the correct number of elements based on the XPath expression `//bio:FingerAcquisition/bio:FingerQuality/bio:FingerPrintQuality/@result`. Exactly 1 element is expected in accordance to the acquisition process.
- If the image is not expected to return an error code and has a resolution of 500ppi: The NFIQ value (i.e. `//bio:FingerAcquisition/bio:FingerQuality/bio:FingerPrintQuality/@result`) returned by the IUT for the current fingerprint image equals the corresponding Ground Truth value.
- If the image is expected to return an error code: The IUT returns the error-code '255'.
- If the image is not expected to return an error code and has a resolution of 500ppi: The image returned by the IUT (i.e. `//bio:FingerAcquisition/bio:Records/bio:BinaryRecord[@type='bmp']/tradd:BinaryData/tradd:Value`) is not resampled.
- If the image is not expected to return an error code and has a resolution of 1000ppi: The image returned by the IUT (`//bio:FingerAcquisition/bio:Records/bio:BinaryRecord[@type='bmp']/tradd:BinaryData/tradd:Value`) is resampled to 500ppi.

*Evaluation***Evaluation Step Description / Evaluation Result**

1

Description:

- Repeat the TC for all applicable images from the database.
- Examine the response returned by the IUT via the test interface.

Expected Result:

- Overall, less than 5 percent of the requested images are not handled as expected.

Table 2.123 Test Case ID: TC-QA-FP-APP-003**Test Case ID: TC-QA-FP-APP-004***Scope*

Examination of correctness of the Quality Assessment module for fingerprint acquisition of a 4-finger slap.

Preconditions

- The IUT is in operation, required modules are loaded.
- The IUT implements an interface for conformance testing where a single fingerprint image can be provided and the computed quality values and configuration data are returned.
- The following conditions apply regarding the database:
 - The image data that is sent to the IUT is of the following biometric type: FINGER.
 - The database contains segmented fingerprint images including corresponding finger codes.
 - The database may contain Fingerprint images with 500 ppi. The resolution may be provided via the DIB-Header of each image. By default (i.e. if not specified); the resolution shall be assumed to be 500ppi.
 - The database does not contain images expected to return an error code during quality assessment (e.g. corrupt BMP).
 - The database contains corresponding NFIQ2 [NFIQ2] quality values as Ground Truth for images, that are not expected to return an error code.

Test Case ID: TC-QA-FP-APP-004*CTS Mode*

- automatic
- The image data that is sent to the IUT may have the following format:
 - BMP
- HTTP method: POST
- test case variants:

/TR03122/TC-QA-FP-APP-004/1

Table 2.126 TC-QA-FP-APP-004 Test Case Variants*Description*

Step	Description / Expected Result
1	<i>Description:</i> • The IUT is called via the conformance test interface and an image from the database is provided. The image meets the properties described in the preconditions. • The images provided via the conformance test interface are sequences of fingerprint images including the corresponding finger codes from the database simulating 4-4-2 slaps. The respective finger code is provided in the attribute '@subtype' as part of the request. The same fingercode (i.e. '@subtype') occurs three times to simulate repeated captures. Thereby, the i-th occurrence of /biocts:biocts-testsetup/biocts:Parameter[@subtype="n"] is defined as the capture of the finger "n" from the i-th repetition of the slap capture. Among all these synthesized repetitions, there is exactly one slap capture that is the best according to the metric defined in TR-03121. The images meet the properties described in the preconditions. • Examination of correct identification and comparison of the IUTs selection of the best capture out of multiple captures with the Ground Truth database.

Test Case ID: TC-QA-FP-APP-004

Expected Result:

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- If the request sequence corresponds to a right hand slap: The response contains capture-information corresponding to the repeated acquisition of a right hand slap.
- If the request sequence corresponds to a left hand slap: The response contains capture-information corresponding to the repeated acquisition of a left hand slap.
- For each slap: The information contained in the response correctly describes the repeated capture attempts and identifies the best capture attempt in the request (according to the metric defined in FM QA-FP-APP). I.e. the base64 encoded images (bio:BinaryRecord/tradd:BinaryData/tradd:Value) of segmented fingerprints that are identified as the best attempt for each slap are bitwise identical to the corresponding input images.
- The response contains qa-information for every fingerprint image from the best capture for the attempt from the request sequence (according to the metric defined in FM QA-FP-APP). I.e. for each reference to a record block x in //bio:FingerCapture/@selected of each capture the response contains elements bio:FingerPrintQuality where bio:FingerPrintQuality/@ref matches x for every fingercode relevant @fc.
- The qa-information contained in the response for every finger from the best capture matches the groundtruth. I.e. for each reference to a record block x in /bio:FingerCapture/@selected of each capture the value of bio:FingerPrintQuality[@ref= x][@fc= y]/@result is equal to the groundtruth value for the image corresponding to the capture of finger y .
- The qa-thresholds for every finger contained in the response match the values required for the specific scenario (enrolment, plain control verification/identification or rolled acquisition). I.e. each value //bio:FingerQuality/bio:FingerPrintQuality[@fc=" x "]/@threshold shall match the required value for the finger x .

Evaluation

Evaluation Step Description / Evaluation Result

1

Description:

- Repeat the TC for all applicable images from the database.
- Examine the response returned by the IUT via the test interface.

Expected Result:

- Overall, less than 5 percent of the requested images are not handled as expected.

Table 2.125 Test Case ID: TC-QA-FP-APP-004

Test Case ID: TC-QA-FP-APP-005

Scope

Examination of correctness of NFIQ2 algorithm implementation. Note, that this testcase is applicable for IUTs that are only able to process fingerprints in 500ppi).

Test Case ID: TC-QA-FP-APP-005*Preconditions*

- The IUT is in operation, required modules are loaded.
- The IUT implements an interface for conformance testing where a single fingerprint image can be provided and the computed quality values and configuration data are returned.
- The following conditions apply regarding the database:
 - The image data that is sent to the IUT is of the following biometric type: FINGER.
 - The database contains Fingerprint images including corresponding finger codes. The respective finger code is provided in the attribute '@subtype' as part of the request.
 - The database contains Fingerprint images with 500 ppi. I.e. for all images the resolution may be assumed to be 500ppi (regardless of DIB-Header)
 - The database contains images expected to return an error code during quality assessment (e.g. corrupt BMP).
 - The database contains corresponding NFIQ2 [NFIQ2] quality values as Ground Truth for images, that are not expected to return an error code.

CTS Mode

- automatic
- The image data that is sent to the IUT may have the following format:
 - BMP
- HTTP method: POST
- test case variants:

/TR03122/TC-QA-FP-APP-005/1

Table 2.128 TC-QA-FP-APP-005 Test Case Variants*Description*

Step	Description / Expected Result
1	<i>Description:</i> • The IUT is called via the conformance test interface and an image from the database is provided. The image meets the properties described in the pre-conditions. • The quality values contained in the IUT-response are checked against the groundtruth. • If the image sent to the IUT is expected to return an error code: Check whether an error code is returned by the IUT. • Check the resolution of the image returned by the IUT. <i>Expected Result:</i> • The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • If the image is not expected to return an error code: The response contains the correct number of elements based on the XPath expression //bio:FingerAcquisition/bio:FingerQuality/bio:FingerPrintQuality/@result. Exactly 1 element is expected in accordance to the acquisition process. • If the image is not expected to return an error code: The NFIQ value (//bio:FingerAcquisition/bio:FingerQuality/bio:FingerPrintQuality/@result) returned by the IUT for the current fingerprint image equals the corresponding Ground Truth value. • If the image is expected to return an error code: The IUT returns the error-code '255'. • If the image is not expected to return an error code: The image returned by the IUT (i.e. //bio:FingerAcquisition/bio:Records/bio:BinaryRecord[@type=' bmp ']/tradd:BinaryData/tradd:Value) is not resampled.

Evaluation

Evaluation Step	Description / Evaluation Result
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Test Case ID: TC-QA-FP-APP-005

1

Description:

- Repeat the TC for all applicable images from the database.
- Examine the response returned by the IUT via the test interface.

Expected Result:

- Overall, less than 5 percent of the requested images are not handled as expected.

Table 2.127 Test Case ID: TC-QA-FP-APP-005

2.5 Test Cases Presentation Attack Detection

2.5.1 Test Cases FM PAD-FI-APP and APP1

Note, test cases 001 and 002 are interchangeable alternatives to each other.

Test Case ID: TC-PAD-FI-APP-001

<i>Scope</i>	Examination of the Common Criteria certification of the Presentation Attack Detection (PAD) subsystem.	
<i>Preconditions</i>	• The following test resources are necessary: • Product documentation of camera model is at hand (e.g. data sheet, manual). • Common Criteria Agreement certificate of the facial image camera system is at hand.	
<i>CTS Mode</i>	Not supported	
<i>Description</i>	Step	Description / Expected Result
	1	<i>Description:</i> • Consult product documentation and verify that the subsystem certification meets the requirements. <i>Expected Result:</i> • The documentation attest that the system is certified under the Common Criteria Agreement according the following Protection Profile: BSI-CC-PP-0118-2022: Common Criteria Protection Profile - Biometric Mechanisms Protection Profile (BMPP, Version 2.0, base PP and at least the functional package PAD)

Table 2.129 Test Case ID: TC-PAD-FI-APP-001**Test Case ID: TC-PAD-FI-APP-002**

<i>Scope</i>	Examination of availability and correctness of PAD functionality in the context of the acquisition of facial images with and without Presentation Attack Instrument (PAI) in use.
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Test Case ID: TC-PAD-FI-APP-002*Preconditions*

- The IUT is in operation, required modules are loaded.
- The camera system used in the specific application scenario is connected.
- The IUT implements an interface for conformance testing where a single image can be captured and the computed PAD values are returned.
- The PAD functionality of the IUT is activated for performing the test case.
- The IUT is able to return the results for every captured image via the test interface.
- The IUT is able to return the results for the conducted PAD.
- The following test resources are necessary:
 - At least three different biometric subjects SHALL available to perform bona fide presentations.
 - Four different artefacts of each of the eight artefact classes SHALL be created, so that a total of 32 artefacts are obtained. These artefacts SHOULD be usually constructed before testing begins. For every testrun new artefacts SHOULD be constructed.
 - Class 1: Masks created from casted silicon, big (size of a face) and small (smaller than a normal face), in different thicknesses: 3-D presentation.
 - Class 2: Masks created from latex in different thicknesses and styles: 3-D presentation.
 - Class 3: Masks printed on paper in different thicknesses and styles: 3-D presentation, the photograph of the biometric subject is printed on paper but presented in a 3-D manner (used as a mask covering the whole face or only parts of it).
 - Class 4: Masks printed with 3D-printer in different thicknesses and styles: 3-D presentation.
 - Class 5: Photographs displayed on electronic devices, e.g. phones, tablets, etc.: 2-D presentation.
 - Class 6: Videos displayed on electronic devices, e.g. phones, tablets, etc.: 2-D presentation.
 - Class 7: Faces printed on paper with different thicknesses and structures of paper and different structures of printing (colouring, etc.): 2-D presentation, a photograph of the biometric subject is presented to the system as a flat paper copy.
 - Faces printed on fabrics with different thicknesses and structures of the fabrics and different methods of printing (flock print, silk screening, etc.): 2-D or 3-D presentation, a photograph of the biometric subject is presented to the system on fabrics (e.g. on a t-shirt). A 3-D presentation can materialise if pieces of clothing are used by the biometric subject.
 - For the selection of electronic devices for the artefact classes 5 and 6: At least three different devices from three different manufacturers are used.
 - For the selection of electronic devices for the artefact classes 5 and 6: No device is older than five years (calculated from the date of release in Germany).
 - For the selection of electronic devices for the artefact classes 5 and 6: At least one smartphone and one tablet is used.

CTS Mode

- interactive
- no provision of pre-defined input data
- HTTP method: GET
- test case variants:

/TR03122/TC-PAD-FI-APP-002/1

Table 2.131 TC-PAD-FI-APP-002 Test Case Variants*Description*

Step	Description / Expected Result
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Test Case ID: TC-PAD-FI-APP-002

1

Description:

- Initiate test case by calling the IUT via the test interface.
- Perform a facial acquisition using either a test subject (bona-fide) or a PAI presentation. If a PAI presentation is performed, the presentations are performed depending on the PAI by holding the artefact manually into the camera, wearing the artefact as a mask or by using a special construction in the IUTs capture area.
- The response returned by the IUT via the test interface is evaluated.

Expected Result:

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response contains the encoded facial image (bio:FaceAcquisition)
- If a bona-fide presentation was performed: All acceptance conditions for the facial image acquisition are satisfied and achieved.
- The IUT's PAD returns its detection results in the form of a normalized PAD score in the range [0,1] where 0 indicates bona fide and 1 presentation attack to the calling application.
- If a bona-fide presentation was performed: The PAD score returned by the IUT indicates a bona fide face presentation.
- If a presentation using a PAI was performed: The PAD score returned by the IUT indicates a presentation attack.
- If a bona-fide presentation was performed: No PAD alarm is shown to the operator.
- If a PAD presentation was performed: A PAD alarm is shown to the operator.

*Evaluation***Evaluation Step****Description / Evaluation Result**

1

Description:

- For each of the three biometric subjects the test case is repeated 20 times resulting in a total of 60 bona-fide presentations.
- For each of the PAI the test case is repeated 10 times resulting in a total of 320 PAI presentations.
- In each step slight alterations in the presentation are allowed (e.g. covering of integrated lights, wearing additional accessories like glasses).

Expected Result:

- An artefact is considered as detected, if the device shows a PAD alarm in six out of ten attempts. If the IUT (repeatable) does not accept a presentation and delivers the result "no face found" (or alike), the presentation is counted as "detected".
- A device failed the total test, if a single artefact class never triggered a PAD alarm.
- A device failed the total test, if nine or more artefacts in total were not detected as stated above.
- A device failed the total test, if more than three of the bona fide facial images are not recognized as bona fide. Failures of detection from the other test steps do not count for this requirement. Note that during certification a more relaxed false-alarm-rate is tested than specified within the respective Function Module. The requirement of the rate specified in the respective Function Module is tested on operational collected data only.

Table 2.130 Test Case ID: TC-PAD-FI-APP-002

2.5.2 Test Cases FM PAD-FP-APP and APP1

Note, test cases 001 and 002 are interchangeable alternatives to each other.

Test Case ID: TC-PAD-FP-APP-001

Scope Examination of the Common Criteria certification of the Presentation Attack Detection (PAD) subsystem.

Preconditions

- The following test resources are necessary:
 - Product documentation of fingerprint scanner model is at hand (e.g. data sheet, manual).
 - Common Criteria Agreement certificate of the fingerprint scanner model is at hand.

CTS Mode Not supported

Description

Step	Description / Expected Result
1	<i>Description:</i> - Consult product documentation and verify that the subsystem certification meets one of the following requirements. <i>Expected Result:</i> - The documentation attests that the system is certified under the Common Criteria Agreement according the following Protection Profile: BSI-CC-PP-0063-2010: Fingerprint Spoof Detection Protection Profile (FSDPP) OR - The documentation attests that the system is certified under the Common Criteria Agreement according the following Protection Profile: BSI-CC-PP-0118-2022: Common Criteria Protection Profile - Biometric Mechanisms Protection Profile (BMPP, Version 2.0, base PP and at least the functional package PAD) - The documentation attests that the system is certified under the Common Criteria Agreement according the following Protection Profile: BSI-CC-PP-0118-2022: Common Criteria Protection Profile - Biometric Mechanisms Protection Profile (BMPP, Version 2.0, base PP and at least the functional package PAD)

Table 2.132 Test Case ID: TC-PAD-FP-APP-001

Note, if test case 002 is performed during stand alone testing of Presentation Attack Detection (PAD) functionality of finger print scanners, the usage of the CTS is OPTIONAL. When not using the CTS the results (PAD score, binary PAD result, configured thresholds and NIFQ2 score for each acquired finger) are allowed to be provided in other ways than the defined `bio:FingerAcquisition XML`.

Test Case ID: TC-PAD-FP-APP-002

Scope Examination of availability and correctness of PAD functionality and finger codes in the context of the acquisition of fingerprint images with and without PAI in use.

Test Case ID: TC-PAD-FP-APP-002*Preconditions*

- The IUT is in operation, required modules are loaded.
- The fingerprint scanner used in the specific application scenario is connected
- The IUT implements an interface for conformance testing where a fingerprints can be captured and the computed PAD values are returned.
- The PAD functionality of the IUT is activated for performing the test case.
- The IUT is able to return the results for every captured finger via the test interface.
- The IUT is able to return the results for the conducted PAD.
- The following test resources are necessary:
 - At least three different biometric subjects (with generally good fingerprints) SHALL available to perform bona fide presentations.
 - Three different artefacts of each of the seven artefact classes SHALL be created, so that a total of 21 artefacts are obtained. The following artefacts classes (fingertips in different thicknesses and/or massive fingers) SHALL be used during test cases
 - Class 1: Artefacts created from different kinds of silicon, in different colouring.
 - Class 2: Artefacts created from different kinds of latex, in different colouring.
 - Class 3: Artefacts created from different kinds of gelatine, in different colouring.
 - Class 4: Artefacts created from different kinds of wood glue, in different colouring.
 - Class 5: Artefacts created from different kinds of window painting, in different colouring.
 - Class 6: Artefacts created from different kinds of paper, in different colouring.
 - Class 7:Artefacts created from different kinds of latex with window painting (using the following ratios: 2:1, 1:1, 1:2), in different colouring.
 - The materials of the different artefact classes SHALL NOT be mixed. Different kinds of each artefact class MAY be created adding glycerol (0.5 ml per gramm of artefact class material maximum) and/or graphite (5 knife-point per gramm of artefact class material maximum). For each testrun, new artefacts MAY be created.
 - Artefacts SHALL provide a NFIQ-2.2-score of at least 30 (or an equivalent score with other NFIQ versions).

CTS Mode

- interactive
- no provision of pre-defined input data
- HTTP method: GET
- test case variants:

/TR03122/TC-PAD-FP-APP-002/1

Table 2.134 TC-PAD-FP-APP-002 Test Case Variants*Description*

Step	Description / Expected Result
1	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • Perform a fingerprint acquisition using either a test subject (bona-fide) or a PAI. • If a PAI is performed exactly one finger of exactly one hand SHOULD be equipped with the artefact per fingerprint acquisition • The response returned by the IUT via the test interface is evaluated.

Test Case ID: TC-PAD-FP-APP-002

Expected Result:

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response contains the encoded fingerprint image (bio:FingerAcquisition)
- If a bona-fide presentation was performed: All acceptance conditions for the fingerprint image acquisition are satisfied and achieved.
- The IUT's PAD returns its detection results in the form of a normalized PAD score in the range [0,1] where 0 indicates bona fide and 1 presentation attack to the calling application. Also, an overall PAD score is returned.
- If a bona-fide presentation was performed: The PAD score returned by the IUT indicates a bona fide presentation.
- If a PAI presentation was performed: The PAD score returned by the IUT indicates a presentation attack.
- If a bona-fide presentation was performed: No PAD alarm is shown to the operator.
- If a PAD presentation was performed: A PAD alarm is shown to the operator.
- The individual PAD scores returned by the IUT indicate a presentation attack for fingers with artefacts and bona fide for fingers without artefacts.
- Every finger originating from an artefact is correctly assigned to its corresponding finger code. (Abstract example: If there is a hand with four bona fide fingers and one finger originating from an artefact, then the PAD score stating it is a presentation attack is assigned to the corresponding finger. Concrete example: If there is a hand with only the index finger coming from an artefact, then the PAD score stating it is a presentation attack is assigned to the index finger.)

Evaluation

Evaluation Step	Description / Evaluation Result
1	<i>Description:</i> • For each of the three biometric subjects the test case is repeated 20 times resulting in a total of 60 bona-fide presentations. • For each of the PAI the test case is repeated 10 times resulting in a total of 210 PAI presentations. • In each step where a PAI is used a rotation principle should be in place that specifies that the equipment of a finger with an artefact changes or rolls in each step. This is done to ensure that not always the same finger is equipped with an artefact for each fingerprint acquisition. Doing this every artefact SHOULD be present for at least three different fingers of at least one hand. <i>Expected Result:</i> • An artefact is considered as detected, if the device shows a PAD alarm in at least six out of ten attempts. • A device failed the total test, if a single artefact class never triggered a PAD alarm. • A device failed the total test, if four or more artefacts in total were not detected as stated above. • A device failed the total test, if within the bona fide part of the test more than three of the bona fide presentations are not recognized as bona fide. Failures of detection from the PAI part of the test do not count for this requirement. Note that during certification a more relaxed false-alarm-rate is tested than specified within the respective Function Module. The requirement of the rate specified in the respective Function Module is tested on operational collected data only

Table 2.133 Test Case ID: TC-PAD-FP-APP-002

2.6 Test Cases Compression

2.6.1 Test Cases FM COM-FI-BCL

Test Case ID: TC-COM-FI-BCL-001

<i>Scope</i>	Examine the correctness of the compression of live images used within the context of border control.
<i>Preconditions</i>	• The IUT is in operation, required modules are loaded. • As necessary test resources: a test database containing multiple compressed and uncompressed (including losslessly compressed) facial images. The provided formats are BMP, PNG, JPEG and JPEG2000. • The IUT implements an interface for conformance testing where a single image can be provided and the according biometric container including the image data is returned. • The following conditions apply regarding the database: • The image data that is sent to the IUT is of the following biometric type: FACE. • The groundtruth database contains compliant images w.r.t. the required quality values. • Note: Images in BMP, PNG and JPEG2000 formats are considered uncompressed or lossless compressed. • Note: Images in JPEG format are considered lossy compressed.
<i>CTS Mode</i>	• automatic • The image data that is sent to the IUT may have the following formats: • BMP • PNG • JPEG • JPEG2000 • HTTP method: POST • test case variants:

/TR03122/TC-COM-FI-BCL-001/1

Table 2.136 TC-COM-FI-BCL-001 Test Case Variants

<i>Description</i>	Step	Description / Expected Result
	1	<i>Description:</i> • The IUT is called via the conformance test interface and an image from the database is provided. The image meets the properties described in the preconditions. • Extraction of the compressed facial image file from the biometric container.

Test Case ID: TC-COM-FI-BCL-001*Expected Result:*

- The IUT delivers a response that is conformant to the schema file bcl6v1.xsd.
- If the provided image has a different format than JPEG: The resulting image is compressed using JPEG2000 (according to ISO/IEC 15444-1) or JPEG (according to ISO/IEC 10918-1) compression standard.
- If the response from the IUT contains an image: The image is compressed using the compression standard referenced in the respective bio:Binary-Record/@type.
- If the provided image has a different format than JPEG: The actual file size of the compressed image is below 375 kB.
- If the response from the IUT contains an image: The compression level is at most 20:1.
- If the response from the IUT contains an image: The image resolution of the returned image is the same as that of the input image.
- If the provided image has JPEG format: Since multiple lossy compression is not allowed, either an error code that the acceptance conditions are not met is returned, or no additional processing is performed on the input image (i.e. the returned image is bitwise identical to the provided image).

*Evaluation***Evaluation Step Description / Evaluation Result**

1

Description:

- Examine the response returned by the IUT via the test interface.

Expected Result:

- Overall, less than 0 percent of the requested images are not handled as expected.

Table 2.135 Test Case ID: TC-COM-FI-BCL-001**2.6.2 Test Cases FM COM-FI-IMA****Test Case ID: TC-COM-FI-IMA-001**

Scope Examine the correctness of the compression of live images used within the context of immigration authorities.

- Preconditions*
- The IUT is in operation, required modules are loaded.
 - As necessary test resources: a test database containing multiple compressed and uncompressed (including losslessly compressed) facial images. The provided formats are BMP, PNG, JPEG and JPEG2000.
 - The IUT implements an interface for conformance testing where a single image can be provided and the according biometric container including the image data is returned.
 - The following conditions apply regarding the database:
 - The image data that is sent to the IUT is of the following biometric type: FACE.
 - The groundtruth database contains compliant images w.r.t. the required quality values.
 - Note: Images in BMP, PNG and JPEG2000 formats are considered uncompressed or lossless compressed.
 - Note: Images in JPEG format are considered lossy compressed.

Test Case ID: TC-COM-FI-IMA-001*CTS Mode*

- automatic
- The image data that is sent to the IUT may have the following formats:
 - BMP
 - PNG
 - JPEG
 - JPEG2000
- HTTP method: POST
- test case variants:

/TR03122/TC-COM-FI-IMA-001/1

Table 2.138 TC-COM-FI-IMA-001 Test Case Variants*Description*

Step	Description / Expected Result
1	<i>Description:</i> • The IUT is called via the conformance test interface and an image from the database is provided. The image meets the properties described in the pre-conditions. • Extraction of the compressed facial image file from the biometric container. <i>Expected Result:</i> • The IUT delivers a response that is conformant to the schema file ima6v1.xsd. • If the provided image has a different format than JPEG: The resulting image is compressed using JPEG2000 (according to ISO/IEC 15444-1) or JPEG (according to ISO/IEC 10918-1) compression standard. • If the response from the IUT contains an image: The image is compressed using the compression standard referenced in the respective bio:Binary-Record/@type. • If the provided image has a different format than JPEG: The actual file size of the compressed image is below 375 kB. • If the response from the IUT contains an image: The compression level is at most 20:1. • If the response from the IUT contains an image: The image resolution of the returned image is the same as that of the input image. • If the provided image has JPEG format: Since multiple lossy compression is not allowed, either an error code that the acceptance conditions are not met is returned, or no additional processing is performed on the input image (i. e. the returned image is bitwise identical to the provided image).

Evaluation

Evaluation Step	Description / Evaluation Result
1	<i>Description:</i> • Examine the response returned by the IUT via the test interface. <i>Expected Result:</i> • Overall, less than 0 percent of the requested images are not handled as expected.

Table 2.137 Test Case ID: TC-COM-FI-IMA-001

2.6.3 Test Cases FM COM-FI-JPG

Test Case ID: TC-COM-FI-JPG-001*Scope*

Examine correct parametrisation of JPEG compression.

Test Case ID: TC-COM-FI-JPG-001*Preconditions*

- The IUT is in operation, required modules are loaded.
- The IUT implements an interface for conformance testing where a single image can be provided and the according biometric container including the image data is returned.
- The following conditions apply regarding the database:
 - The image data that is sent to the IUT is of the following biometric type: FACE.
 - The groundtruth database contains compliant images w.r.t. the required quality values.
 - Images in small size (600 x 800 pixels) and standard size (1200 x 1600 pixels) are used.
 - Note: All images of formats JPEG, BMP, PNG, and JPEG2000 are expected to be processed.

CTS Mode

- automatic
- The image data that is sent to the IUT may have the following formats:
 - BMP
 - PNG
 - JPEG
 - JPEG2000
- HTTP method: POST
- test case variants:

/TR03122/TC-COM-FI-JPG-001/1

Table 2.140 TC-COM-FI-JPG-001 Test Case Variants*Description*

Step	Description / Expected Result
1	<i>Description:</i> • The IUT is called via the conformance test interface and an image from the database is provided. The image meets the properties described in the pre-conditions. • Extraction of the compressed facial image file from the biometric container. • Check image format after compression of images. • Check minimum file size after compression of images. <i>Expected Result:</i> • The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • The response contains a single facial image. • The returned image is compressed using JPEG (according to ISO/IEC 10918-1) compression standard. • If the provided image has a resolution of 600 x 800 pixels: The actual file size of the compressed image is above 35 KiB. • If the provided image has a resolution of 1200 x 1600 pixels: The actual file size of the compressed image is above 100 KiB. • The compression level is at most 15:1. • The image resolution of the returned image is the same as that of the input image.

Evaluation

Evaluation Step	Description / Evaluation Result
1	<i>Description:</i> • Examine the response returned by the IUT via the test interface.

Test Case ID: TC-COM-FI-JPG-001

Expected Result:

- Overall, less than 0 percent of the requested images are not handled as expected.

Table 2.139 Test Case ID: TC-COM-FI-JPG-001

2.6.4 Test Cases FM COM-FP-BCL

Test Case ID: TC-COM-FP-BCL-001

Scope Examine the correctness of the compression of fingerprint images contained in ANSI/NIST-ITL 1 files within the context of border control

Preconditions

- The IUT is in operation, required modules are loaded.
- The following conditions apply regarding the database:
 - The image data that is sent to the IUT is of the following biometric type: FINGER.
 - The database contains multiple fingerprint images.

CTS Mode

- automatic
- The image data that is sent to the IUT may have the following format:
 - BMP
- HTTP method: POST
- test case variants:

/TR03122/TC-COM-FP-BCL-001/1

Table 2.142 TC-COM-FP-BCL-001 Test Case Variants

Description

Step	Description / Expected Result
1	<i>Description:</i> • The IUT is called via the conformance test interface and an image from the database is provided. The image meets the properties described in the preconditions. • The IUT is called via the conformance test interface and sequences of fingerprint images including the corresponding finger codes from the database are provided. The respective finger code is provided in the attribute '@subtype' as part of the request. Each fingercode (i.e. '@subtype') occurs one time. The images meet the properties described in the preconditions. • Check file size and file compression: initiate test case by loading fingerprint and slap images using the test interface. <i>Expected Result:</i> • The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • The response contains the correct number of elements based on the XPath expression <code>//bio:Records/bio:BinaryRecord[@type="ees-ansi-nist"]/tradd:BinaryData/tradd:Value</code>. Exactly 1 element is expected in accordance to the acquisition process. • If the ANSI/NIST container has an EES purpose: The ANSI/NIST container has a maximum file size of 5332 kB. • If the ANSI/NIST container has an SIS purpose: The ANSI/NIST container has a maximum file size of 3000 kB.

Evaluation

Evaluation Step	Description / Evaluation Result
1	<i>Description:</i> • Examine the response returned by the IUT via the test interface.

Test Case ID: TC-COM-FP-BCL-001

Expected Result:

- Overall, less than 0 percent of the requested images are not handled as expected.

2

Description:

- Examination of the returned fingerprint images by the IUT.

Expected Result:

- The returned fingerprint images show the same fingerprints that were shown on the input images.

Table 2.141 Test Case ID: TC-COM-FP-BCL-001

2.6.5 Test Cases FM COM-FP-IMA

Test Case ID: TC-COM-FP-IMA-001

Scope Examine the correctness of the compression of fingerprint images contained in ANSI/NIST-ITL 1 files within the context of immigration authorities.

Preconditions

- The IUT is in operation, required modules are loaded.
- The following conditions apply regarding the database:
 - The image data that is sent to the IUT is of the following biometric type: FINGER.
 - The database contains multiple fingerprint images.

CTS Mode

- automatic
- The image data that is sent to the IUT may have the following format:
 - BMP
- HTTP method: POST
- test case variants:

/TR03122/TC-COM-FP-IMA-001/1

Table 2.144 TC-COM-FP-IMA-001 Test Case Variants*Description*

Step	Description / Expected Result
1	<i>Description:</i> • The IUT is called via the conformance test interface and an image from the database is provided. The image meets the properties described in the preconditions. • The IUT is called via the conformance test interface and sequences of fingerprint images including the corresponding finger codes from the database are provided. The respective finger code is provided in the attribute '@subtype' as part of the request. Each fingerprint code (i.e. '@subtype') occurs one time. The images meet the properties described in the preconditions. • Check file size and file compression. <i>Expected Result:</i> • The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • The response contains the correct number of elements based on the XPath expression <code>//bio:Records/bio:BinaryRecord[@type="ees-ansi-nist"]/tradd:BinaryData/tradd:Value</code>. Exactly 1 element is expected in accordance to the acquisition process. • If the ANSI/NIST container has an EES purpose: The ANSI/NIST container has a maximum file size of 5332 kB. • If the ANSI/NIST container has an SIS purpose: The ANSI/NIST container has a maximum file size of 3000 kB.

Test Case ID: TC-COM-FP-IMA-001*Evaluation***Evaluation Step****Description / Evaluation Result**

1

Description:

- Examine the response returned by the IUT via the test interface.

Expected Result:

- Overall, less than 0 percent of the requested images are not handled as expected.

2

Description:

- Examination of the returned fingerprint images by the IUT.

Expected Result:

- The returned fingerprint images show the same fingerprints that were shown on the input images.

Table 2.143 Test Case ID: TC-COM-FP-IMA-001

2.6.6 Test Cases FM COM-FP-WSQ

Test Case ID: TC-COM-FP-WSQ-001*Scope*

Examination of the size and compression format of the resulting fingerprint images.

Preconditions

- The IUT is in operation, required modules are loaded.
- The implementation number of the used Wavelet Scalar Quantisation (WSQ) algorithm is available.
- The IUT implements an interface for conformance testing where a single image can be provided and the according biometric container including the image data is returned.
- The following conditions apply regarding the database:
 - The image data that is sent to the IUT is of the following biometric type: FINGER.
 - The database contains fingerprint images.
 - The database contains corresponding Ground Truth values of WSQ compressed images with a bit-rate of 0.75.

CTS Mode

- automatic
- The image data that is sent to the IUT may have the following format:
 - BMP
- HTTP method: POST
- test case variants:

/TR03122/TC-COM-FP-WSQ-001/1

Table 2.146 TC-COM-FP-WSQ-001 Test Case Variants*Description***Step****Description / Expected Result**

1

Description:

- The IUT is called via the conformance test interface and an image from the database is provided. The image meets the properties described in the preconditions.
- Examination of the size of the WSQ image data.
- Examination of the software implementation number (sf) from the WSQ header of the image data.

Test Case ID: TC-COM-FP-WSQ-001

Expected Result:

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response contains the correct number of elements based on the XPath expression `//bio:Records/bio:BinaryRecord[@type="wsq"]/tradd:BinaryData/tradd:Value`. Exactly 1 element is expected in accordance to the acquisition process.
- The size of the compressed WSQ image is between 1/10 and 1/20 of the original raw image.
- The image resolution of the returned image is the same as that of the input image.
- The sf value in the WSQ header is not 0.
- The sf value equals the value from the preconditions.

Evaluation

Evaluation Step	Description / Evaluation Result
1	<i>Description:</i> • A device has failed the total test if the compression rate is outside the specified limits for more than 5% of the images submitted. <i>Expected Result:</i> • Overall, less than 5 percent of the requested images are not handled as expected.
2	<i>Description:</i> • Examination of the certification of the WSQ algorithm used by the IUT. <i>Expected Result:</i> • The WSQ implementation referenced by the software implementation number (sf) is certified by the FBI. • The WSQ implementation is version 3.1 and based on NBIS Version 5.0. • All returned fingerprint images can be decoded using a certified WSQ implementation.

Table 2.145 Test Case ID: TC-COM-FP-WSQ-001

2.6.7 Test Cases FM COM-FP-WSQE

Test Case ID: TC-COM-FP-WSQE-001

<i>Scope</i>	Examination of the resulting size and compression format of the WSQ generation from uncompressed fingerprint image data.
<i>Preconditions</i>	• The IUT is in operation, required modules are loaded. • The implementation number of the used Wavelet Scalar Quantisation (WSQ) algorithm is available. • The IUT implements an interface for conformance testing where a single image can be provided and the according biometric container including the image data is returned. • The following conditions apply regarding the database: • The image data that is sent to the IUT is of the following biometric type: FINGER. • The database contains uncompressed fingerprint images. • The database contains corresponding Ground Truth values of WSQ compressed images with a bit-rate of 0.75.

Test Case ID: TC-COM-FP-WSQE-001*CTS Mode*

- automatic
- The image data that is sent to the IUT may have the following format:
 - BMP
- HTTP method: POST
- test case variants:

/TR03122/TC-COM-FP-WSQE-001/2

Table 2.148 TC-COM-FP-WSQE-001 Test Case Variants*Description*

Step	Description / Expected Result
1	<i>Description:</i> • The IUT is called via the conformance test interface and an image from the database is provided. The image meets the properties described in the pre-conditions. • Examination of the size of the generated WSQ image data. • Examination of the software implementation number (sf) from the WSQ header of the image data. <i>Expected Result:</i> • The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • The response contains the correct number of elements based on the XPath expression //bio:Records/bio:BinaryRecord[@type="wsq"]/tradded:BinaryData/tradd:Value. Exactly 1 element is expected in accordance to the acquisition process. • The size of the compressed WSQ image is between 1/10 and 1/20 of the original raw image. • The image resolution of the returned image is the same as that of the input image. • The sf value in the WSQ header is not 0. • The sf value equals the value from the preconditions.

Evaluation

Evaluation Step	Description / Evaluation Result
1	<i>Description:</i> • Examine the response returned by the IUT via the test interface. <i>Expected Result:</i> • Overall, less than 0 percent of the requested images are not handled as expected.
2	<i>Description:</i> • Examination of the certification of the WSQ algorithm used by the IUT. <i>Expected Result:</i> • The WSQ implementation referenced by the software implementation number (sf) is certified by the FBI. • The WSQ implementation is version 3.1 and based on NBIS Version 5.0. • All returned fingerprint images can be decoded using a certified WSQ implementation.

Table 2.147 Test Case ID: TC-COM-FP-WSQE-001

2.6.8 Test Cases FM COM-FP-WSQR

Test Case ID: TC-COM-FP-WSQR-001

Scope	Examination of the size and compression format of the resulting fingerprint images				
Preconditions	- The IUT is in operation, required modules are loaded. - The following test resources are necessary: - An FBI certificate for the used Wavelet Scalar Quantisation (WSQ) implementation of the Implementation Under Test (IUT) is available.				
CTS Mode	Not supported				
Description	<table border="1"> <thead> <tr> <th>Step</th><th>Description / Expected Result</th></tr> </thead> <tbody> <tr> <td>1</td><td> <i>Description:</i> - Acquire a fingerprint using the IUT. - Extract the compressed fingerprint image file from the biometric container. <i>Expected Result:</i> - The size of the compressed WSQ image is smaller than or equal to 18 kB. - The sf value in the WSQ header is not equal 0. - The sf value is referenced in the FBI certificate for the WSQ implementation of the module. - All returned fingerprint images can be decoded with the WSQ algorithm. </td></tr> </tbody> </table>	Step	Description / Expected Result	1	<i>Description:</i> - Acquire a fingerprint using the IUT. - Extract the compressed fingerprint image file from the biometric container. <i>Expected Result:</i> - The size of the compressed WSQ image is smaller than or equal to 18 kB. - The sf value in the WSQ header is not equal 0. - The sf value is referenced in the FBI certificate for the WSQ implementation of the module. - All returned fingerprint images can be decoded with the WSQ algorithm.
Step	Description / Expected Result				
1	<i>Description:</i> - Acquire a fingerprint using the IUT. - Extract the compressed fingerprint image file from the biometric container. <i>Expected Result:</i> - The size of the compressed WSQ image is smaller than or equal to 18 kB. - The sf value in the WSQ header is not equal 0. - The sf value is referenced in the FBI certificate for the WSQ implementation of the module. - All returned fingerprint images can be decoded with the WSQ algorithm.				

Table 2.149 Test Case ID: TC-COM-FP-WSQR-001

2.7 Test Cases Operation

2.7.1 Test Cases FM O-ALL-USV

Test Case ID: TC-O-ALL-USV-001

Scope	Examination of the observability of the self-service system (SSS) used for to obtain digitised facial images.				
Applicability	This test case only applies to volume GID. The TC is only applicable if the IUT includes a physical back wall with the purpose of providing a uniform background behind the biometric subject.				
Preconditions	- The IUT is in operation, required modules are loaded. - The IUT implements a graphical user interface.				
CTS Mode	Not supported				
Description	<table border="1"> <thead> <tr> <th>Step</th><th>Description / Expected Result</th></tr> </thead> <tbody> <tr> <td>1</td><td> <i>Description:</i> - Position the test subject looking frontally into the camera at a distance of 70 cm from the camera system. This distance is to be understood as the length of the optical path. - Position the test subject with a body height of 140 cm in front of the camera system. Capture a facial image. - Calculate the frontal obstruction angle β_1 at the nasion of the biometric subject looking at the device. The frontal obstruction angle is calculated using the outermost edges of the device in the area 20 cm below and above the height of the "Frankfurt Horizon" of the test subject. - Calculate the backward obstruction angle β_2 at the nasion of the biometric subject looking at the device. The backward obstruction angle is calculated using the outermost edges of the back wall in the area 20 cm below and above the height of the "Frankfurt Horizon" of the test subject. </td></tr> </tbody> </table>	Step	Description / Expected Result	1	<i>Description:</i> - Position the test subject looking frontally into the camera at a distance of 70 cm from the camera system. This distance is to be understood as the length of the optical path. - Position the test subject with a body height of 140 cm in front of the camera system. Capture a facial image. - Calculate the frontal obstruction angle β_1 at the nasion of the biometric subject looking at the device. The frontal obstruction angle is calculated using the outermost edges of the device in the area 20 cm below and above the height of the "Frankfurt Horizon" of the test subject. - Calculate the backward obstruction angle β_2 at the nasion of the biometric subject looking at the device. The backward obstruction angle is calculated using the outermost edges of the back wall in the area 20 cm below and above the height of the "Frankfurt Horizon" of the test subject.
Step	Description / Expected Result				
1	<i>Description:</i> - Position the test subject looking frontally into the camera at a distance of 70 cm from the camera system. This distance is to be understood as the length of the optical path. - Position the test subject with a body height of 140 cm in front of the camera system. Capture a facial image. - Calculate the frontal obstruction angle β_1 at the nasion of the biometric subject looking at the device. The frontal obstruction angle is calculated using the outermost edges of the device in the area 20 cm below and above the height of the "Frankfurt Horizon" of the test subject. - Calculate the backward obstruction angle β_2 at the nasion of the biometric subject looking at the device. The backward obstruction angle is calculated using the outermost edges of the back wall in the area 20 cm below and above the height of the "Frankfurt Horizon" of the test subject.				

Test Case ID: TC-O-ALL-USV-001

Expected Result:

- The entire face is visible on the image, in the required resolution.
- The IUTs quality assessment of the captured image is positive.
- The calculated frontal obstruction angle β_1 is less than or equal to the maximum acceptable angle.
- The calculated backward obstruction angle β_2 is less than or equal to the maximum acceptable angle.

2

Description:

- Position the test subject looking frontally into the camera at a distance of 70 cm from the camera system. This distance is to be understood as the length of the optical path.
- Position the test subject with a body height of 170 cm in front of the camera system. Capture a facial image.
- Calculate the frontal obstruction angle β_1 at the nasion of the biometric subject looking at the device. The frontal obstruction angle is calculated using the outermost edges of the device in the area 20 cm below and above the height of the "Frankfurt Horizon" of the test subject.
- Calculate the backward obstruction angle β_2 at the nasion of the biometric subject looking at the device. The backward obstruction angle is calculated using the outermost edges of the back wall in the area 20 cm below and above the height of the "Frankfurt Horizon" of the test subject.

Expected Result:

- The entire face is visible on the image, in the required resolution.
- The IUTs quality assessment of the captured image is positive.
- The calculated frontal obstruction angle β_1 is less than or equal to the maximum acceptable angle.
- The calculated backward obstruction angle β_2 is less than or equal to the maximum acceptable angle.

3

Description:

- Position the test subject looking frontally into the camera at a distance of 70 cm from the camera system. This distance is to be understood as the length of the optical path.
- Position the test subject with a body height of 200 cm in front of the camera system. Capture a facial image.
- Calculate the frontal obstruction angle β_1 at the nasion of the biometric subject looking at the device. The frontal obstruction angle is calculated using the outermost edges of the device in the area 20 cm below and above the height of the "Frankfurt Horizon" of the test subject.
- Calculate the backward obstruction angle β_2 at the nasion of the biometric subject looking at the device. The backward obstruction angle is calculated using the outermost edges of the back wall in the area 20 cm below and above the height of the "Frankfurt Horizon" of the test subject.

Expected Result:

- The entire face is visible on the image, in the required resolution.
- The IUTs quality assessment of the captured image is positive.
- The calculated frontal obstruction angle β_1 is less than or equal to the maximum acceptable angle.
- The calculated backward obstruction angle β_2 is less than or equal to the maximum acceptable angle.

Table 2.150 Test Case ID: TC-O-ALL-USV-001

Test Case ID: TC-O-ALL-USV-002

Scope Examination of the observability of the self-service system (SSS) in regard to side panels.

Applicability This test case only applies to volume GID.

Preconditions

- The IUT is in operation, required modules are loaded.
- The IUT implements a graphical user interface.
- The following test resources are necessary:
 - One test subject is available to perform the acquisition with.

CTS Mode Not supported

Description

Step	Description / Expected Result
1	<i>Description:</i> • Verify that the capture area in front of the camera is accessible from both sides. <i>Expected Result:</i> • The capture area in front of the camera is accessible from both sides.
2	<i>Description:</i> • Position the test subject looking frontally into the camera at a distance of 70 cm from the camera system. This distance is to be understood as the length of the optical path. • Verify that no curtains, doors, or similar items are part of or attached to the IUT that could restrict visibility into the capture area. <i>Expected Result:</i> • No curtains, doors, or similar objects are part of or attached to the IUT that could restrict visibility into the capture area.
3	<i>Description:</i> • Position the biometric subject looking frontally into the camera at the maximal possible distance (according to the product documentatio) from the camera system. This distance is to be understood as the length of the optical path. • Verify that no curtains, doors, or similar items are part of or attached to the IUT that could restrict visibility into the capture area. <i>Expected Result:</i> • No curtains, doors, or similar objects are part of or attached to the IUT that could restrict visibility into the capture area.

Table 2.151 Test Case ID: TC-O-ALL-USV-002

Test Case ID: TC-O-ALL-USV-003

Scope Examination of specifications regarding conditions for installation in the authorities.

Applicability This test case only applies to volume GID.

Preconditions

- The following test resources are necessary:
 - Product documentation of IUT is at hand (e.g. data sheet, manual).

CTS Mode Not supported

Description

Step	Description / Expected Result
1	<i>Description:</i> • Consult product documentation of the device.

Test Case ID: TC-O-ALL-USV-003

Expected Result:

- The product documentation indicates the conditions for installation in the authorities.

Table 2.152 Test Case ID: TC-O-ALL-USV-003

2.8 Test Cases User Interface

2.8.1 Test Cases FM UI-FI-BSJ

Test Case ID: TC-UI-FI-BSJ-001

Scope	Note that the requirement of presence of a feedback screen depends on the application context. If the acquisition system is required to have a feedback screen for the facial image acquisition within the specific application context in question, or if the vendor decided to implement a feedback screen although it is not mandatory in the specific application context in question, the feedback screen shall be examined according to the following specifications: Examination of availability of feedback screen, digital mirror or physical mirror image and user guidance in the user interface shown to the biometric subject during facial image acquisition.	
Preconditions	• The IUT is in operation, required modules are loaded. • The integrated camera system used in the specific application scenario is connected. • The IUT implements a user interface of facial image acquisition shown to the biometric subject. • The IUT is able to display a digital mirror or physical mirror image and information for guidance to help the biometric subject with the correct positioning. • The acquisition system is required by another Function Module to have a feedback screen for the facial acquisition • Fake facial images (e.g. created with a 3D-printer) are available as necessary test resource.	
CTS Mode	Not supported	
Description	Step	Description / Expected Result
	1	<i>Description:</i> A biometric subject performs a compliant face presentation in the IUT's capture area. <i>Expected Result:</i> • A digital mirror or physical mirror image is shown by the IUT for guidance to help the biometric subject with the correct positioning. • A compliant facial image is acquired.
	2	<i>Description:</i> A biometric subject performs a non-compliant face presentation in the IUT's capture area where positional requirements are not met. <i>Expected Result:</i> • A digital mirror or physical mirror image is shown by the IUT for guidance to help the biometric subject with the correct positioning. • The IUT displays user guidance information to help the biometric subject with the correct positioning in front of the camera. • The facial image acquisition is omitted by the IUT.
	3	<i>Description:</i> • A biometric subject initially performs a non-compliant face presentation in the IUT's capture area where positional requirements are not met. • The biometric subject follows the user guidance information provided by the IUT until a compliant face presentation is performed where positional requirements are met.

Test Case ID: TC-UI-FI-BSJ-001

4	<i>Expected Result:</i>
	• A digital mirror or physical mirror image is shown by the IUT for guidance to help the biometric subject with the correct positioning. • A compliant facial image is acquired.
	<i>Description:</i>
	A biometric subject initially performs a facial image presentation attack in the IUT's capture area.
	<i>Expected Result:</i>
	• A facial image is captured and the resulting image is displayed by the IUT. • Both the PAD result and the PAD score are only displayed to the operator. Neither the PAD result nor the PAD score are displayed to the biometric subject.

Table 2.153 Test Case ID: TC-UI-FI-BSJ-001

2.8.2 Test Cases FM UI-FI-OP

Test Case ID: TC-UI-FI-OP-001

<i>Scope</i>	Examination of the correct user interface for facial images with sufficient quality	
<i>Preconditions</i>	• IUT is running, required modules are loaded • If the IUT does not provide a graphical user interface (e.g. it is implemented as a library), a rudimentary example user interface has to be provided for evaluation purposes	
<i>CTS Mode</i>	Not supported	
<i>Description</i>	Step	Description / Expected Result
	1	<i>Description:</i> An image with sufficient quality is captured using the IUT. <i>Expected Result:</i> • Any relevant quality components is displayed in the GUI (each resulting in OK) • The overall quality is indicated visually by the IUT (OK) • The IUT provides the option to put in veto before release of the image (veto is put in via UI, i.e. image is rejected despite positive Quality Assessment (QA) result computed by IUT) • The IUT provides a warning message to the operator if PAD has been performed and a presentation attack alarm has been raised. All facial images within an acquisition where at least one facial image caused a PAD alarm SHALL be displayed to the operator. Those facial images for which a PAD alarm is present SHALL be highlighted (e.g. by a red frame).

Table 2.154 Test Case ID: TC-UI-FI-OP-001**Test Case ID: TC-UI-FI-OP-002**

<i>Scope</i>	Examination of the correct user interface for facial images with insufficient quality.	
<i>Preconditions</i>	• IUT is running, required modules are loaded • If the IUT does not provide a graphical user interface (e.g. it is implemented as a library), a rudimentary example user interface has to be provided for evaluation purposes	
<i>CTS Mode</i>	Not supported	
<i>Description</i>	Step	Description / Expected Result

Test Case ID: TC-UI-FI-OP-002

1

Description:

An image with insufficient quality is captured using the IUT e.g. the eye are closed test person closes the eyes and rotates the head.

Initiate test case calling the IUT via the test interface

Expected Result:

- Any relevant quality components is displayed in the GUI (at least one resulting in NOK)
- The overall quality is indicated visually by the IUT (NOK)
- The cause for the negative overall quality assessment is displayed
- The IUT provides the option to put in veto before release of the image (veto is put in via UI, i.e. image is accepted despite negative QA result computed by IUT)
- The IUT provides a warning message to the operator if PAD has been performed and a presentation attack alarm has been raised. All facial images within an acquisition where at least one facial image caused a PAD alarm SHALL be displayed to the operator. Those facial images for which a PAD alarm is present SHALL be highlighted (e.g. by a red frame).

Table 2.155 Test Case ID: TC-UI-FI-OP-002

2.8.3 Test Cases FM UI-FP-BSJ

Test Case ID: TC-UI-FP-BSJ-001

Scope Examination of availability of real-time feedback and user guidance in the user interface shown to the biometric subject during fingerprint acquisition.

- Preconditions*
- The IUT is in operation, required modules are loaded.
 - The fingerprint scanner used in the specific application scenario is connected.
 - The IUT implements a user interface of fingerprint acquisition shown to the biometric subject.
 - The IUT is able to display feedback and information for guidance to help the biometric subject with the correct positioning.
 - Fake fingerprints (e.g. made of silicone) are available as necessary test resource.

CTS Mode Not supported

Description

Step	Description / Expected Result
1	<i>Description:</i> A biometric subject performs a compliant fingerprint presentation on the IUT's fingerprint scanner. <i>Expected Result:</i> • A digital feedback is shown by the IUT in real-time including the acquisition process status and guidance to help the biometric subject with the fingerprint selection and correct positioning. • A compliant fingerprint image is acquired.
2	<i>Description:</i> A biometric subject performs a non-compliant fingerprint presentation on the IUT's fingerprint scanner where positional requirements are not met.

Test Case ID: TC-UI-FP-BSJ-001

3	<i>Expected Result:</i>
	• A digital feedback is shown by the IUT in real-time including the acquisition process status and guidance to help the biometric subject with the fingerprint selection and correct positioning. • The IUT displays user guidance information to help the biometric subject with the correct positioning on the sensor. • The fingerprint image acquisition is omitted by the IUT.
4	<i>Description:</i>
	• A biometric subject initially performs a non-compliant fingerprint presentation on the IUT's fingerprint scanner where positional requirements are not met. • The biometric subject follows the user guidance information provided by the IUT until a compliant fingerprint presentation is performed where positional requirements are met.
4	<i>Expected Result:</i>
	• A digital feedback is shown by the IUT in real-time including the acquisition process status and guidance to help the biometric subject with the fingerprint selection and correct positioning. • A compliant fingerprint image is acquired.
4	<i>Description:</i>
	A biometric subject initially performs a fingerprint presentation attack on the IUT's fingerprint scanner.
4	<i>Expected Result:</i>
	• A fingerprint or slap image is captured and the resulting image is displayed by the IUT. • Both the PAD result and the PAD score are only displayed to the operator. Neither the PAD result nor the PAD score are displayed to the biometric subject.

Table 2.156 Test Case ID: TC-UI-FP-BSJ-001

2.8.4 Test Cases FM UI-FP-OP

Test Case ID: TC-UI-FP-OP-001

<i>Scope</i>	Examination of the correct user interface for capturing fingerprints.	
<i>Preconditions</i>	• The IUT is in operation, required modules are loaded. • The fingerprint scanner used in the specific application scenario is connected. • The IUT implements a user interface of fingerprint acquisition shown to the biometric subject. • The IUT is able to display a digital feedback and information for guidance to help the biometric subject with the correct positioning. • If the IUT does not provide a graphical user interface (e.g. it is implemented as a library), a rudimentary example user interface has to be provided for evaluation purposes.	
<i>CTS Mode</i>	Not supported	
<i>Description</i>	Step	Description / Expected Result
	1	<i>Description:</i> A biometric subject performs a compliant fingerprint presentation on the IUT's fingerprint scanner.

Test Case ID: TC-UI-FP-OP-001*Expected Result:*

- Guidance to help the biometric subject with the fingerprint selection and correct positioning is shown by the IUT.
- Visual feedback of the actual fingerprints is given by the IUT (at least by displaying the resulting captured images).
- The IUT provides the following information to the operator:
 - Visual feedback of all segmented fingerprints.
 - Feedback on the quality level of captured fingerprints.
 - A warning message if a slap acquisition is in place, a slap classifier is in use and the classification result mismatches with the expected slap of the current acquisition.
 - A warning message if PAD has been performed and a presentation attack alarm has been raised (for each finger individually and in form of an overall result).
- The IUT provides veto power for the operator for the enrolment.

2

Description:

A biometric subject performs a non-compliant fingerprint presentation on the IUT's fingerprint scanner.

Expected Result:

- Guidance to help the biometric subject with the fingerprint selection and correct positioning is shown by the IUT.
- Visual feedback of the actual fingerprints is given by the IUT (at least by displaying the resulting captured images).
- The IUT provides the following information to the operator:
 - Visual feedback of all segmented fingerprints.
 - Feedback on the quality level of captured fingerprints.
 - A warning message if a slap acquisition is in place, a slap classifier is in use and the classification result mismatches with the expected slap of the current acquisition.
 - A warning message if PAD has been performed and a presentation attack alarm has been raised (for each finger individually and in form of an overall result).
- The IUT provides veto power for the operator for the enrolment.

Table 2.157 Test Case ID: TC-UI-FP-OP-001**Test Case ID: TC-UI-FP-OP-002**

<i>Scope</i>	Examination of the correct user interface for capturing fingerprints if a uniqueness check error occurs.				
<i>Preconditions</i>	• The IUT is in operation, required modules are loaded. • The fingerprint scanner used in the specific application scenario is connected. • The IUT implements a user interface of fingerprint acquisition shown to the biometric subject. • If the IUT does not provide a graphical user interface (e.g. it is implemented as a library), a rudimentary example user interface has to be provided for evaluation purposes.				
<i>CTS Mode</i>	Not supported				
<i>Description</i>	<table border="1"> <thead> <tr> <th>Step</th><th>Description / Expected Result</th></tr> </thead> <tbody> <tr> <td></td><td></td></tr> </tbody> </table>	Step	Description / Expected Result		
Step	Description / Expected Result				

Test Case ID: TC-UI-FP-OP-002

1

Description:

A biometric subject performs a compliant fingerprint presentation on the IUT's fingerprint scanner.

The biometric subject performs a fingerprint presentation on the IUT's fingerprint scanner using the same finger for a different finger position.

Expected Result:

- The IUT displays a warning message that a uniqueness check error has been detected.
- The fingers involved in the unexpected successful comparisons SHALL be pictorially displayed to the operator.

Table 2.158 Test Case ID: TC-UI-FP-OP-002

2.8.5 Test Cases FM UI-FP-VER

Test Case ID: TC-UI-FP-VER-001

Scope Examination of availability of user interface for operator during fingerprint verification.

Preconditions

- The IUT is in operation, required modules are loaded.
- The fingerprint scanner used in the specific application scenario is connected.
- The IUT implements a user interface of fingerprint acquisition shown to the operator.
- The IUT is able to display a visual feedback of the fingerprint verification to the operator.

CTS Mode Not supported

*Description***Step Description / Expected Result**

1

Description:

- A biometric subject performs a compliant fingerprint presentation on the IUT's fingerprint scanner.
- Depending on the used scanner a compliant presentation of a single finger, multiple fingers or slap is conducted, respectively.

Expected Result:

A compliant fingerprint image is acquired.

2

Description:

The operator initiates a fingerprint verification process.

Expected Result:

A visual feedback about the result of the verification process is shown by the IUT to the operator.

Table 2.159 Test Case ID: TC-UI-FP-VER-001

2.9 Test Cases Biometric Comparison

2.9.1 Test Cases FM CMP-FI-VER

Test Case ID: TC-CMP-FI-VER-001

Scope Examination of the configured security level for biometric comparison.

Preconditions

- The following test resources are necessary:
 - The vendor of the biometric comparison algorithm provides a documentation which gives information about the working point of the comparison algorithm.

CTS Mode Not supported

*Description***Step Description / Expected Result**

Test Case ID: TC-CMP-FI-VER-001

1

Description:

- Examine the documentation provided by the vendor of the comparison algorithm.

Expected Result:

- The documentation gives information (typically a DET curve) of the performance of the algorithm based on a reference database.
- The documentation gives evidence that the comparison algorithm ensures a working point of a maximum false-match-rate (FMR) of 0.1 % and a false-non-match-rate (FNMR) below 2 %. Stronger settings are allowed (lower FMR and/or FNMR).
- The documentation gives evidence that the comparison algorithm can be calibrated for the security level set within this specific scenario of verification.

Table 2.160 Test Case ID: TC-CMP-FI-VER-001**2.9.2 Test Cases FM CMP-FP-CV****Test Case ID: TC-CMP-FP-CC-001**

Scope Examination of the configured security level for biometric comparison.

Preconditions

- The following test resources are necessary:
- The vendor of the biometric comparison algorithm provides a documentation which gives information about the operating point of the comparison algorithm.

CTS Mode Not supported

Description

Step Description / Expected Result

1

Description:

- Examine the documentation provided by the vendor of the comparison algorithm.

Expected Result:

- The documentation gives information (typically a DET curve) of the performance of the algorithm based on a reference database.
- The documentation gives evidence that the comparison algorithm ensures an operating point of a maximum false match rate (FMR) of 0.1 % and a false non-match rate (FNMR) below 2 %. Stronger settings are allowed (lower FMR and/or FNMR).
- The documentation gives evidence that the comparison algorithm can be calibrated for the security level set within this specific scenario of verification.

Table 2.161 Test Case ID: TC-CMP-FP-CC-001**Test Case ID: TC-CMP-FP-CC-002**

Scope Examination of proper crossmatching functionality of plain and rolled fingerprints.

Preconditions

- The IUT is in operation, required modules are loaded.
- The multi fingerprint scanner used in the specific application scenario is connected
- The following test resources are necessary:
 - One test subject is available to perform the acquisition with.

Test Case ID: TC-CMP-FP-CC-002

- CTS Mode*
- interactive
 - no provision of pre-defined input data
 - HTTP method: GET
 - test case variants:

/TR03122/TC-CMP-FP-CC-002/1

Table 2.163 TC-CMP-FP-CC-002 Test Case Variants*Description*

Step	Description / Expected Result
1	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • Capture 10 fingerprints, each fingerprint plain and rolled (in identical, mated order) resulting in a total of 10 captured fingerprint pairs (plain, rolled) of matching fingerprints <i>Expected Result:</i> • The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • The response contains information about both the flat and rolled FingerAcquisition. • The IUT returns a comparison score per pair of prints and each finger and a result of each verification (element bio:Verification). • Each single comparison score is above the threshold and each verification result is positive.
2	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • Capture 10 fingerprints, each fingerprint plain and rolled in non-mated order (e.g. by permutating the fingers or using a second testperson) resulting in a total of 10 captured fingerprint pairs (plain, rolled) of non-matching fingerprints <i>Expected Result:</i> • The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • The response contains information about both the flat and rolled FingerAcquisition. • The IUT returns a comparison score per pair of prints and each finger and a result of each verification (element bio:Verification). • Each single comparison score is below the threshold and each verification result is negative.

Table 2.162 Test Case ID: TC-CMP-FP-CC-002

2.9.3 Test Cases FM CMP-FP-VER

Test Case ID: TC-CMP-FP-VER-001

- Scope* Examination of the configured security level for biometric comparison.
- Preconditions*
- The following test resources are necessary:
 - The vendor of the biometric comparison algorithm provides a documentation which gives information about the working point of the comparison algorithm.

CTS Mode Not supported

Description

Step	Description / Expected Result
------	-------------------------------

Test Case ID: TC-CMP-FP-VER-001

1

Description:

- Examine the documentation provided by the vendor of the comparison algorithm.

Expected Result:

- The documentation gives information (typically a DET curve) of the performance of the algorithm based on a reference database.
- The documentation gives evidence that the comparison algorithm ensures a working point of a maximum false match rate (FMR) of 0.1 % and a false non-match rate (FNMR) below 2 %. Stronger settings are allowed (lower FMR and/or FNMR).
- The documentation gives evidence that the comparison algorithm can be calibrated for the security level set within this specific scenario of verification.

Table 2.164 Test Case ID: TC-CMP-FP-VER-001

2.10 Test Cases Logging

2.10.1 Test Cases FM LOG-ALL-GENERIC

Test Case ID: TC-LOG-ALL-GENERIC-001

Scope Examination of correct syntax and semantics for the logging of information regarding generic processes and transactions.

Preconditions

- Five sets of logging data (encoded as XML) each, e.g. produced by performing five deviating processes using IUT
- The performed steps for each log data set are documented.
- TR-03121 XML schema definition files and documentation.
- Vendor specific documentation for error codes and further information.

CTS Mode File based

Description

Step	Description / Expected Result
------	-------------------------------

1

Description:

Validate each log file against the TR-03121 schema definition "trbio6v1.xsd".

Expected Result:

Each file passes the validation process without errors.

2

Description:

Verify the content of each log file and check the contained information.

Test Case ID: TC-LOG-ALL-GENERIC-001*Expected Result:*

- Each log file contains at least one element of type `type.transaction`.
- The transaction UUIDs are globally unique.
- The value of `Location` is compliant to the semantic of the relevant profile.
- If this transaction is dependent or derived from another transaction, the `TransactionReference` is set to the corresponding transaction UUID.
- In case of any abnormal termination of the transaction or any of its sub-processes, the `ErrorCode` is set.
- Each during the transaction performed enrolment process is represented by an `Enrolment` element.
 - If a central system replies directly with enrolment statues information, the `SubmitTime` is set.
 - A `ControlVerification` element is logged for any control verification performed during the enrolment process.
- Each during the transaction performed identification process is represented by an `Identification` element.

If a central system provides detailed scoring information, the list of candidates is contained.
- Each during the transaction performed verification process is represented by a `Verification` element.
 - All performed comparisons are logged.
 - For each comparison the vendor specific score as well as the threshold are logged.

Table 2.165 Test Case ID: TC-LOG-ALL-GENERIC-001**Test Case ID: TC-LOG-ALL-GENERIC-002**

Scope Examination of error code list for abnormal termination.

Preconditions Error code list for IUT is at hand.

CTS Mode Not supported

Description

Step	Description / Expected Result
1	<i>Description:</i> Check availability of detailed error code list. <i>Expected Result:</i> A detailed list of error codes used with complete semantic descriptions is available for any error code in case of abnormal termination of the transaction or any of its sub-processes.

Table 2.166 Test Case ID: TC-LOG-ALL-GENERIC-002

2.10.2 Test Cases FM LOG-ALL-ARE

Test Case ID: TC-LOG-ALL-ARE-001

Scope Examination of correct syntax and semantics for the logging of information in the context of alien register enrolment.

Test Case ID: TC-LOG-ALL-ARE-001*Preconditions*

- Five sets of logging data (encoded as XML) containing are-log (or fragment elements of biotypes6v1, where are-log is not required) are available.
- Each logging data set is produced by performing four deviating processes using the IUT, which include but are not limited to:
 - Bona-fide acquisition of at least one modality with fulfilled quality components.
 - Acquisition of at least one modality with unfulfilled quality components.
 - Rejected acquisition result via veto by operator and repeated acquisition.
 - Failed acquisition process based on incorrect behaviour of the biometric subject.
- The performed steps for each log data set are documented.
- TR-03121 XML schema definition files and documentation are at hand.

CTS Mode

File based

*Description***Step** **Description / Expected Result**

1

Description:

Validate each log file against the TR-03121 schema definition "are6v1.xsd".

Expected Result:

Each file passes the validation process without errors.

2

Description:

Verify the content of each log file and check the contained information.

Expected Result:

- The process flow and information described by the contents of the log file matches the actually performed process using the IUT.
- If information about existing facial images of the applicant exist in the CIR, the number of images, whether an acceptable image is available and whether any of the acceptable image was used are logged.
- Details for each performed control verification are stored.
- The name of the city or village where the biometric process takes place is set as Location.
- Depending on the implemented Application Profile the expected root element and sub-elements are present. Deviations are only permissible if justified by the process being carried out (e.g. process cancelled).

Table 2.167 Test Case ID: TC-LOG-ALL-ARE-001

2.10.3 Test Cases FM LOG-ALL-BCL

Test Case ID: TC-LOG-ALL-BCL-001*Scope*

Examination of correct syntax and semantics for the logging of information in the context of border control.

Test Case ID: TC-LOG-ALL-BCL-001

- Preconditions*
- Five sets of logging data (encoded as XML) containing `bc1-log` (or fragment elements of `bioty-pes6v1`, where `bc1-log` is not required) are available.
 - Each logging data set is produced by performing five deviating processes using the IUT, which include but are not limited to:
 - Bona-fide acquisition of at least one modality with fulfilled quality components.
 - Acquisition of at least one modality with unfulfilled quality components.
 - Cancelled acquisition process.
 - Rejected image acquisition result via veto by operator and repeated acquisition.
 - Failed acquisition process based on incorrect behaviour of the biometric subject.
 - The performed steps for each log data set are documented.
 - TR-03121 XML schema definition files and documentation are at hand.

CTS Mode

File based

Description

Step	Description / Expected Result
1	<i>Description:</i> Validate each log file against the TR-03121 schema definition "<code>bcl6v1.xsd</code>". <i>Expected Result:</i> Each file passes the validation process without errors.
2	<i>Description:</i> Verify the content of each log file and check the contained information. <i>Expected Result:</i> • The process flow and information described by the contents of the log file matches the actually performed process using the IUT. • For externally stored images the embedded <code>type.data.record</code> elements refer to the image via the attribute <code>externalReference</code> and furthermore do not contain <code>Data</code> elements. • Depending on the implemented Application Profile the expected root element and sub-elements are present. Deviations are only permissible if justified by the process being carried out (e.g. process cancelled).

Table 2.168 Test Case ID: TC-LOG-ALL-BCL-001

2.10.4 Test Cases FM LOG-ALL-BDS

Test Case ID: TC-LOG-ALL-BDS-001

- Scope*
- Examination of correct syntax and semantics for the logging of information for biometric data selections in the context of identity documents.

Test Case ID: TC-LOG-ALL-BDS-001

<i>Preconditions</i>	• Five sets of logging data (encoded as XML) containing <code>gid-log</code> elements are available. • Each logging data set is produced by performing five deviating processes using the IUT, which include but are not limited to: • Bona-fide acquisition of at least one modality with fulfilled quality components. • Acquisition of at least one modality with unfulfilled quality components. • A transaction for a biometric subject with legal basis to take fingerprints. • A transaction for a biometric subject with legal basis to take fingerprints where no fingerprints could be captured. • A transaction for a biometric subject without a legal basis to take fingerprints (i.e. under age). • The performed steps for each log data set are documented. • TR-03121 XML schema definition files and documentation are at hand. • Note: If the IUT is a biometric component only, this test case shall not be applied as the relevant data are not acquired by the IUT itself.
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CTS Mode File based

Description

Step	Description / Expected Result
1	<i>Description:</i> Validate each log file against the TR-03121 schema definition "<code>gid6v1.xsd</code>". <i>Expected Result:</i> Each file passes the validation process without errors.
2	<i>Description:</i> Verify the content of each log file and check the contained information. <i>Expected Result:</i> • Information whether fingerprint are legally allowed to be captured are represented by <code>FingerprintCaptureAllowed</code>. • If there was no legal basis to take fingerprints, the information is set to <code>false</code> and no fingerprint images are contained. • If there was a legal basis to take fingerprints for a person without fingers or hands, the information is set to <code>true</code>. • The element <code>Location</code> contains the "Behördenkennziffer".

Table 2.169 Test Case ID: TC-LOG-ALL-BDS-001

2.10.5 Test Cases FM LOG-ALL-GID

Test Case ID: TC-LOG-ALL-GID-001

<i>Scope</i>	Examination of correct syntax and semantics for the logging of information in the context of identity documents.		
<i>Preconditions</i>	• One set of logging data (encoded as XML) after completing the process of the respective implemented Application Profile(s). • If multiple Application Profiles are implemented (e.g. a Self-Service System with both fingerprint and facial image acquisition) the entire process has to be performed. • The performed steps for each log data set are documented. • TR-03121 XML schema definition files and documentation are at hand.		
<i>CTS Mode</i>	File based		
<i>Description</i>	<table> <tr> <th>Step</th><th>Description / Expected Result</th></tr> </table>	Step	Description / Expected Result
Step	Description / Expected Result		

Test Case ID: TC-LOG-ALL-GID-001

1

Description:

Validate each log file against the TR-03121 schema definition "gid6v1.xsd" and verify the stated Application Profile within the contained ApplicationProfile-elements matches the actual implemented Application Profile(s) according to the Table definitions of FM LOG-ALL-GID.

Expected Result:

- Each file passes the validation process without errors.
- The created logging data by the IUT has the ApplicationProfile-elements filled with the matching Application Profile(s).
- For logging data that has only been processed by the IUT the ApplicationProfile-elements have not been modified in the logging data extended by the IUT.
- Depending on the implemented Application Profile the expected root element and sub-elements are present. Deviations are only permissible if justified by the process being carried out (e.g. process cancelled).

Table 2.170 Test Case ID: TC-LOG-ALL-GID-001

2.10.6 Test Cases FM LOG-ALL-IMA

Test Case ID: TC-LOG-ALL-IMA-001

Scope Examination of correct syntax and semantics for the logging of information in the context of border control.

Preconditions

- Five sets of logging data (encoded as XML) containing ima-log elements are available.
- Each logging data set is produced by performing five deviating processes using the IUT, which include but are not limited to:
 - bona-fide acquisition of at least one modality with fulfilled quality components
 - acquisition of at least one modality with unfulfilled quality components
 - cancelled acquisition process
 - rejected image acquisition result via veto by operator and repeated acquisition
 - failed acquisition process based on incorrect behaviour of the subject.
- The performed steps for each log data set are documented.
- TR-03121 XML schema definition files and documentation are at hand.

CTS Mode File based

*Description***Step Description / Expected Result**

1

Description:

Validate each log file against the TR-03121 schema definition "trbio6v1.xsd".

Expected Result:

Each file passes the validation process without errors.

2

Description:

Verify the content of each log file and check the contained information.

Expected Result:

- The process flow and information described by the contents of the log file matches the actually performed process using the IUT.
- The element ApplicationProfile is filled with a valid application profile of volume IMA.

Table 2.171 Test Case ID: TC-LOG-ALL-IMA-001

2.10.7 Test Cases FM LOG-FI-GENERIC

Test Case ID: TC-LOG-FI-GENERIC-001

<i>Scope</i>	Examination of correct syntax and semantics for the logging of information regarding generic facial image acquisitions.
<i>Preconditions</i>	• Five sets of logging data (encoded as XML) containing FaceAcquisition elements are available. • Each logging data set is produced by performing five deviating processes using the IUT, which include but are not limited to: • Bona-fide facial image acquisition with fulfilled quality components. • Facial image acquisition with unfulfilled quality components. • Cancelled acquisition process. (This is only applicable in the case of application profiles within the volume border control.) • Rejected image acquisition result via veto by operator and repeated acquisition. (This is not applicable to self-service systems.) • Failed acquisition process based on incorrect behaviour of the biometric subject. • The performed steps for each log data set are documented. • TR-03121 XML schema definition files and documentation are at hand.
<i>CTS Mode</i>	File based
<i>Description</i>	

Step	Description / Expected Result
1	<i>Description:</i> Validate each log file against the TR-03121 schema definition "trbio6v1.xsd". <i>Expected Result:</i> Each file passes the validation process without errors.
2	<i>Description:</i> Verify the content of each log file and check the contained information. <i>Expected Result:</i> • For each performed capture a FaceCapture element exists. • If a veto was put by the operator, the type of veto is logged. • Quality information are provided at least for the selected capture. • For each quality component the identifier, upper and lower value bound as well as the upper and lower threshold are logged. • If more than one FaceCapture element exists, the capture attribute references the related capture within each FaceQuality element. • The quality values are logged for each performed PAD. • If an user interface was available during the face acquisition, the displayed information are stored in an UserInterface element.

Table 2.172 Test Case ID: TC-LOG-FI-GENERIC-001

Test Case ID: TC-LOG-FI-GENERIC-002

<i>Scope</i>	Examination of error code list for abnormal termination.
<i>Preconditions</i>	Error code list for IUT is at hand.
<i>CTS Mode</i>	Not supported
<i>Description</i>	

Step	Description / Expected Result
1	<i>Description:</i> Check availability of detailed error code list.

Test Case ID: TC-LOG-FI-GENERIC-002*Expected Result:*

A detailed list of error codes used with complete semantic descriptions is available for any error code in case of abnormal termination of the transaction or any of its sub-processes.

Table 2.173 Test Case ID: TC-LOG-FI-GENERIC-002

2.10.8 Test Cases FM LOG-FI-BCL

Test Case ID: TC-LOG-FI-BCL-001

Scope Examination of correct syntax and semantics for the logging of facial image acquisitions information in the context of border control.

Preconditions

- Five sets of logging data (encoded as XML) containing bc1-log elements are available.
- Each logging data set is produced by performing five deviating processes using the IUT, which include but are not limited to:
 - Bona-fide acquisition of at least one modality with fulfilled quality components.
 - Acquisition of at least one face image with unfulfilled quality components.
 - Cancelled acquisition process.
 - Rejected image acquisition result via veto by operator and repeated acquisition.
 - Failed acquisition process based on incorrect behaviour of the biometric subject.
- The performed steps for each log data set are documented.
- TR-03121 XML schema definition files and documentation are at hand.

CTS Mode File based

Description

Step	Description / Expected Result
1	<i>Description:</i> Validate each log file against the TR-03121 schema definition "bcl6v1.xsd". <i>Expected Result:</i> Each file passes the validation process without errors.
2	<i>Description:</i> Verify the content of each log file and check the contained information. <i>Expected Result:</i> • The process flow and information described by the contents of the log file matches the actually performed process using the IUT. • For each performed PAD the detailed PAD quality values are logged.

Table 2.174 Test Case ID: TC-LOG-FI-BCL-001

2.10.9 Test Cases FM LOG-FI-GID

Test Case ID: TC-LOG-FI-GID-001

Scope Check of consistency of logged values

Preconditions

- Scanning of application form or capturing of the facial image and provision of a selected single image
- Provision of the according biometric container including the image data

Description

Step	Description / Expected Result
1	<i>Description:</i> Examination of relation of the range and/or tolerance if values are available

Test Case ID: TC-LOG-FI-GID-001

2

Expected Result:

If min, max, tmin or tmax are present the following relation has to be valid:
 $\text{min} \leq \text{tmin} \leq \text{tmax} \leq \text{max}$

Description:

Examination of the XML Logging

Expected Result:

- The XML-element FaceAcquisition and/or FaceAcquisition MAY exist multiple times (including their Records/BinaryRecord children).
-
- The XML-element FaceAcquisition/Records/BinaryRecord/@type and/or FaceDelivery/Records/BinaryRecord/@type is jpeg when coming from a face acquisition or delivery process.
- The XML-element FaceAcquisition/Records/BinaryRecord/BinaryData/Value or FaceDelivery/Records/BinaryRecord/BinaryData/Value occurs exactly once before being sent to document production after the selection procedure. The output image fulfills the requirements to be sent to the document production (e.g. FaceAcquisition/Records/BinaryRecord/@type or FaceDelivery/Records/BinaryRecord/@type is jpeg2000). All other binary data values of face deliveries or acquisitions have been deleted. However, multiple appearances of these elements are permitted before being fed to the selection procedure (e.g. from any acquisition or delivery application).
- Neither the XML-element FaceAcquisition/Records/XMLRecord nor the XML-element FaceDelivery/Records/XMLRecord is used.

Table 2.175 Test Case ID: TC-LOG-FI-GID-001

2.10.10 Test Cases FM LOG-FP-GENERIC

Test Case ID: TC-LOG-FP-GENERIC-001

Scope	Examination of correct syntax and semantics for the logging of generic fingerprint information.	
Preconditions	• Five sets of logging data (encoded as XML) containing FingerAcquisition elements are available. • Each logging data set is produced by performing five deviating processes using the IUT, which include but are not limited to: • Bona-fide acquisition of at least one modality with fulfilled quality components. • Acquisition of at least one fingerprint with unfulfilled quality components. • Cancelled acquisition process. (This is only applicable in the case of application profiles within the volume border control.) • Fingerprint acquisitions with missing fingers or slaps. • Failed acquisition process based on incorrect behaviour of the biometric subject. • The performed steps for each log data set are documented. • TR-03121 XML schema definition files and documentation are at hand.	
CTS Mode	File based	
Description	Step	Description / Expected Result
	1	<i>Description:</i> Validate each log file against the TR-03121 schema definition "trbio6v1.xsd". <i>Expected Result:</i> Each file passes the validation process without errors.

Test Case ID: TC-LOG-FP-GENERIC-001

2

Description:

Verify the content of each log file and check the contained information.

Expected Result:

- The process flow and information described by the contents of the log file matches the actually performed process using the IUT.
- The reference to the corresponding record is logged if available
- For each performed PAD the detailed PAD quality values are logged.
- Details of the uniqueness check are contained if performed and duplicates are found.
- In unsupervised acquisition scenarios the surveillance information are stored and linked to the fingerprint capture attempt.
- The `MultiplePersonDetectionInformation` specifies whether multiple persons have been detected during the fingerprint acquisition process or not.
- If a slap classification is performed during the acquisition process, the details are be logged as `FingerClassifierInformation`.

Table 2.176 Test Case ID: TC-LOG-FP-GENERIC-001**2.10.11 Test Cases FM LOG-FP-BCL****Test Case ID: TC-LOG-FP-BCL-001**

<i>Scope</i>	Verification of semantics of logging data and information contained therein.				
<i>Preconditions</i>	• At least 2 biometric subjects Five sets of logging data (encoded as XML) containing <code>FingerAcquisition</code> elements are available. • Each logging data set is produced by performing five deviating processes using the IUT, which include but are not limited to: • Bona-fide acquisition of at least one modality with fulfilled quality components. • Acquisition of at least one fingerprint with unfulfilled quality components. • Cancelled acquisition process. • Fingerprint acquisitions with missing fingers or slaps. • Failed acquisition process based on incorrect behaviour of the biometric subject. • The performed steps for each log data set are documented. • TR-03121 XML schema definition files and documentation are at hand.				
<i>CTS Mode</i>	Not supported				
<i>Description</i>	<table> <tr> <th>Step</th><th>Description / Expected Result</th></tr> <tr> <td>1</td><td> <i>Description:</i> Verify the content of each log file and check the contained information. </td></tr> </table>	Step	Description / Expected Result	1	<i>Description:</i> Verify the content of each log file and check the contained information.
Step	Description / Expected Result				
1	<i>Description:</i> Verify the content of each log file and check the contained information.				

Test Case ID: TC-LOG-FP-BCL-001*Expected Result:*

- The process flow and information described by the contents of the log file matches the actually performed process using the IUT.
- The reference to the corresponding record is logged if available
- For each performed PAD the detailed PAD quality values are logged.
- Details of the uniqueness check are contained if performed and duplicates are found.
- In unsupervised acquisition scenarios the surveillance information are stored and linked to the fingerprint capture attempt.
- The `MultiplePersonDetectionInformation` is specified when multiple persons have been present during the fingerprint acquisition process.
- If a slap classification is performed during the acquisition process, the details are be logged as `FingerClassifierInformation`.

Table 2.177 Test Case ID: TC-LOG-FP-BCL-001**2.10.12 Test Cases FM LOG-FP-GID****Test Case ID: TC-LOG-FP-GID-001**

Scope Check of consistency of logged values

Preconditions

- Acquisition of the fingerprints of two different fingers
- Provision of the according biometric container including the image data

Description

Step	Description / Expected Result
1	<i>Description:</i> For both elements of record in <code>bio:FingerAcquisition</code>: Examination of the values in the field Biometric Data Block (BDB) Biometric Subtype in the Common Biometric Exchange Formats Framework (CBEFF) Header and finger position code in the finger image header <i>Expected Result:</i> For each record: the BDB Biometric Subtype contains the value for the same finger as the finger position code in the finger image header record
2	<i>Description:</i> For both elements record in <code>bio:FingerAcquisition</code>: Examination of the finger position code in the finger image header and for both elements <code>bio:FingerCapture</code> (or if a slap acquisition on a multifinger scanner has been performed for the one <code>bio:FingerCapture</code> with the respective two <code>bio:FingerRecordReference</code>): Examination of the value in the attribute <code>fc</code> <i>Expected Result:</i> For both finger position codes in record there is a corresponding <code>fc</code> in <code>qa_finger</code>
3	<i>Description:</i> Check if the fingerprint positioned on the fingerprint sensor (as demanded in the IUT graphical user interface (GUI)) is logged in the correct value in <code>fc</code> within <code>bio:FingerAcquisition</code>. <i>Expected Result:</i> Value of <code>fc</code> is equal to the fingerprint positioned on the fingerprint sensor
4	<i>Description:</i> Examination of the XML Logging

Test Case ID: TC-LOG-FP-GID-001*Expected Result:*

- If the XML-element `FingerprintCaptureAllowed` is true the XML-element `FingerAcquisition` occurs at least once for applicants applying for:
 - Residence Permit
 - Passport
 - Identity Card
- The XML-element `FingerAcquisition` does not occur if the XML-element `FingerprintCaptureAllowed` is false.
- The XML-elements of `FingerAcquisition/Records/BinaryRecord` may be missing e.g. if fingers are physically impossible to acquire.
- The XML-element `FingerAcquisition/Records` occurs at maximum once in one of the `FingerAcquisition` XML-elements.
- The XML-element `FingerAcquisition/Records/XMLRecord` is not used.
- The XML-element `FingerAcquisition/Records/BinaryRecord/BinaryData/Value` occurs at least once and at maximum two times.
- The XML-element `FingerAcquisition/Records/BinaryRecord/@type` is `icao-cbeff-bit-bdb-19794-4`
- The XML-element `FingerAcquisition/Records/BinaryRecord/Data` occurs exactly once per record.

Table 2.178 Test Case ID: TC-LOG-FP-GID-001

2.11 Test Cases Coding

2.11.1 Test Cases FM COD-ALL-ARE

Test Case ID: TC-COD-ALL-ARE-001

<i>Scope</i>	Examine proper overall encoding of input data (considering biographic, biometric and transaction specific data) for identification purposes.
<i>Preconditions</i>	• The IUT is in operation, required modules are loaded. • The hardware used in the specific application scenario is connected • The multi fingerprint scanner used in the specific application scenario is connected • IUT is configured to perform the identification process including the manual acquisition of ten plain fingerprints using the IUT. • The IUT is able to return the results for every captured image via the test interface. • The following test resources are necessary: • One test subject is available to perform the acquisition with. • Provision of any other than biometric data (e.g. biographic, agency or process related) may be done in form of static dummy values by IUT, so that the process can be finalized and the resulting data is returned by IUT as GSAT3 XML.
<i>CTS Mode</i>	• interactive • no provision of pre-defined input data • HTTP method: GET • test case variants:

/TR03122/TC-COD-ALL-ARE-001/1

Table 2.180 TC-COD-ALL-ARE-001 Test Case Variants

Description	Step	Description / Expected Result
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Test Case ID: TC-COD-ALL-ARE-001

1

Description:

- Initiate test case by calling the IUT via the test interface.
- Perform the whole acquisition process using the IUT in the identification scenario with subsequent examination of returned result data

Expected Result:

- The IUT delivers a response that is conformant to the schema file are6v1.xsd.
- The response contains the correct number of elements based on the XPath expression `/are:are-log/bio:Records/bio:XMLRecord[@type='g-sat+xml']/tradd:XMLData/tradd:Value`. Exactly 1 element is expected in accordance to the acquisition process.
- The XML embedded in the element `"/are:are-log/bio:Records/bio:XMLRecord[@type='gsat+xml']/tradd:XMLData/tradd:Value"` of the response XML is GSAT3-xml.

Table 2.179 Test Case ID: TC-COD-ALL-ARE-001

2.11.2 Test Cases FM COD-ALL-BCL

Test Case ID: TC-COD-ALL-BCL-001

Scope

Examine requirements and interfaces for the overall coding of biometric (facial image and fingerprints) and biographic data used within the context of border control.

Preconditions

- The IUT is in operation, required modules are loaded.
- The multi fingerprint scanner used in the specific application scenario is connected
- The hardware used in the specific application scenario is connected
- The IUT implements an interface for conformance testing where biometric data can be acquired.
- The following test resources are necessary:
 - One test subject is available to perform the acquisition with.
 - Provision of any other than biometric data (e.g. biographic, agency or process related) may be done in form of static and fictitious dummy values by the IUT. The goal is to focus on the encoding of biometric data so that the process can be finalized and the resulting data is returned by IUT in TR-03121 XML format, i.e. a FaceAcquisition element is returned for facial data respectively a FingerAcquisition for fingerprint data.

CTS Mode

- interactive
- no provision of pre-defined input data
- HTTP method: GET
- test case variants:

/TR03122/TC-COD-ALL-BCL-001/1

Table 2.182 TC-COD-ALL-BCL-001 Test Case Variants

Description

Step Description / Expected Result

1

Description:

- Initiate test case by calling the IUT via the test interface.
- Perform the acquisition process of all desired biometric elements using the IUT.
- Examine the response returned by the IUT via the test interface.

Test Case ID: TC-COD-ALL-BCL-001*Expected Result:*

- The IUT delivers a response that is conformant to the schema file bcl6v1.xsd.
- The response contains the correct number of elements based on the XPath expression /bc1:bc1-log. Exactly 1 element is expected in accordance to the acquisition process.
- The data encoded in the log matches the data acquired in the acquisition process.
- Optional attributes and elements of the schema are considered as far as possible.

Table 2.181 Test Case ID: TC-COD-ALL-BCL-001

2.11.3 Test Cases FM COD-ALL-GID

Test Case ID: TC-COD-ALL-GID-001

Scope	Examine requirements and interfaces for the overall coding of biometric (facial image and fingerprints) and biographic data used within the context of German Identity Document.
Preconditions	• The IUT is in operation, required modules are loaded. • A fingerprint scanner and a facial camera are connected • The IUT implements an interface for conformance testing where biometric data can be acquired. • The following test resources are necessary: • A test person to perform the biometric acquisition with • Provision of any other than biometric data (e.g. biographic, agency or process related) may be done in form of static and fictitious dummy values by the IUT. The goal is to focus on the encoding of biometric data so that the process can be finalized and the resulting data is returned by IUT in TR-03121 XML format, i.e. a FaceAcquisition element is returned for facial data respectively a FingerAcquisition for fingerprint data.
CTS Mode	• interactive • no provision of pre-defined input data • HTTP method: GET • test case variants:

/TR03122/TC-COD-ALL-GID-001/1

Table 2.184 TC-COD-ALL-GID-001 Test Case Variants

<i>Description</i>	<table> <tr> <th>Step</th><th>Description / Expected Result</th></tr> <tr> <td>1</td><td> <i>Description:</i> • Initiate test case by calling the IUT via the test interface. • Perform the acquisition process of all desired biometric elements using the IUT with subsequent examination of returned result data </td></tr> </table>	Step	Description / Expected Result	1	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • Perform the acquisition process of all desired biometric elements using the IUT with subsequent examination of returned result data
Step	Description / Expected Result				
1	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • Perform the acquisition process of all desired biometric elements using the IUT with subsequent examination of returned result data				

Test Case ID: TC-COD-ALL-GID-001*Expected Result:*

- The IUT delivers a response that is conformant to the schema file gi-d6v1.xsd.
- The response contains the correct number of elements based on the XPath expression `//bio:Records/bio:BinaryRecord/tradd:BinaryData/tradd:Value`. Any number of elements (including none) is possible in accordance to the acquisition process.
- The data encoded in the log matches the data acquired in the acquisition process.

Table 2.183 Test Case ID: TC-COD-ALL-GID-001

2.11.4 Test Cases FM COD-ALL-IMA

Test Case ID: TC-COD-ALL-IMA-001

<i>Scope</i>	Examine requirements and interfaces for the overall coding of biometric (facial image and fingerprints) and biographic data used within the context of Immigration Authorities.
<i>Preconditions</i>	• The IUT is in operation, required modules are loaded. • All required acquisition devices, i.e. a fingerprint scanner respectively a digital camera are at hand and connected to the IUT. • The IUT implements an interface for conformance testing where biometric data can be acquired. • The following test resources are necessary: • One test subject is available to perform the acquisition with. • Provision of any other than biometric data (e.g. biographic, agency or process related) may be done in form of static and fictitious dummy values by the IUT. The goal is to focus on the encoding of biometric data so that the process can be finalized and the resulting data is returned by IUT in TR-03121 XML format, i.e. a FaceAcquisition element is returned for facial data respectively a FingerAcquisition for fingerprint data.
<i>CTS Mode</i>	• interactive • no provision of pre-defined input data • HTTP method: GET • test case variants:

/TR03122/TC-COD-ALL-IMA-001/1

Table 2.186 TC-COD-ALL-IMA-001 Test Case Variants

<i>Description</i>	Step	Description / Expected Result
	1	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • Perform the acquisition process of all desired biometric elements using the IUT with subsequent examination of returned result data • Examine the response returned by the IUT via the test interface.

Test Case ID: TC-COD-ALL-IMA-001

Expected Result:

- The IUT delivers a response that is conformant to the schema file ima6v1.xsd.
- The response contains the correct number of elements based on the XPath expression `/ima:ima-log//bio:FingerAcquisition`. Any number of elements (including none) is possible in accordance to the acquisition process.
- The response contains the correct number of elements based on the XPath expression `/ima:ima-log//bio:FaceAcquisition`. Any number of elements (including none) is possible in accordance to the acquisition process.
- The response contains the correct number of elements based on the XPath expression `/ima:ima-log/bio:Records/bio:XMLRecord[@type ne 'gsat' and @type ne 'gsat-xml']`. No elements are expected to appear in accordance to the acquisition process.
- The data encoded in the log matches the data acquired in the acquisition process.
- Optional attributes and elements of the schema are considered as far as possible (e.g. error codes only need to be logged, in case an error occurred; an acquisition element is only required, in case an acquisition process has at least been started).

Table 2.185 Test Case ID: TC-COD-ALL-IMA-001

2.11.5 Test Cases FM COD-FI-GENERIC

Test Case ID: TC-COD-FI-GENERIC-001

<i>Scope</i>	Examine requirements and interfaces for the overall coding used during the acquisition process of facial images.
<i>Preconditions</i>	• The IUT is in operation, required modules are loaded. • The integrated camera system used in the specific application scenario is connected. • The IUT implements an interface for conformance testing where a single image can be captured. • The IUT is able to return the results as TR-LOG for every captured image via the test interface. • The following test resources are necessary: • A test person to perform the biometric acquisition with. • Provision of any other than biometric data (e.g. biographic, agency or process related) may be done in form of static and fictitious dummy values by the IUT. The goal is to focus on the encoding of biometric data so that the process can be finalized and the resulting data is returned by IUT in TR-03121 XML format, i.e. a FaceAcquisition element is returned for facial data.
<i>CTS Mode</i>	• interactive • no provision of pre-defined input data • HTTP method: GET • test case variants:

/TR03122/TC-COD-FI-GENERIC-001/1

Table 2.188 TC-COD-FI-GENERIC-001 Test Case Variants

<i>Description</i>	Step	Description / Expected Result
	1	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • Perform a successful facial acquisition process. • Examine the response returned by the IUT via the test interface.

Test Case ID: TC-COD-FI-GENERIC-001*Expected Result:*

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response contains the correct number of elements based on the XPath expression //bio:FaceAcquisition/bio:Records/bio:BinaryRecord/tradd:BinaryData/tradd:Value. Exactly 1 element is expected in accordance to the acquisition process.
- The data encoded in the log matches the data acquired in the acquisition process.

Table 2.187 Test Case ID: TC-COD-FI-GENERIC-001

2.11.6 Test Case FM COD-FI-GID

Test Case ID: TC-COD-FI-GID-001

Scope This test case checks the requirements and interfaces for the coding of facial images used for application of GID

Preconditions • As necessary test resources: Facial images intended to be stored on a GID chip card.

CTS Mode Not supported

Description

Step	Description / Expected Result
1	<i>Description:</i> • Examine the resolution of the processed image. • Examine the aspect ratio of the processed image. • Examine the colour depth and colour space of the processed image. • Examine the encoded file format of the processed image. <i>Expected Result:</i> • The resolution of the facial images is 1244 x 1600 pixels for live capture- or delivered facial images respectively 622 x 800 pixels for scanned facial images. • The facial image was not upscaled and the aspect ratio of the image was kept. • The resulting facial image is a color image (24 bit sRGB) or a grey scale image (8 bit sRGB). • The facial images are encoded as JPEG.

Table 2.189 Test Case ID: TC-COD-FI-GID-001

2.11.7 Test Case FM COD-FI-PRD

Test Case ID: TC-COD-FI-PRD-001

Scope This test case checks the requirements and interfaces for the coding of facial images used for document production.

Preconditions As necessary test resources: Facial images intended to be used for document production.

CTS Mode File based

Description

Step	Description / Expected Result
1	<i>Description:</i> Examine the processed image, i.e.: • the resolution of the facial image, • the colour depth and colour space, • the encoded file format of the image.

Test Case ID: TC-COD-FI-PRD-001*Expected Result:*

The facial images meet the following criteria:

- The resolution of the facial images is 622 x 800 pixels without any up-scaling.
- The resulting facial image is a color image (24 bit sRGB) or a grey scale image (8 bit sRGB).
- The facial images are encoded as JPEG 2000.

Table 2.190 Test Case ID: TC-COD-FI-PRD-001**2.11.8 Test Case FM COD-FI-ROD****Test Case ID: TC-COD-FI-ROR-001**

Scope This test case checks the requirements and interfaces for the coding of facial images used for Register of Documents (Pass- und Personalausweisregister).

Preconditions As necessary test resources: Facial images intended to be stored Register of Documents.

CTS Mode File based

Description

Step	Description / Expected Result
1	<i>Description:</i> Examine the processed image, i.e.: • the resolution of the facial image, • the aspect ratio, • the colour depth and colour space, • the encoded file format of the image. <i>Expected Result:</i> The facial images meet the following criteria: • The resolution of the facial images is 1244 x 1600 pixels without any up-scaling. • The resulting facial image is a color image (24 bit sRGB) or a grey scale image (8 bit sRGB). • The facial images are encoded as JPEG.

Table 2.191 Test Case ID: TC-COD-FI-ROD-001**2.11.9 Test Cases FM COD-FI-VER****Test Case ID: TC-COD-FI-VER-001**

Scope Examine requirements for the coding used during the verification process of facial images.

Preconditions

- The IUT is in operation, required modules are loaded.
- The integrated camera system used in the specific application scenario is connected
- The IUT implements an interface for conformance testing where facial images can be acquired.
- The IUT is able to return the results as TR XML for every verification process of facial images.
- The following test resources are necessary:
 - One test subject is available to perform the acquisition with.

Test Case ID: TC-COD-FI-VER-001

- CTS Mode*
- interactive
 - no provision of pre-defined input data
 - HTTP method: GET
 - test case variants:

/TR03122/TC-COD-FI-VER-001/1

Table 2.193 TC-COD-FI-VER-001 Test Case Variants*Description*

Step	Description / Expected Result
1	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • Perform the facial image acquisition process using the IUT. • Initiate a face verification process. • Examine the response returned by the IUT via the test interface. <i>Expected Result:</i> • The IUT delivers a response that is conformant to the schema file bcl6v1.xsd. • The response contains the correct number of elements based on the XPath expression <code>/bcl:bcl-log/bio:Verification</code>. Exactly 1 element is expected in accordance to the acquisition process. • The verification process was conducted for the modality "face". • The response contains correctly encoded data resulting from the verification process.

Table 2.192 Test Case ID: TC-COD-FI-VER-001**2.11.10 Test Cases FM COD-FP-GID****Test Case ID: TC-COD-FP-GID-001**

Scope Correct coding of the CBEFF container

- Preconditions*
- IUT is running, required modules are loaded
 - A multi fingerprint scanner and facial camera is connected
 - As necessary test resources: a test person to perform the biometric acquisition with
 - IUT is configured to perform the identification process including following steps
 - Manual acquisition of two plain fingerprints using the IUT
 - Provision of any other than biometric data (e.g. biographic, agency or process related) may be done in form of static dummy values by IUT, so that the process can be finalized and the resulting data is returned by IUT as CBEFF container

- CTS Mode*
- interactive
 - no provision of pre-defined input data
 - HTTP method: GET
 - test case variants:

/TR03122/TC-COD-FP-GID-001/1

Table 2.195 TC-COD-FP-GID-001 Test Case Variants*Description*

Step	Description / Expected Result
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Test Case ID: TC-COD-FP-GID-001

1

Description:

Decoding of the CBEFF Header, Biometric Header Template (BHT) (XML element <data>) for each fingerprint

Expected Result:

Provision of the according biometric container including the fingerprint data:

- CBEFF Patron Header Version = '0101'
- BDB Biometric Type: value = '08' (Fingerprint)
- BDB Biometric Subtype: one of the following values:
 - 0000 0101 (thumb right)
 - 0000 0110 (thumb left)
 - 0000 1001 (index right)
 - 0000 1010 (index left)
 - 0000 1101 (middle finger right)
 - 0000 1110 (middle finger left)
 - 0001 0001 (ring finger right)
 - 0001 0010 (ring finger left)
- BDB Format Owner: value = '0x0101' (ISO/IEC JTC 1 SC 37-Biometrics)
- BDB Format Type: value = '0x0007' (ISO/IEC JTC 1 SC 37-Biometrics)

2

Description:

Decoding of the general record header

Expected Result:

- Format Identifier = 0x46495200 ("FIR" - Finger Image Record)
- Version Number = 0x30313000 ("010")
- Record Length = 32+ 1 * (14 bytes + Data length)
- Capture device ID = 2 bytes (Vendor specified)
- Image acquisition level = '0x001F' (Level 31) or '0x0029' (Level 41)
- Number of fingers = 1
- Scale units = 1 (pixel/inch)
- Scan resolution (horiz) = 01F4 (500ppi)
- Scan resolution (vert) = 01F4 (500ppi)
- Image resolution (horiz) <= Scan resolution (horiz)
- Image resolution (vert) <= Scan resolution (vert)
- Pixel depth = 08 (256 grae levels)
- Image compression Algorithm = 02 (WSQ)
- Reserved = 00 00

3

Description:

Decoding of the finger image header record

Test Case ID: TC-COD-FP-GID-001*Expected Result:*

- Length of finger data block (bytes) = includes header and the image data block
- Finger position = finger code {1,2,3,4,6,7,8,9}
- Count of views = '01'
- View number = '01'
- Finger image quality {0-100} or 255 in case of error
- Impression type = 0 (Live-scan plain)
- Horizontal line length >0
- Vertical line length >0
- Reserved = 00

4

Description:

Examination of the finger image data

Expected Result:

- Finger image data size < 18432 bytes
- Encoded as WSQ image
- sf value coded in the image header unequal 0

Table 2.194 Test Case ID: TC-COD-FP-GID-001**2.11.11 Test Cases FM COD-FP-VER****Test Case ID: TC-COD-FP-VER-001**

Scope Examine requirements for the coding used during the verification process of fingerprint images.

- Preconditions*
- The IUT is in operation, required modules are loaded.
 - The fingerprint scanner used in the specific application scenario is connected
 - The IUT implements an interface for conformance testing where fingerprint images can be captured.
 - The IUT is able to return the results as TR XML for every verification process of fingerprint images.
 - The following test resources are necessary:
 - One test subject is available to perform the acquisition with.

- CTS Mode*
- interactive
 - no provision of pre-defined input data
 - HTTP method: GET
 - test case variants:

/TR03122/TC-COD-FP-VER-001/1

Table 2.197 TC-COD-FP-VER-001 Test Case Variants*Description***Step Description / Expected Result**

- | Step | Description / Expected Result |
|------|---|
| 1 | <i>Description:</i> <ul style="list-style-type: none"> • Initiate test case by calling the IUT via the test interface. • Perform the fingerprint image acquisition process using the IUT. • Initiate a fingerprint verification process. • Examine the response returned by the IUT via the test interface. |

Test Case ID: TC-COD-FP-VER-001*Expected Result:*

- The IUT delivers a response that is conformant to the schema file bcl6v1.xsd.
- The response contains the correct number of elements based on the XPath expression `/bcl:bcl-log/bio:Verification`. At least 1 element is expected in accordance to the acquisition process.
- The verification process was conducted for the modality "fingerprint".
- The response contains correctly encoded data resulting from the verification process.

Table 2.196 Test Case ID: TC-COD-FP-VER-001

3 Test Cases Partial Application Processes

The following sections define test cases for Partial Application Processes.

3.1 Test Cases PAP ACQ-FI-SV-5: Supervised Facial Image Acquisition System

Test Case ID: TC-PAP-ACQ-FI-SV-5-001

<i>Scope</i>	Examination of the availability and the correctness of veto power of the operator in the automated and manual capturing process for facial image acquisition.
<i>Applicability</i>	Note that the TC is separated in different groups of steps: Steps 1-4 test the manual rejection succeeded by acceptance during automated facial image acquisition. Steps 5-11 test manual rejection succeeded by acceptance, including switching from manual to automated facial image acquisition. Note that the support of a manual service mode is optional within the application context German Identity Documents (GID). If the IUT does not supports a manual mode, the second group of steps are omitted.
<i>Preconditions</i>	• The IUT is in operation, required modules are loaded. • The integrated camera system used in the specific application scenario is connected • If HLBS is used by the system, the HLBS service definition for "Facial Image Acquisition" specified in TR-03121-2.2 is implemented. • The following test resources are necessary: • One test subject is available to perform the acquisition with. • Documentation of the configured timeout is at hand.
<i>CTS Mode</i>	• interactive • no provision of pre-defined input data • HTTP method: GET • test case variants:

/TR03122/TC-PAP-ACQ-FI-SV-5-001/1

Table 3.2 TC-PAP-ACQ-FI-SV-5-001 Test Case Variants

<i>Description</i>	Step	Description / Expected Result
	1	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • A single biometric subject is present in the IUT's capture area. • The operator starts an automated acquisition process of a facial image for enrolment or verification/identification. • The operator starts the capturing process of a facial image. <i>Expected Result:</i> • The IUT executes the process autonomously, i.e. the system handles the entire process execution, including configuration of the camera system to the height of the biometric subject. • An image is captured and shown to the Operator. Corresponding QA results are shown.
	2	<i>Description:</i> • The operator conducts a manual QA of the acquired facial image and reviews the software-based QA result. The operator rejects the acquired facial image.

Test Case ID: TC-PAP-ACQ-FI-SV-5-001

	<i>Expected Result:</i> • The IUT starts a new automated acquisition process of a facial image for enrolment or verification/identification.
3	<i>Description:</i> • The operator starts the capturing process of a facial image.
	<i>Expected Result:</i> • The IUT executes the process autonomously, i.e. the system handles the entire process execution, including configuration of the camera system to the height of the biometric subject. • An image is captured and shown to the Operator. Corresponding QA results are shown.
4	<i>Description:</i> • The operator conducts a manual QA of the acquired facial image and reviews the software-based QA result. The operator accepts the acquired facial image.
	<i>Expected Result:</i> • The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • The response contains the encoded facial image (bio:FaceAcquisition). • The image contained in the response matches the image that was displayed and accepted.
5	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • A single biometric subject is present in the IUT's capture area. • The operator starts a manual acquisition process of a facial image for enrolment or verification/identification.
	<i>Expected Result:</i> • The IUT executes the process where each sub-process requires manual handling of the operator. • The IUT's test interface offers the operator the option to manually adjust the camera system to the height of the biometric subject
6	<i>Description:</i> • The operator starts the automatic height configuration process of a facial image for enrolment or verification/identification.
	<i>Expected Result:</i> • The IUT adjusts the configuration of the camera system to the height of the biometric subject.
7	<i>Description:</i> • The operator starts the capturing process of a facial image.
	<i>Expected Result:</i> • The IUT executes the face image capturing process. An image is captured and shown to the Operator. Corresponding QA results are shown.
8	<i>Description:</i> • The operator conducts a manual QA of the acquired facial image and reviews the software-based QA result. The operator rejects the acquired facial image.

Test Case ID: TC-PAP-ACQ-FI-SV-5-001

	<i>Expected Result:</i>
	• The IUT starts a new manual acquisition process of a facial image for enrolment or verification/identification. • The IUT executes the process where each sub-process requires manual handling of the operator.
9	<i>Description:</i> • The operator starts the automatic height configuration process of a facial image for enrolment or verification/identification.
	<i>Expected Result:</i>
	• The IUT adjusts the configuration of the camera system to the height of the biometric subject.
10	<i>Description:</i> • The operator starts the capturing process of a facial image.
	<i>Expected Result:</i>
	• The IUT executes the face image capturing process. An image is captured and shown to the Operator. Corresponding QA results are shown.
11	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • The operator conducts a manual QA of the acquired facial image and reviews the software-based QA result. The operator accepts the acquired facial image.
	<i>Expected Result:</i>
	• The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • The response contains the encoded facial image (bio:FaceAcquisition). • The image contained in the response matches the image that was displayed and accepted.

Table 3.1 Test Case ID: TC-PAP-ACQ-FI-SV-5-001

Test Case ID: TC-PAP-ACQ-FI-SV-5-002

<i>Scope</i>	Examination of the availability and the correctness of the camera adjustment in the manual capturing process for facial image acquisition.
<i>Preconditions</i>	• The IUT is in operation, required modules are loaded. • The integrated camera system used in the specific application scenario is connected • If HLBS is used by the system, the HLBS service definition for "Facial Image Acquisition" specified in TR-03121-2.2 is implemented. • The following test resources are necessary: • One test subject is available to perform the acquisition with.
<i>CTS Mode</i>	• interactive • no provision of pre-defined input data • HTTP method: GET • test case variants:

/TR03122/TC-PAP-ACQ-FI-SV-5-002/1

Table 3.4 TC-PAP-ACQ-FI-SV-5-002 Test Case Variants

Description

Step	Description / Expected Result
------	-------------------------------

Test Case ID: TC-PAP-ACQ-FI-SV-5-002

- | | |
|---|--|
| 1 | <p><i>Description:</i></p> <ul style="list-style-type: none"> • Initiate test case by calling the IUT via the test interface. • A single biometric subject is present in the IUT's capture area. • The operator starts a manual acquisition process of a facial image for enrolment or verification/identification. <p><i>Expected Result:</i></p> <ul style="list-style-type: none"> • The IUT executes the process where each sub-process requires manual handling of the operator. • The IUT's test interface offers the operator the option to manually adjust the camera system to the height of the biometric subject. |
| 2 | <p><i>Description:</i></p> <ul style="list-style-type: none"> • The operator starts the automatic height configuration process of a facial image for enrolment or verification/identification. <p><i>Expected Result:</i></p> <ul style="list-style-type: none"> • The IUT adjusts the configuration of the camera system to the height of the biometric subject. |
| 3 | <p><i>Description:</i></p> <ul style="list-style-type: none"> • The operator starts the capturing process of a facial image. <p><i>Expected Result:</i></p> <ul style="list-style-type: none"> • The IUT executes the face image capturing process. An image is captured and shown to the Operator. Corresponding QA results are shown. |
| 4 | <p><i>Description:</i></p> <ul style="list-style-type: none"> • The operator conducts a manual QA of the acquired facial image and reviews the software-based QA result. The operator rejects the acquired facial image. <p><i>Expected Result:</i></p> <ul style="list-style-type: none"> • The IUT starts a new manual acquisition process of a facial image for enrolment or verification/identification. • The IUT executes the process where each sub-process requires manual handling of the operator. • The IUT's test interface offers the operator the option to manually adjust the camera system to the height of the biometric subject |
| 5 | <p><i>Description:</i></p> <ul style="list-style-type: none"> • The operator manually increases the height configuration of the camera system up to the maximum height configuration. <p><i>Expected Result:</i></p> <ul style="list-style-type: none"> • The IUT adjusts the configuration of the camera system to the height configured by the operator. |
| 6 | <p><i>Description:</i></p> <ul style="list-style-type: none"> • The operator manually decreases the height configuration of the camera system up to the minimum height configuration. <p><i>Expected Result:</i></p> <ul style="list-style-type: none"> • The IUT adjusts the configuration of the camera system to the height configured by the operator. |
| 7 | <p><i>Description:</i></p> <ul style="list-style-type: none"> • The operator starts the capturing process of a facial image. |

Test Case ID: TC-PAP-ACQ-FI-SV-5-002

8

Expected Result:

- The IUT executes the face image capturing process. An image is captured and shown to the Operator. Corresponding QA results are shown.

Description:

- The operator conducts a manual QA of the acquired facial image and re-views the software-based QA result. The operator accepts the acquired facial image.

Expected Result:

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response contains the encoded facial image (bio:FaceAcquisition).
- The image contained in the response matches the image that was displayed and accepted.

Table 3.3 Test Case ID: TC-PAP-ACQ-FI-SV-5-002

Test Case ID: TC-PAP-ACQ-FI-SV-5-003

Scope Examination of the availability and the correctness of the switching functionality between the automated and the manual capturing process for facial image acquisition.

- Preconditions*
- The IUT is in operation, required modules are loaded.
 - The integrated camera system used in the specific application scenario is connected
 - If HLBS is used by the system, the HLBS service definition for "Facial Image Acquisition" specified in TR-03121-2.2 is implemented.
 - The following test resources are necessary:
 - One test subject is available to perform the acquisition with.

- CTS Mode*
- interactive
 - no provision of pre-defined input data
 - HTTP method: GET
 - test case variants:

/TR03122/TC-PAP-ACQ-FI-SV-5-003/1

Table 3.6 TC-PAP-ACQ-FI-SV-5-003 Test Case Variants

Description

Step	Description / Expected Result
1	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • A single biometric subject is present in the IUT's capture area. • The operator starts a manual acquisition process of a facial image for enrolment or verification/identification. • The operator switches from the manual acquisition process to an automated acquisition process. • The operator starts the capturing process of a facial image. <i>Expected Result:</i> • The IUT executes the process autonomously, i.e. the system handles the entire process execution, including configuration of the camera system to the height of the biometric subject. • An image is captured and shown to the Operator. Corresponding QA results are shown.

Test Case ID: TC-PAP-ACQ-FI-SV-5-003

2	<i>Description:</i> - The operator conducts a manual QA of the acquired facial image and re-views the software-based QA result. The operator rejects the acquired facial image. <i>Expected Result:</i> - The IUT starts a new automated acquisition process of a facial image for enrolment or verification/identification.
3	<i>Description:</i> - The operator switches from the automated acquisition process to a manual acquisition process. <i>Expected Result:</i> - The IUT executes the process where each sub-process requires manual handling of the operator. - The IUT's test interface offers the operator the option to manually adjust the camera system to the height of the biometric subject
4	<i>Description:</i> - The operator starts the automatic height configuration process of a facial image for enrolment or verification/identification. <i>Expected Result:</i> - The IUT adjusts the configuration of the camera system to the height of the biometric subject.
5	<i>Description:</i> - The operator starts the facial image capturing process.. <i>Expected Result:</i> - The IUT executes the face image capturing process. An image is captured and shown to the Operator. Corresponding QA results are shown.
6	<i>Description:</i> - The operator conducts a manual QA of the acquired facial image and re-views the software-based QA result. The operator accepts the acquired facial image. <i>Expected Result:</i> - The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. - The response contains the encoded facial image (bio:FaceAcquisition). - The image contained in the response matches the image that was displayed and accepted.

Table 3.5 Test Case ID: TC-PAP-ACQ-FI-SV-5-003

3.2 Test Cases PAP ACQ-FI-AUTO-1: Automated Facial Image Acquisition

Test Case ID: TC-PAP-ACQ-FI-AUTO-1-001

Scope	Examination of the correctness and duration of the unsupervised automated face image acquisition processes for optimal face presentations (bona fide).
-------	--

Test Case ID: TC-PAP-ACQ-FI-AUTO-1-001*Preconditions*

- The IUT is in operation, required modules are loaded.
- The integrated camera system used in the specific application scenario is connected
- If HLBS is used by the system, the HLBS service definition for "Facial Image Acquisition System" specified in TR-03121-2.2 is implemented.
- Depending on the scenario the IUT is used in (supervised or non-supervised), PAD functionality is to be activated or disabled for performing the test case. The active setting has to be regarded when the results of this test case are evaluated.
- The following test resources are necessary:
 - One test subject is available to perform the acquisition with.

CTS Mode

- interactive
- no provision of pre-defined input data
- HTTP method: GET
- test case variants:

/TR03122/TC-PAP-ACQ-FI-AUTO-1-001/1

Table 3.8 TC-PAP-ACQ-FI-AUTO-1-001 Test Case Variants*Description*

Step	Description / Expected Result
1	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • A single biometric subject is present in the IUT's capture area. • The process for automated facial image acquisition is started. <i>Expected Result:</i> • The IUT guides the biometric subject to present a compliant face presentation according to FM AS-FI-ICS3. • The IUT adjusts the configuration of the camera system to the height of the biometric subject.
2	<i>Description:</i> • If applicable: The BSJ adjusts their distance to the IUT to perform a compliant capture w.r.t. to the requirements according to FM AS-FI-ICS3. <i>Expected Result:</i> • The IUT indicates, that all acceptance conditions (no multiple faces and optimal distance) are met according to FM AS-FI-ICS3.
3	<i>Description:</i> • The biometric subject performs an optimal (bona-fide) face presentation w.r.t. to all QA-components.

Test Case ID: TC-PAP-ACQ-FI-AUTO-1-001*Expected Result:*

- The IUT captures a face image.
- Depending on the PAD configuration, the IUT performs a PAD. In this case, all acceptance conditions are met according to FM PAD-FI-APP.
- The IUT assesses the quality of the captured face image: The acceptance conditions are not met according to the respective FM QA-FI-*.
- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response contains information regarding the process conduction as well as the encoded facial image (bio:FaceAcquisition).
- Within the application context Border Control (BCL), the time required for the overall facial image acquisition process did not exceed ten seconds if PAD is performed or seven seconds if no PAD is performed.
- Within the application context German Identity Documents (GID), the time required for the overall facial image acquisition process did not exceed thirty seconds (including PAD and, if applicable, excluding background replacement).

Table 3.7 Test Case ID: TC-PAP-ACQ-FI-AUTO-1-001**Test Case ID: TC-PAP-ACQ-FI-AUTO-1-002**

<i>Scope</i>	Examination of the correctness and availability of timeout of the unsupervised automated face image acquisition processes for non-compliant face presentations.
<i>Preconditions</i>	• The IUT is in operation, required modules are loaded. • The integrated camera system used in the specific application scenario is connected • If HLBS is used by the system, the HLBS service definition for "Facial Image Acquisition System" specified in TR-03121-2.2 is implemented. • Depending on the scenario the IUT is used in (supervised or non-supervised), PAD functionality is to be activated or disabled for performing the test case. The active setting has to be regarded when the results of this test case are evaluated. • The following test resources are necessary: • One test subject is available to perform the acquisition with.
<i>CTS Mode</i>	• interactive • no provision of pre-defined input data • HTTP method: GET • test case variants:

/TR03122/TC-PAP-ACQ-FI-AUTO-1-002/1

Table 3.10 TC-PAP-ACQ-FI-AUTO-1-002 Test Case Variants

<i>Description</i>	Step	Description / Expected Result
	1	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • A single biometric subject is present in the IUT's capture area. • The process for automated facial image acquisition is started. <i>Expected Result:</i> • The IUT guides the biometric subject to present a compliant face presentation according to FM AS-FI-ICS3. • The IUT adjusts the configuration of the camera system to the height of the biometric subject.

Test Case ID: TC-PAP-ACQ-FI-AUTO-1-002

2	<i>Description:</i> - If applicable: The BSJ adjusts their distance to the IUT to perform a compliant capture w.r.t. to the requirements according to FM AS-FI-ICS3. <i>Expected Result:</i> - The IUT indicates, that all acceptance conditions (no multiple faces and optimal distance) are met according to FM AS-FI-ICS3.
3	<i>Description:</i> - The biometric subject performs a non-conformant face presentation w.r.t. to supported QA-components, e.g. the biometric subject does not look in the camera. <i>Expected Result:</i> - The IUT captures a face image. - Depending on the PAD configuration, the IUT performs a PAD. In this case, all acceptance conditions are met according to FM PAD-FI-APP. Note: The result might be dependant from the specific BSJ behaviour. In case of doubt, the TC shall be repeated. - The IUT assesses the quality of the captured face image: The acceptance conditions are not met according to the respective FM QA-FI-*
4	<i>Description:</i> - The biometric subject repeats Step 3 for the duration of the configured timeout. <i>Expected Result:</i> - The IUT repeats the automated face image acquisition process until the timeout exceeds. - After the timeout is exceeded the IUT aborts the face image acquisition process. - The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. - The response contains information regarding the process conduction as well as the encoded facial image exhibiting the best quality of all captured images (bio:FaceAcquisition).

Table 3.9 Test Case ID: TC-PAP-ACQ-FI-AUTO-1-002

3.3 Test Cases PAP ACQ-FI-SV-4: Supervised Facial Image Acquisition

Test Case ID: TC-PAP-ACQ-FI-SV-4-001

<i>Scope</i>	Verify correct behaviour of facial image acquisition process using live enrolment equipment operated by an operator.
<i>Preconditions</i>	- The IUT is in operation, required modules are loaded. - The hardware used in the specific application scenario is connected - If HLBS is used by the system, the HLBS service definition for "Facial Image Acquisition" specified in TR-03121-2.2 is implemented. - The following test resources are necessary: - One test subject is available to perform the acquisition with. - Documentation of the configured timeout is at hand.

Test Case ID: TC-PAP-ACQ-FI-SV-4-001

CTS Mode

- interactive
- no provision of pre-defined input data
- HTTP method: GET
- test case variants:

/TR03122/TC-PAP-ACQ-FI-SV-4-001/1

Table 3.12 TC-PAP-ACQ-FI-SV-4-001 Test Case Variants

Description

Step	Description / Expected Result
1	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • Place the biometric test subject properly in front of the digital camera. • Follow the instructions of the IUT for the biometric test subject regarding automatic facial acquisition. • Wait for an automatic captured image. • Verify the displayed image and accept it. • Examine the response returned by the IUT via the test interface. <i>Expected Result:</i> • The IUT captured a facial image automatically. • The image is displayed by the IUT. • The displayed images are cropped and derotated. • A quality assessment of the captured image has been performed. The IUT shows QA-results to the operator. • The image is released to the requesting application after being accepted. • The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • The response contains an image. • The image contained in the response matches the image that was displayed.
2	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • Place the biometric test subject improperly in front of the digital camera, i.e. such that the quality assessment of the captured images is negative for every capture. At least two images SHALL be captured. • Perform insufficient captures until the timeout is exceeded. • Accept the displayed image. • Examine the response returned by the IUT via the test interface.

Test Case ID: TC-PAP-ACQ-FI-SV-4-001

Expected Result:

- The IUT captured facial images automatically.
- The images are displayed by the IUT.
- A quality assessment of the captured image has been performed. The IUT shows negative QA-results to the operator.
- The IUT gives the option to crop and derotate the displayed images.
- The image is released to the requesting application after being accepted.
- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response contains an image.
- The image contained in the response matches the image that was displayed and accepted.

3

Description:

- Initiate test case by calling the IUT via the test interface.
- Verify the option to capture an image manually: Trigger the image capture manually. Note, that the quality of the captured image is not relevant for this test.
- Verify the option to crop and rotate the captured image manually: Crop and rotate the captured image manually.
- Accept the cropped and derotated image.
- Examine the response returned by the IUT via the test interface.

Expected Result:

- The IUT displays an option to take an image manually.
- Upon manual capture: An image was captured and displayed by the IUT.
- The IUT displays an option to crop and derotate an image manually.
- The IUT displays the cropped and derotated image.
- The image is released to the requesting application after being accepted.
- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response contains an image.
- The image contained in the response matches the image that was displayed, cropped, derotated and accepted.

4

Description:

- Initiate test case by calling the IUT via the test interface.
- Do not place a biometric test subject in front of the camera.
- Wait for the timeout to trigger
- Accept the displayed image.
- Examine the response returned by the IUT via the test interface.

Expected Result:

- No image was captured
- The acquisition process ended.
- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response does not contain an image but information about the timeout (attribute "FaceAcquisition/@result").

Test Case ID: TC-PAP-ACQ-FI-SV-4-001

5

Description:

- Initiate test case by calling the IUT via the test interface.
- No biometric test subject is placed in front of the camera.
- Wait roughly 70 % of the timeout time.
- Trigger the image capture manually.
- Reject the captured image.
- Perform a recapture without restarting the capturing process manually. Wait roughly 70 % of the timeout time.
- Trigger the image capture manually.
- Accept the captured image.
- Examine the response returned by the IUT via the test interface.

Expected Result:

- The IUT displays the manually captured image.
- The IUT gives the option to manually reject the image.
- Upon rejection, the image capture process was restarted automatically.
- Upon rejection, the timeout was reset. Waiting 70% of the timeout again did not end the capturing process prematurely.
- A new image could be captured manually.
- The newly captured image could be accepted.
- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response contains an image.
- The image contained in the response matches the image that was displayed and accepted.

Table 3.11 Test Case ID: TC-PAP-ACQ-FI-SV-4-001

3.4 Test Cases PAP DEL-FI-SV-1: Supervised Facial Image Digital Delivery

Test Case ID: TC-PAP-DEL-FI-SV-1-001

<i>Scope</i>	Verify correct behaviour of the image delivery process when an admissible image is available from Central Identity Register (CIR)
<i>Preconditions</i>	• The IUT is in operation, required modules are loaded. • The IUT implements the test interface for conformance testing where a single image can be provided. • The following conditions apply regarding the database: • The image data that is sent to the IUT is of the following biometric type: FACE. • The groundtruth database contains compliant images w.r.t. the required quality values.

Test Case ID: TC-PAP-DEL-FI-SV-1-001

- CTS Mode*
- interactive
 - The image data that is sent to the IUT may have the following formats:
 - JPEG
 - BMP
 - HTTP method: POST
 - test case variants:

/TR03122/TC-PAP-DEL-FI-SV-1-001/1

Table 3.14 TC-PAP-DEL-FI-SV-1-001 Test Case Variants*Description*

Step	Description / Expected Result
1	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • A single compliant facial image is loaded and sent to the IUT. A timestamp (not older than 6 months) may be provided via EXIF-tag "DateTimeOriginal". • Check display of loaded images in IUT. <i>Expected Result:</i> • The resource image used as input is displayed within the GUI of the IUT, if applicable cropped and aligned.
2	<i>Description:</i> • Check quality indication of loaded image. <i>Expected Result:</i> • The quality assessment result for the displayed image is indicated and a hint regarding re-usability of the image is given by the IUT.

Table 3.13 Test Case ID: TC-PAP-DEL-FI-SV-1-001

3.5 Test Cases PAP ACQ-FPS-SV-1: Supervised Acquisition Single Slap

Test Case ID: TC-PAP-ACQ-FPS-SV-1-001

- Scope* Verification of the overall supervised acquisition process of a single slap.
- Preconditions*
- The IUT is in operation, required modules are loaded.
 - The multi fingerprint scanner used in the specific application scenario is connected
 - If HLBS is used by the system, the HLBS service definition for "Fingerprint Acquisition" specified in TR-03121-2.2 is implemented.
 - The following test resources are necessary:
 - One test subject is available to perform the acquisition with.
 - Product documentation of IUT is at hand (e.g. data sheet, manual).
- CTS Mode*
- interactive
 - no provision of pre-defined input data
 - HTTP method: GET
 - test case variants:

/TR03122/TC-PAP-ACQ-FPS-SV-1-001/1

Table 3.16 TC-PAP-ACQ-FPS-SV-1-001 Test Case Variants*Description*

Step	Description / Expected Result
------	-------------------------------

Test Case ID: TC-PAP-ACQ-FPS-SV-1-001

1

Description:

- Initiate test case by calling the IUT via the test interface.
- Select one finger as missing.
- Follow the instructions of the IUT for the slap placement on the scanner for the biometric test subject and acquire a slap (bona-fide).
- Choose to use the acquired slap
- Choose not to capture the second slap.
- Examine the response returned by the IUT via the test interface.

Expected Result:

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The IUT displays acquired FP images, an indication of the achieved quality values and results of PAD.
- The response contains information regarding the performed slap capture
- The response contains information regarding the missing finger.
- PAD has been performed for each single fingerprint image.
- The PAD system didn't report an error.
- The PAD system did report no warning of an attack.
- No Uniqueness check error was reported.
- QA values have been calculated for each single fingerprint image.
- The response contains the encoded segmented fingerprint images.

2

Description:

- Initiate test case by calling the IUT via the test interface.
- Follow the instructions of the IUT for the slap placement on the scanner for the biometric test subject and acquire a slap (bona-fide).
- Choose to use the acquired slap
- Choose to capture the second slap.
- Examine the response returned by the IUT via the test interface.

Expected Result:

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The IUT displays acquired FP images, an indication of the achieved quality values and results of PAD, uniquenesscheck and slap classification.
- The response contains information regarding the performed slap capture
- The response contains no information regarding missing fingers.
- PAD has been performed for each single fingerprint image.
- The PAD system didn't report an error.
- The PAD system did report no warning of an attack.
- No Uniqueness check error was reported.
- The slap classification did not report any deviations.
- QA values have been calculated for each single fingerprint image.
- The response contains the encoded segmented fingerprint images.

Table 3.15 Test Case ID: TC-PAP-ACQ-FPS-SV-1-001

Test Case ID: TC-PAP-ACQ-FPS-SV-1-002

<i>Scope</i>	Verification of the correct acquisition process for a single slap.
<i>Applicability</i>	Only for Volume ARE: Test step 3 is skipped if the recommended FM AS-FP-SLP is not implemented.
<i>Preconditions</i>	• The IUT is in operation, required modules are loaded. • The multi fingerprint scanner used in the specific application scenario is connected • If HLBS is used by the system, the HLBS service definition for "Fingerprint Acquisition" specified in TR-03121-2.2 is implemented. • For step 3: The IUT offers the option to toggle the slap classifier between off and evaluation mode. • The following test resources are necessary: • One test subject is available to perform the acquisition with. • Product documentation of IUT is at hand (e.g. data sheet, manual).
<i>CTS Mode</i>	• interactive • no provision of pre-defined input data • HTTP method: GET • test case variants:

/TR03122/TC-PAP-ACQ-FPS-SV-1-002/1

Table 3.18 TC-PAP-ACQ-FPS-SV-1-002 Test Case Variants

<i>Description</i>	Step	Description / Expected Result
	1	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • Follow the instructions of the IUT for the slap placement on the scanner for the biometric test subject and acquire a slap (bona-fide). • Choose to use the acquired slap. • Choose not to capture the second slap. • Examine the response returned by the IUT via the test interface. <i>Expected Result:</i> • The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • The IUT displays acquired FP images, an indication of the achieved quality values and results of PAD. • The IUT does not display error messages or warnings (e.g. of duplicates). • The response contains information regarding the performed slap capture • The response contains no information regarding a missing finger. • PAD has been performed for each single fingerprint image. • The PAD system didn't report an error. • The PAD system did report no warning of an attack. • No Uniqueness check error and no duplicate fingerprints were reported. • QA values have been calculated for each single fingerprint image. • The response contains the encoded segmented fingerprint images.

Test Case ID: TC-PAP-ACQ-FPS-SV-1-002

2

Description:

- Initiate test case by calling the IUT via the test interface.
- Follow the instructions of the IUT for the slap placement on the scanner for the biometric test subject and acquire a slap (bona-fide).
- Choose to use the acquired slap.
- Choose to capture the second slap.
- Place the same hand again on the finger scanner area.
- Follow the instructions of the IUT for the slap placement on the scanner for the biometric test subject and acquire the slap.
- Choose to use the acquired slap.
- Examine the response returned by the IUT via the test interface.

Expected Result:

- The IUT displays acquired FP images, an indication of the achieved quality values and results of PAD.
- The IUT displays a warning or error for duplicated fingerprint images.
- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response contains information regarding the performed slap capture
- The response contains no information regarding missing fingers.
- PAD has been performed for each single fingerprint image.
- The PAD system didn't report an error.
- The PAD system did report no warning of an attack.
- A uniqueness check warning regarding the repeated capture was reported.
- QA values have been calculated for each single fingerprint image.
- The response contains the encoded segmented fingerprint images.

3

Description:

- Set the slap classifier to evaluation mode.
- Initiate test case by calling the IUT via the test interface.
- Follow the instructions of the IUT for the slap placement on the scanner for the biometric test subject and acquire a slap (bona-fide).
- Choose to use the acquired slap.
- Choose to capture the second slap.
- Place the same hand again on the finger scanner area.
- Choose to use the acquired slap.
- Examine the response returned by the IUT via the test interface.

Test Case ID: TC-PAP-ACQ-FPS-SV-1-002*Expected Result:*

- The IUT displays acquired FP images, an indication of the achieved quality values and results of PAD.
- The IUT does not display a warning or error for duplicated fingerprint images.
- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response contains information regarding the performed slap capture
- The response contains no information regarding missing fingers.
- PAD has been performed for each single fingerprint image.
- The PAD system didn't report an error.
- The PAD system did report no warning of an attack.
- A uniqueness check warning regarding the repeated capture was reported.
- QA values have been calculated for each single fingerprint image.
- The response contains the encoded segmented fingerprint images.

Table 3.17 Test Case ID: TC-PAP-ACQ-FPS-SV-1-002**Test Case ID: TC-PAP-ACQ-FPS-SV-1-003**

<i>Scope</i>	Verification of the correct acquisition process for a single finger. This test case covers PAP Task ACQ-FPP-SV-2.
<i>Preconditions</i>	• The IUT is in operation, required modules are loaded. • The multi fingerprint scanner used in the specific application scenario is connected • If HLBS is used by the system, the HLBS service definition for "Fingerprint Acquisition" specified in TR-03121-2.2 is implemented. • The following test resources are necessary: • One test subject is available to perform the acquisition with. • Product documentation of IUT is at hand (e.g. data sheet, manual).
<i>CTS Mode</i>	• interactive • no provision of pre-defined input data • HTTP method: GET • test case variants:

/TR03122/TC-PAP-ACQ-FPS-SV-1-003/1

Table 3.20 TC-PAP-ACQ-FPS-SV-1-003 Test Case Variants

<i>Description</i>	Step	Description / Expected Result
	1	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • Start the acquisition process and switch to the single finger acquisition. • Follow the instructions of the IUT for the single finger placement on the scanner for the biometric test subject and successively acquire four fingers in single finger mode (bona-fide). • Choose not to capture the second slap. • Examine the response returned by the IUT via the test interface.

Test Case ID: TC-PAP-ACQ-FPS-SV-1-003*Expected Result:*

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The IUT displays acquired FP images, an indication of the achieved quality values and results of PAD.
- The IUT does not display error messages or warnings (e.g. of duplicates).
- The response contains information regarding the performed capture.
- The response contains no information regarding a missing finger.
- PAD has been performed for each single fingerprint image.
- The PAD system didn't report an error.
- The PAD system did report no warning of an attack.
- No Uniqueness check error and no duplicate fingerprints were reported.
- QA values have been calculated for each single fingerprint image.
- The response contains the encoded fingerprint images.

2

Description:

- Initiate test case by calling the IUT via the test interface.
- Select one finger as missing.
- Start the acquisition process and switch to the single finger acquisition.
- Follow the instructions of the IUT for the single finger placement on the scanner for the biometric test subject and successively capture three fingers in single finger mode. Present the same finger repeatedly to introduce a uniqueness check error.
- Choose not to capture the second slap.
- Examine the response returned by the IUT via the test interface.

Expected Result:

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The IUT displays acquired FP images, an indication of the achieved quality values and results of PAD.
- The IUT displays error messages or warnings regarding the result of the uniquenesscheck identifying potential duplicates.
- The response contains information regarding the performed capture.
- The response contains information regarding a missing finger.
- PAD has been performed for each single fingerprint image.
- The PAD system didn't report an error.
- The PAD system did report no warning of an attack.
- A Uniqueness check error and duplicate fingerprints were reported.
- QA values have been calculated for each single fingerprint image.
- The response contains the encoded fingerprint images.

Table 3.19 Test Case ID: TC-PAP-ACQ-FPS-SV-1-003**Test Case ID: TC-PAP-ACQ-FPS-SV-1-004**

Scope

Verification of the correct acquisition process for a single slap where the capture is forced manually.

Test Case ID: TC-PAP-ACQ-FPS-SV-1-004

- Preconditions*
- The IUT is in operation, required modules are loaded.
 - The multi fingerprint scanner used in the specific application scenario is connected
 - If HLBS is used by the system, the HLBS service definition for "Fingerprint Acquisition" specified in TR-03121-2.2 is implemented.
 - For step 3: The IUT offers the option to toggle the slap classifier between off and evaluation mode.
 - The following test resources are necessary:
 - One test subject is available to perform the acquisition with.
 - Product documentation of IUT is at hand (e.g. data sheet, manual).
- CTS Mode*
- interactive
 - no provision of pre-defined input data
 - HTTP method: GET
 - test case variants:

/TR03122/TC-PAP-ACQ-FPS-SV-1-004/1

Table 3.22 TC-PAP-ACQ-FPS-SV-1-004 Test Case Variants*Description*

Step	Description / Expected Result
1	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • Follow the instructions of the IUT for the slap placement on the scanner for the biometric test subject. • As the operator: Acquire a slap (bona-fide) by forcing the capture manually. • Choose to use the acquired slap. • Choose not to capture the second slap. • Examine the response returned by the IUT via the test interface. <i>Expected Result:</i> • The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • The instructions were displayed and correct for the captured fingers. • The process ended without capturing the second slap. • The IUT displays acquired FP images, an indication of the achieved quality values and results of PAD. • In case the quality of the captured slap was too bad and therefore a segmentation was not possible, an error message is shown. In this case a TR-03121 XML consistent with the bad QA is required. This might affect some of the results expected otherwise (see below). • The response contains information regarding the performed slap capture • The response contains no information regarding a missing finger. • PAD has been performed for each single fingerprint image. • The PAD system didn't report an error. • The PAD system did report no warning of an attack. • No Uniqueness check error and no duplicate fingerprints were reported. • QA values have been calculated for each single fingerprint image. • The response contains the encoded segmented fingerprint images.

Table 3.21 Test Case ID: TC-PAP-ACQ-FPS-SV-1-004

Test Case ID: TC-PAP-ACQ-FPS-SV-1-005

<i>Scope</i>	Verification of the correct acquisition process for a single finger where the capture is forced manually. This test case covers PAP Task ACQ-FPP-SV-2.
<i>Preconditions</i>	• The IUT is in operation, required modules are loaded. • The multi fingerprint scanner used in the specific application scenario is connected • If HLBS is used by the system, the HLBS service definition for "Fingerprint Acquisition" specified in TR-03121-2.2 is implemented. • The following test resources are necessary: • One test subject is available to perform the acquisition with. • Product documentation of IUT is at hand (e.g. data sheet, manual).
<i>CTS Mode</i>	• interactive • no provision of pre-defined input data • HTTP method: GET • test case variants:

/TR03122/TC-PAP-ACQ-FPS-SV-1-005/1

Table 3.24 TC-PAP-ACQ-FPS-SV-1-005 Test Case Variants

<i>Description</i>	Step	Description / Expected Result
	1	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • Start the acquisition process and switch to the single finger acquisition. • Follow the instructions of the IUT for the single finger placement on the scanner for the biometric test subject, but place the fingers in such a manner (with bad quality) that the pre-qualification is insufficient. • As the operator: successively acquire four fingers in single finger mode by forcing the capture manually. • Choose not to capture the second slap. • Examine the response returned by the IUT via the test interface. <i>Expected Result:</i> • The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • The IUT displays acquired FP images, an indication of the achieved quality values and results of PAD. • The IUT does not display error messages or warnings (e.g. of duplicates). • The response contains information regarding the performed capture. • The response contains no information regarding a missing finger. • PAD has been performed for each single fingerprint image. • The PAD system didn't report an error. • The PAD system did report no warning of an attack. • No Uniqueness check error and no duplicate fingerprints were reported. • QA values have been calculated for each single fingerprint image. • The QA-results indicate insufficient quality. • The response contains the encoded fingerprint images.

Table 3.23 Test Case ID: TC-PAP-ACQ-FPS-SV-1-005

3.6 Test Cases PAP ACQ-FP442-SV-1: Supervised Acquisition 4-4-2 for Enrolment

Test Case ID: TC-PAP-ACQ-FP442-SV-1-001

Scope Examination of the correct process for the supervised acquisition of slaps in the order of 4-4-2. Note: PAP ACQ-FP442-SV-1 uses PAP Task ACQ-FPS-SV-1 and PAP Task ACQ-FPP-SV-2 which have been defined as part of PAP ACQ-FPS-SV-1. Therefore, this TC directly calls TC-PAP-ACQ-FPS-SV-1-* in step 2.

Preconditions

- The IUT is in operation, required modules are loaded.
- The multi fingerprint scanner used in the specific application scenario is connected
- If HLBS is used by the system, the HLBS service definition for "Fingerprint Acquisition" specified in TR-03121-2.2 is implemented.
- The following test resources are necessary:
 - One test subject is available to perform the acquisition with.
 - Product documentation of IUT is at hand (e.g. data sheet, manual).

CTS Mode

- interactive
- no provision of pre-defined input data
- HTTP method: GET
- test case variants:

/TR03122/TC-PAP-ACQ-FP442-SV-1-001/1

Table 3.26 TC-PAP-ACQ-FP442-SV-1-001 Test Case Variants

Description

Step	Description / Expected Result
1	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • Follow the instructions of the IUT for the biometric test subject regarding slap placement on the scanner. • Present slap consecutively enforcing sufficient quality and compliant behaviour. • Check the correct execution of the acquisition process with slaps of sufficient quality. • Examine the response returned by the IUT via the test interface. <i>Expected Result:</i> • The IUT instructs the user to acquire the slaps in the following order: right slap, left slap, thumb slap. • The IUT displays an option to mark missing fingers prior to capturing the respective slap. • The IUT finishes without any error message. • The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • The response contains an element //bio:FingerAcquisition and relevant sub-elements containing the information corresponding to the performed acquisition process. • The response contains elements //bio:Records/bio:BinaryRecord/tradd:BinaryData/tradd:Value containing base64 encoded fingerprint images of all ten fingers at least. • The images contained in the response correspond to the fingers, that were captured.

Test Case ID: TC-PAP-ACQ-FP442-SV-1-001

2

Description:

- The PAP ACQ-FP442-SV-1 uses PAP Task ACQ-FPS-SV-1 and PAP Task ACQ-FPP-SV-2 which have been defined as part of PAP ACQ-FPS-SV-1. Therefore, Test Cases TC-PAP-ACQ-FPS-SV-1-*: Supervised Acquisition Single Slap have to be tested separately for conformance.

Expected Result:

- Test Cases TC-PAP-ACQ-FPS-SV-1-* for PAP ACQ-FPS-SV-1: Supervised Acquisition Single Slap are passed.

Table 3.25 Test Case ID: TC-PAP-ACQ-FP442-SV-1-001

3.7 Test Cases PAP ACQ-FP4141-SV-1: Supervised Acquisition 4-1-4-1 for Enrolment

Test Case ID: TC-PAP-ACQ-FP4141-SV-1-001

<i>Scope</i>	Examination of the correct process for the supervised slap acquisition of ten fingers in the order of 4-1-4-1. Note: PAP ACQ-FP4141-SV-1 uses PAP Task ACQ-FPS-SV-1 and PAP Task ACQ-FPP-SV-2 which have been defined as part of PAP ACQ-FPS-SV-1. Therefore, this TC directly calls TC-PAP-ACQ-FPS-SV-1-* in step 2.
<i>Preconditions</i>	• The IUT is in operation, required modules are loaded. • The multi fingerprint scanner used in the specific application scenario is connected • If HLBS is used by the system, the HLBS service definition for "Fingerprint Acquisition" specified in TR-03121-2.2 is implemented. • The following test resources are necessary: • One test subject is available to perform the acquisition with. • Product documentation of IUT is at hand (e.g. data sheet, manual).
<i>CTS Mode</i>	• interactive • no provision of pre-defined input data • HTTP method: GET • test case variants:

/TR03122/TC-PAP-ACQ-FP4141-SV-1-001/1

Table 3.28 TC-PAP-ACQ-FP4141-SV-1-001 Test Case Variants

<i>Description</i>	<table> <tr> <th>Step</th><th>Description / Expected Result</th></tr> <tr> <td>1</td><td> <i>Description:</i> • Initiate test case by calling the IUT via the test interface. • Follow the instructions of the IUT for the biometric test subject regarding slap placement on the scanner. • Present slap consecutively enforcing sufficient quality and compliant behaviour. • Check the correct execution of the acquisition process with slaps of sufficient quality. • Examine the response returned by the IUT via the test interface. </td></tr> </table>	Step	Description / Expected Result	1	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • Follow the instructions of the IUT for the biometric test subject regarding slap placement on the scanner. • Present slap consecutively enforcing sufficient quality and compliant behaviour. • Check the correct execution of the acquisition process with slaps of sufficient quality. • Examine the response returned by the IUT via the test interface.
Step	Description / Expected Result				
1	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • Follow the instructions of the IUT for the biometric test subject regarding slap placement on the scanner. • Present slap consecutively enforcing sufficient quality and compliant behaviour. • Check the correct execution of the acquisition process with slaps of sufficient quality. • Examine the response returned by the IUT via the test interface.				

Test Case ID: TC-PAP-ACQ-FP4141-SV-1-001*Expected Result:*

- The IUT instructs the user to acquire the fingers in the following order: right slap, right thumb, left slap, left thumb.
- The IUT displays an option to mark missing fingers prior to capturing the respective finger or slap.
- The IUT finishes without any error message.
- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response contains an element `//bio:FingerAcquisition` and relevant sub-elements containing the information corresponding to the performed acquisition process.
- The response contains elements `//bio:Records/bio:BinaryRecord/tradd:BinaryData/tradd:Value` containing base64 encoded fingerprint images of all ten fingers at least.
- The images contained in the response correspond to the fingers, that were captured.

2

Description:

- The PAP ACQ-FP4141-SV-1 uses PAP Task ACQ-FPS-SV-1 and PAP Task ACQ-FPP-SV-2 which have been defined as part of PAP ACQ-FPS-SV-1. Therefore, Test Cases TC-PAP-ACQ-FPS-SV-1-*: Supervised Acquisition Single Slap have to be tested separately for conformance.

Expected Result:

- Test Cases TC-PAP-ACQ-FPS-SV-1-* for PAP ACQ-FPS-SV-1: Supervised Acquisition Single Slap are passed.

Table 3.27 Test Case ID: TC-PAP-ACQ-FP4141-SV-1-001

3.8 Test Cases PAP ACQ-FP2P-SV-1: Supervised Acquisition of Two Plain Fingers on Multi-Finger Hardware for Enrolment

Test Case ID: TC-PAP-ACQ-FP2P-SV-1-001

<i>Scope</i>	Examination of the correct process for the acquisition of up to two fingerprints using the two finger capture mode with a multi fingerprint scanner
<i>Preconditions</i>	• The IUT is in operation, required modules are loaded. • The multi fingerprint scanner used in the specific application scenario is connected • If HLBS is used by the system, the HLBS service definition for "Fingerprint Acquisition" specified in TR-03121-2.2 is implemented. • The following test resources are necessary: • One test subject is available to perform the acquisition with. • Product documentation of IUT is at hand (e.g. data sheet, manual).
<i>CTS Mode</i>	• interactive • no provision of pre-defined input data • HTTP method: GET • test case variants:

/TR03122/TC-PAP-ACQ-FP2P-SV-1-001/1

Table 3.30 TC-PAP-ACQ-FP2P-SV-1-001 Test Case Variants*Description*

Step	Description / Expected Result
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Test Case ID: TC-PAP-ACQ-FP2P-SV-1-001

1

Description:

- Initiate test case by calling the IUT via the test interface.
- Present two index fingers simultaneously enforcing sufficient quality and compliant behaviour.
- Follow the instructions of the IUT for the biometric test subject regarding finger placement on the scanner.
- Check the correct execution of the acquisition process with fingerprints of sufficient quality.
- Examine the response returned by the IUT via the test interface.

Expected Result:

- The acquisition is triggered by placing the finger(s) on the acquisition hardware in the two finger mode (right index and left index).
- The IUT finishes without any error message.
- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response contains an element //bio:FingerAcquisition and relevant sub-elements containing the information corresponding to the performed acquisition process.
- PAD has been performed for both fingerprint images.
- The PAD system didn't report an error.
- The PAD system did report no warning of an attack.
- No Uniqueness check error or warning was reported.
- QA values have been calculated for both fingerprint images individually.
- The response contains elements //bio:Records/bio:BinaryRecord/tradd:BinaryData/tradd:Value containing base64 encoded fingerprint images of the right and left index fingers.
- The images contained in the response correspond to the fingers, that were captured.

2

Description:

- Initiate test case by calling the IUT via the test interface.
- Mark the right index finger as handicapped.
- Present right thumb and left index finger subsequently enforcing sufficient quality.
- Follow the instructions of the IUT for the biometric test subject regarding finger placement on the scanner.
- Check the correct execution of the acquisition process with fingerprints of sufficient quality
- Examine the response returned by the IUT via the test interface.

Test Case ID: TC-PAP-ACQ-FP2P-SV-1-001

Expected Result:

- The IUT allows marking handicapped fingers before acquisition.
- The acquisition is triggered by placing the finger(s) on the acquisition hardware (right thumb and left index).
- The marked finger is excluded from the acquisition process.
- The IUT switches to single finger mode.
- The IUT finishes without any error message.
- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response contains an element `//bio:FingerAcquisition` and relevant sub-elements containing the information corresponding to the performed acquisition process.
- PAD has been performed for both fingerprint images.
- The PAD system didn't report an error.
- The PAD system did report no warning of an attack.
- No Uniqueness check error or warning was reported.
- QA values have been calculated for both fingerprint images individually.
- The response contains elements `//bio:Records/bio:BinaryRecord/tradd:BinaryData/tradd:Value` containing base64 encoded fingerprint images of the right thumb and left index finger.
- The images contained in the response correspond to the fingers, that were captured.

3

Description:

- Initiate test case by calling the IUT via the test interface.
- Mark the left index finger as handicapped.
- Present right index finger and left thumb subsequently enforcing sufficient quality.
- Follow the instructions of the IUT for the biometric test subject regarding finger placement on the scanner.
- Check the correct execution of the acquisition process with fingerprints of sufficient quality
- Examine the response returned by the IUT via the test interface.

Test Case ID: TC-PAP-ACQ-FP2P-SV-1-001

Expected Result:

- The IUT allows marking handicapped fingers before acquisition.
- The acquisition is triggered by placing the finger(s) on the acquisition hardware (right index and left thumb).
- The marked finger is excluded from the acquisition process.
- The IUT switches to single finger mode.
- The IUT finishes without any error message.
- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response contains an element `//bio:FingerAcquisition` and relevant sub-elements containing the information corresponding to the performed acquisition process.
- PAD has been performed for both fingerprint images.
- The PAD system didn't report an error.
- The PAD system did report no warning of an attack.
- No Uniqueness check error or warning was reported.
- QA values have been calculated for both fingerprint images individually.
- The response contains elements `//bio:Records/bio:BinaryRecord/tradd:BinaryData/tradd:Value` containing base64 encoded fingerprint images of the right index finger and left thumb.
- The images contained in the response correspond to the fingers, that were captured.

4

Description:

- Initiate test case by calling the IUT via the test interface.
- Mark a whole hand (right or left) as not available/handicapped
- Present index finger and thumb of either the right or left hand subsequently enforcing sufficient quality.
- Follow the instructions of the IUT for the biometric test subject regarding finger placement on the scanner.
- Check the correct execution of the acquisition process with fingerprints with a missing hand
- Examine the response returned by the IUT via the test interface.

Test Case ID: TC-PAP-ACQ-FP2P-SV-1-001

Expected Result:

- The IUT allows marking a hand (or all fingers of one hand individually) as not available before acquisition
- The acquisition is triggered by placing the finger(s) on the acquisition hardware (index finger and thumb of either left or right hand).
- The marked hand is excluded from the acquisition process.
- The IUT switches to single finger mode for the existing hand.
- The IUT finishes without any error message.
- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- If the left hand was marked as handicapped: The response contains an element `//bio:FingerAcquisition` and relevant sub-elements containing the information corresponding to the performed acquisition process of the right index finger and thumb.
- If the right hand was marked as handicapped: The response contains an element `//bio:FingerAcquisition` and relevant sub-elements containing the information corresponding to the performed acquisition process of the left index finger and thumb.
- If the left hand was marked as handicapped: PAD has been performed for fingerprint images of right index and thumb.
- If the right hand was marked as handicapped: PAD has been performed for fingerprint images of right index and thumb.
- The PAD system didn't report an error.
- The PAD system did report no warning of an attack.
- No Uniqueness check error or warning was reported.
- QA values have been calculated for both fingerprint images individually.
- The response contains elements `//bio:Records/bio:BinaryRecord/tradd:BinaryData/tradd:Value` containing base64 encoded fingerprint images of the right and left index fingers.
- The images contained in the response correspond to the fingers, that were captured.

5

Description:

- Initiate test case by calling the IUT via the test interface.
- Present two index fingers simultaneously enforcing insufficient quality for one selected finger.
- Follow the instructions of the IUT for the biometric test subject regarding finger placement on the scanner.
- Check the correct execution of the acquisition process with fingerprints of insufficient quality.
- Examine the response returned by the IUT via the test interface.

Test Case ID: TC-PAP-ACQ-FP2P-SV-1-001

Expected Result:

- The acquisition is triggered by placing the finger(s) on the acquisition hardware in the two finger mode (right index and left index).
- The IUT detects the insufficient quality and restarts the acquisition of the respective finger.
- The acquisition of the slap of insufficient fingerprint quality is repeated twice (i.e. total of three performed attempts).
- After the third attempt the IUT switches to the acquisition of the thumb of the respective hand.
- The IUT finishes without any error message (apart from the bad quality).
- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response contains an element `//bio:FingerAcquisition` and relevant sub-elements containing the information corresponding to the performed acquisition process.
- If insufficient quality was enforced for the left index finger: PAD has been performed for the fingerprint images of right and left index fingers and left thumb.
- If insufficient quality was enforced for the right index finger: PAD has been performed for the fingerprint images of right index finger and thumb and left index finger.
- The PAD system didn't report an error.
- The PAD system did report no warning of an attack.
- No Uniqueness check error or warning was reported.
- QA values have been calculated for all fingerprint images individually.
- If insufficient quality was enforced for the left index finger: The response contains elements `//bio:Records/bio:BinaryRecord/tradd:BinaryData/tradd:Value` containing base64 encoded fingerprint images of the right index and left thumb.
- If insufficient quality was enforced for the right index finger: The response contains elements `//bio:Records/bio:BinaryRecord/tradd:BinaryData/tradd:Value` containing base64 encoded fingerprint images of the right thumb and left index.
- The images contained in the response correspond to the fingers, that were captured.

6

Description:

- Initiate test case by calling the IUT via the test interface.
- Check whether the IUT supports the execution of the mandatory fallback single-finger acquisition process for cases where a slap capture is not practical.
- Select the option to capture single fingers subsequently.
- Present two index fingers subsequently enforcing sufficient quality and compliant behaviour.
- Follow the instructions of the IUT for the biometric test subject regarding finger placement on the scanner.
- Check the correct execution of the acquisition process with fingerprints of sufficient quality.
- Examine the response returned by the IUT via the test interface.

Test Case ID: TC-PAP-ACQ-FP2P-SV-1-001

Expected Result:

- The acquisition is triggered by placing the finger(s) on the acquisition hardware in the two finger mode (right index and left index).
- Upon switching to the mandatory fallback single-finger acquisition process, the IUT allows to capture the Fingers subsequently.
- The IUT finishes without any error message.
- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response contains an element `//bio:FingerAcquisition` and relevant sub-elements containing the information corresponding to the performed acquisition process.
- PAD has been performed for both fingerprint images.
- The PAD system didn't report an error.
- The PAD system did report no warning of an attack.
- No Uniqueness check error or warning was reported.
- QA values have been calculated for both fingerprint images individually.
- The response contains elements `//bio:Records/bio:BinaryRecord/tradd:BinaryData/tradd:Value` containing base64 encoded fingerprint images of the right and left index fingers.
- The images contained in the response correspond to the fingers, that were captured.

7

Description:

- Initiate test case by calling the IUT via the test interface.
- Present two index fingers simultaneously enforcing insufficient quality.
- Check whether the IUT supports the execution of the mandatory fallback single-finger acquisition process for fingers of insufficient quality.
- Select the option to capture single fingers subsequently.
- Present two index fingers subsequently enforcing sufficient quality and compliant behaviour.
- Follow the instructions of the IUT for the biometric test subject regarding finger placement on the scanner.
- Check the correct execution of the acquisition process with fingerprints of sufficient quality.
- Examine the response returned by the IUT via the test interface.

Test Case ID: TC-PAP-ACQ-FP2P-SV-1-001*Expected Result:*

- The acquisition is triggered by placing the finger(s) on the acquisition hardware in the two finger mode (right index and left index).
- Upon switching to the mandatory fallback single-finger acquisition process, the IUT allows to capture the Fingers subsequently.
- The IUT finishes without any error message.
- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response contains an element `//bio:FingerAcquisition` and relevant sub-elements containing the information corresponding to the performed acquisition process.
- PAD has been performed for all fingerprint images.
- The PAD system didn't report an error.
- The PAD system did report no warning of an attack.
- No Uniqueness check error or warning was reported.
- QA values have been calculated for both fingerprint images individually.
- The response contains elements `//bio:Records/bio:BinaryRecord/tradd:BinaryData/tradd:Value` containing base64 encoded fingerprint images of the right and left index fingers.
- The images contained in the response correspond to the fingers, that were captured.

Table 3.29 Test Case ID: TC-PAP-ACQ-FP2P-SV-1-001**Test Case ID: TC-PAP-ACQ-FP2P-SV-1-002**

<i>Scope</i>	Examination of the correct process for the acquisition of up to two fingerprints using the two finger capture mode with a multi fingerprint scanner where the capture is forced manually.
<i>Preconditions</i>	• The IUT is in operation, required modules are loaded. • The multi fingerprint scanner used in the specific application scenario is connected • If HLBS is used by the system, the HLBS service definition for "Fingerprint Acquisition" specified in TR-03121-2.2 is implemented. • The following test resources are necessary: • One test subject is available to perform the acquisition with. • Product documentation of IUT is at hand (e.g. data sheet, manual).
<i>CTS Mode</i>	• interactive • no provision of pre-defined input data • HTTP method: GET • test case variants:

/TR03122/TC-PAP-ACQ-FP2P-SV-1-002/1

Table 3.32 TC-PAP-ACQ-FP2P-SV-1-002 Test Case Variants*Description***Step** **Description / Expected Result**

Test Case ID: TC-PAP-ACQ-FP2P-SV-1-002

1

Description:

- Initiate test case by calling the IUT via the test interface.
- As the biometric subject: Present two index fingers simultaneously, but place the fingers in such a manner (with bad quality) that the pre-qualification is insufficient.
- As the operator: Force the capture of the two fingerprints manually.
- Repeat the insufficient presentation and the forced capture twice (i.e. a total of three times).
- As the operator: Accept the capture that is identified as the best by the IUT.
- Examine the response returned by the IUT via the test interface.

Expected Result:

- The acquisition starts by placing the finger(s) on the acquisition hardware in the two finger mode (right index and left index).
- The acquisition is triggered not until the operator forced the capture manually.
- The capture is repeated twice before the IUT gives the option to accept one of the slaps.
- The IUT finishes without any error message unrelated to the insufficient QA.
- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response contains an element `//bio:FingerAcquisition` and relevant sub-elements containing the information corresponding to the performed acquisition process of the index finger slap.
- PAD has been performed for both fingerprint images.
- No Uniqueness check error or warning was reported.
- For each capture attempt: QA values have been calculated for both fingerprint images individually.
- For each capture attempt: QA values are below the defined threshold.
- The IUT selected the fingerprint capture attempt with the best QA-score (via `@selected`, confirmed by the operator `@veto`).
- The response contains elements `//bio:Records/bio:BinaryRecord/tradd:BinaryData/tradd:Value` containing base64 encoded fingerprint images of the right and left index fingers.
- The images contained in the response correspond to the fingers, that were captured and selected.

2

Description:

- Initiate test case by calling the IUT via the test interface.
- As the biometric subject: Present two index fingers simultaneously, but place the fingers in such a manner (with bad quality) that the pre-qualification is insufficient.
- As the operator: Force the capture of the two fingerprints manually.
- Repeat the insufficient presentation and the forced capture twice (i.e. a total of three times).
- As the operator: Accept the capture that is identified as the best by the IUT.
- Examine the response returned by the IUT via the test interface.

Test Case ID: TC-PAP-ACQ-FP2P-SV-1-002*Expected Result:*

- The acquisition starts by placing the finger(s) on the acquisition hardware in the two finger mode (right index and left index).
- The acquisition is triggered not until the operator forced the capture manually.
- The capture is repeated twice before the IUT gives the option to accept one of the slaps.
- The IUT offers an option to start acquisition in single finger mode after each capture with insufficient QA.
- The IUT finishes without any error message unrelated to the insufficient QA.
- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response contains an element `//bio:FingerAcquisition` and relevant sub-elements containing the information corresponding to the performed acquisition process of the index finger slap.
- PAD has been performed for both fingerprint images.
- No Uniqueness check error or warning was reported.
- For each capture attempt: QA values have been calculated for both fingerprint images individually.
- For each capture attempt: QA values are below the defined threshold.
- The IUT selected the fingerprint capture attempt with the best QA-score (via @selected).
- The response contains elements `//bio:Records/bio:BinaryRecord/tradd:BinaryData/tradd:Value` containing base64 encoded fingerprint images of the right and left index fingers.
- The images contained in the response correspond to the fingers, that were captured and selected.

Table 3.31 Test Case ID: TC-PAP-ACQ-FP2P-SV-1-002

3.9 Test Cases PAP ACQ-FP2P-SV-2: Supervised Acquisition of Two Plain Fingers on Single-Finger Hardware for Enrolment

Test Case ID: TC-PAP-ACQ-FP2P-SV-2-001

<i>Scope</i>	Examination of the correct process for the acquisition of up to two fingerprints using the two finger capture mode with a single fingerprint scanner
<i>Preconditions</i>	• The IUT is in operation, required modules are loaded. • The single fingerprint scanner used in the specific application scenario is connected • If HLBS is used by the system, the HLBS service definition for "Fingerprint Acquisition" specified in TR-03121-2.2 is implemented. • The following test resources are necessary: • One test subject is available to perform the acquisition with. • Product documentation of IUT is at hand (e.g. data sheet, manual).

Test Case ID: TC-PAP-ACQ-FP2P-SV-2-001*CTS Mode*

- interactive
- no provision of pre-defined input data
- HTTP method: GET
- test case variants:

/TR03122/TC-PAP-ACQ-FP2P-SV-2-001/1

Table 3.34 TC-PAP-ACQ-FP2P-SV-2-001 Test Case Variants*Description*

Step	Description / Expected Result
1	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • Present two index fingers subsequently enforcing sufficient quality. • Follow the instructions of the IUT for the biometric test subject regarding finger placement on the scanner. • Check the correct execution of the acquisition process with fingerprints of sufficient quality • Examine the response returned by the IUT via the test interface. <i>Expected Result:</i> • The acquisition is triggered by placing the finger(s) on the acquisition hardware in the two finger mode (right index and left index). • The IUT finishes without any error message. • The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • The response contains an element //bio:FingerAcquisition and relevant sub-elements containing the information corresponding to the performed acquisition process. • PAD has been performed for both fingerprint images. • The PAD system didn't report an error. • The PAD system did report no warning of an attack. • No Uniqueness check error or warning was reported. • QA values have been calculated for both fingerprint images individually. • The response contains elements //bio:Records/bio:BinaryRecord/tradd:BinaryData/tradd:Value containing base64 encoded fingerprint images of the right and left index fingers. • The images contained in the response correspond to the fingers, that were captured.
2	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • Mark the right index finger as handicapped. • Present right thumb and left index finger subsequently enforcing sufficient quality. • Follow the instructions of the IUT for the biometric test subject regarding finger placement on the scanner. • Check the correct execution of the acquisition process with fingerprints of sufficient quality • Examine the response returned by the IUT via the test interface.

Test Case ID: TC-PAP-ACQ-FP2P-SV-2-001

Expected Result:

- The acquisition is triggered by placing the finger(s) on the acquisition hardware (right thumb and left index).
- The IUT finishes without any error message.
- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response contains an element //bio:FingerAcquisition and relevant sub-elements containing the information corresponding to the performed acquisition process.
- PAD has been performed for both fingerprint images.
- The PAD system didn't report an error.
- The PAD system did report no warning of an attack.
- No Uniqueness check error or warning was reported.
- QA values have been calculated for both fingerprint images individually.
- The response contains elements //bio:Records/bio:BinaryRecord/tradd:BinaryData/tradd:Value containing base64 encoded fingerprint images of the right thumb and left index finger.
- The images contained in the response correspond to the fingers, that were captured.

3

Description:

- Initiate test case by calling the IUT via the test interface.
- Mark the left index finger as handicapped.
- Present right index finger and left thumb subsequently enforcing sufficient quality.
- Follow the instructions of the IUT for the biometric test subject regarding finger placement on the scanner.
- Check the correct execution of the acquisition process with fingerprints of sufficient quality
- Examine the response returned by the IUT via the test interface.

Expected Result:

- The acquisition is triggered by placing the finger(s) on the acquisition hardware (right index and left thumb).
- The IUT finishes without any error message.
- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response contains an element //bio:FingerAcquisition and relevant sub-elements containing the information corresponding to the performed acquisition process.
- PAD has been performed for both fingerprint images.
- The PAD system didn't report an error.
- The PAD system did report no warning of an attack.
- No Uniqueness check error or warning was reported.
- QA values have been calculated for both fingerprint images individually.
- The response contains elements //bio:Records/bio:BinaryRecord/tradd:BinaryData/tradd:Value containing base64 encoded fingerprint images of the right index finger and left thumb.
- The images contained in the response correspond to the fingers, that were captured.

Test Case ID: TC-PAP-ACQ-FP2P-SV-2-001

4

Description:

- Initiate test case by calling the IUT via the test interface.
- Mark a whole hand (right or left) as not available/handicapped
- Present index finger and thumb of either the right or left hand subsequently enforcing sufficient quality.
- Follow the instructions of the IUT for the biometric test subject regarding finger placement on the scanner.
- Check the correct execution of the acquisition process with fingerprints with a missing hand
- Examine the response returned by the IUT via the test interface.

Expected Result:

- The IUT allows marking a hand (or all fingers of one hand individually) as not available before acquisition
- The acquisition is triggered by placing the finger(s) on the acquisition hardware (index finger and thumb of either left or right hand).
- The marked hand is excluded from the acquisition process.
- The IUT finishes without any error message.
- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- If the left hand was marked as handicapped: The response contains an element `//bio:FingerAcquisition` and relevant sub-elements containing the information corresponding to the performed acquisition process of the right index finger and thumb.
- If the right hand was marked as handicapped: The response contains an element `//bio:FingerAcquisition` and relevant sub-elements containing the information corresponding to the performed acquisition process of the left index finger and thumb.
- If the left hand was marked as handicapped: PAD has been performed for fingerprint images of right index and thumb.
- If the right hand was marked as handicapped: PAD has been performed for fingerprint images of right index and thumb.
- The PAD system didn't report an error.
- The PAD system did report no warning of an attack.
- No Uniqueness check error or warning was reported.
- QA values have been calculated for both fingerprint images individually.
- The response contains elements `//bio:Records/bio:BinaryRecord/tradd:BinaryData/tradd:Value` containing base64 encoded fingerprint images of the right and left index fingers.
- The images contained in the response correspond to the fingers, that were captured.

5

Description:

- Initiate test case by calling the IUT via the test interface.
- Present two index fingers subsequently enforcing insufficient quality for one selected finger.
- Follow the instructions of the IUT for the biometric test subject regarding finger placement on the scanner.
- Check the correct execution of the acquisition process with fingerprints of insufficient quality.
- Examine the response returned by the IUT via the test interface.

Test Case ID: TC-PAP-ACQ-FP2P-SV-2-001*Expected Result:*

- The acquisition is triggered by placing the finger(s) on the acquisition hardware in the two finger mode (right index and left index).
- The IUT detects the insufficient quality and restarts the acquisition of the respective finger.
- The acquisition of the slap of insufficient fingerprint quality is repeated twice (i.e. total of three performed attempts).
- After the third attempt the IUT switches to the acquisition of the thumb of the respective hand.
- The IUT finishes without any error message (apart from the bad quality).
- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response contains an element `//bio:FingerAcquisition` and relevant sub-elements containing the information corresponding to the performed acquisition process.
- If insufficient quality was enforced for the left index finger: PAD has been performed for the fingerprint images of right and left index fingers and left thumb.
- If insufficient quality was enforced for the right index finger: PAD has been performed for the fingerprint images of right index finger and thumb and left index finger.
- The PAD system didn't report an error.
- The PAD system did report no warning of an attack.
- No Uniqueness check error or warning was reported.
- QA values have been calculated for all fingerprint images individually.
- If insufficient quality was enforced for the left index finger: The response contains elements `//bio:Records/bio:BinaryRecord/tradd:BinaryData/tradd:Value` containing base64 encoded fingerprint images of the right index and left thumb.
- If insufficient quality was enforced for the right index finger: The response contains elements `//bio:Records/bio:BinaryRecord/tradd:BinaryData/tradd:Value` containing base64 encoded fingerprint images of the right thumb and left index.
- The images contained in the response correspond to the fingers, that were captured.

Table 3.33 Test Case ID: TC-PAP-ACQ-FP2P-SV-2-001**Test Case ID: TC-PAP-ACQ-FP2P-SV-2-002**

<i>Scope</i>	Examination of the correct process for the acquisition of up to two fingerprints using the two finger capture mode with a single fingerprint scanner where the capture is forced manually.
<i>Preconditions</i>	• The IUT is in operation, required modules are loaded. • The single fingerprint scanner used in the specific application scenario is connected • If HLBS is used by the system, the HLBS service definition for "Fingerprint Acquisition" specified in TR-03121-2.2 is implemented. • The following test resources are necessary: • One test subject is available to perform the acquisition with. • Product documentation of IUT is at hand (e.g. data sheet, manual).

Test Case ID: TC-PAP-ACQ-FP2P-SV-2-002*CTS Mode*

- interactive
- no provision of pre-defined input data
- HTTP method: GET
- test case variants:

/TR03122/TC-PAP-ACQ-FP2P-SV-2-002/1

Table 3.36 TC-PAP-ACQ-FP2P-SV-2-002 Test Case Variants*Description*

Step	Description / Expected Result
1	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • Present two index fingers subsequently placing the fingers in such a manner (with bad quality) that the pre-qualification is insufficient. • As the operator: force the capture of the two fingerprints manually. • Repeat the previous steps such that the acquisition is repeated a total of three times of each index finger. • If the IUT switches to the acquisition of a thumb: Perform a compliant acquisition of the thumb of the respective hand enforcing good quality. • Check the correct execution of the acquisition process with forced fingerprint acquisition. • Examine the response returned by the IUT via the test interface. <i>Expected Result:</i> • The acquisition is triggered by placing the finger(s) on the acquisition hardware in the two finger mode (right index and left index). • The acquisition is triggered not until the operator forced the capture manually. • The acquisition of the slap of insufficient fingerprint quality is repeated twice (i.e. total of three performed attempts) for each index finger • After the third attempt the IUT switches to the acquisition of the thumb of the respective hand. • The IUT finishes without any error message (apart from the bad quality). • The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • The response contains an element //bio:FingerAcquisition and relevant sub-elements containing the information corresponding to the performed acquisition process. • PAD has been performed for the fingerprint images of right and left index fingers and left thumb. • The PAD system didn't report an error. • The PAD system did report no warning of an attack. • No Uniqueness check error or warning was reported. • QA values have been calculated for all fingerprint images individually. The results match the performed process • The response contains elements //bio:Records/bio:BinaryRecord/tradd:BinaryData/tradd:Value containing base64 encoded fingerprint images of right and left thumbs. • The images contained in the response correspond to the fingers that were captured.

Table 3.35 Test Case ID: TC-PAP-ACQ-FP2P-SV-2-002

3.10 Test Cases PAP ACQ-FP10R-SV-1: Ten Finger Rolled Supervised Acquisition for Enrolment

Test Case ID: TC-PAP-ACQ-FP10R-SV-1-001

Scope Examination of the correct process for the acquisition of up to ten rolled fingerprints.

Preconditions

- The IUT is in operation, required modules are loaded.
- The fingerprint scanner providing rolled fingerprints used in the specific application scenario is connected
- If HLBS is used by the system, the HLBS service definition for "Rolled Fingerprint Acquisition" specified in TR-03121-2.2 is implemented.
- The following test resources are necessary:
 - One test subject is available to perform the acquisition with.
 - Product documentation of IUT is at hand (e.g. data sheet, manual).

CTS Mode

- interactive
- no provision of pre-defined input data
- HTTP method: GET
- test case variants:

/TR03122/TC-PAP-ACQ-FP10R-SV-1-001/1

Table 3.38 TC-PAP-ACQ-FP10R-SV-1-001 Test Case Variants

Description

Step	Description / Expected Result
1	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • Perform the flat acquisition process of all fingers in order to obtain the necessary reference data for the control verification. Ensure sufficient quality and correct execution throughout the process. • Perform the rolled acquisition process of all fingers, beginning with the thumb of the right hand and ending with the little finger of the left hand. Ensure sufficient quality and correct execution throughout the process. • Examine the response returned by the IUT via the test interface. <i>Expected Result:</i> • The acquisition of the reference slaps is performed by the IUT. • The IUT successively performs the acquisition of the rolled fingerprints. • The IUT finishes without any error message. • The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • The response includes the information corresponding to the performed acquisition process (bio:FingerAcquisition and relevant sub-elements). The raw fingerprint data may be stored either as sub-element of this element or as descendant of the sibling element bio:Records • The images contained in the response correspond to the fingers, that were captured.

Test Case ID: TC-PAP-ACQ-FP10R-SV-1-001

2

Description:

- Initiate test case by calling the IUT via the test interface.
- Mark one finger as handicapped in the IUT.
- Perform the flat acquisition process of the fingers, that are not marked as handicapped, in order to obtain the necessary reference data for the control verification. Ensure sufficient quality and correct execution throughout the process.
- Perform the rolled acquisition process of the fingers, that are not marked as handicapped, beginning with the thumb of the right hand (if it is available) and ending with the little finger of the left hand (if it is available). Ensure sufficient quality and correct execution throughout the process.
- Examine the response returned by the IUT via the test interface.

Expected Result:

- The IUT allows marking handicapped fingers before acquisition.
- The acquisition of the reference slaps is performed by the IUT. The marked finger is excluded.
- The IUT successively performs the acquisition of the rolled fingerprints. The marked finger is excluded.
- The IUT finishes without any error message.
- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response includes the information corresponding to the performed acquisition process (bio:FingerAcquisition and relevant sub-elements). The raw fingerprint data may be stored either as sub-element of this element or as descendant of the sibling element bio:Records
- The images contained in the response correspond to the fingers, that were captured.

3

Description:

- Initiate test case by calling the IUT via the test interface.
- Mark multiple fingers as handicapped in the IUT.
- Perform the flat acquisition process of the fingers, that are not marked as handicapped, in order to obtain the necessary reference data for the control verification. Ensure sufficient quality and correct execution throughout the process.
- Perform the rolled acquisition process of the fingers, that are not marked as handicapped, beginning with the thumb of the right hand (if it is available) and ending with the little finger of the left hand (if it is available). Ensure sufficient quality and correct execution throughout the process.
- Examine the response returned by the IUT via the test interface.

Test Case ID: TC-PAP-ACQ-FP10R-SV-1-001*Expected Result:*

- The IUT allows marking multiple handicapped fingers before acquisition.
- The acquisition of the reference slaps is performed by the IUT. The marked fingers are excluded.
- The IUT successively performs the acquisition of the rolled fingerprints. The marked fingers are excluded.
- The IUT finishes without any error message.
- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response includes the information corresponding to the performed acquisition process (bio:FingerAcquisition and relevant sub-elements). The raw fingerprint data may be stored either as sub-element of this element or as descendant of the sibling element bio:Records
- The response contains correct information about the fingers that were marked as handicapped (elements bio:FingerMissing).
- The images contained in the response correspond to the fingers, that were captured.

4

Description:

- Initiate test case by calling the IUT via the test interface.
- Perform the flat acquisition process of all fingers in order to obtain the necessary reference data for the control verification. Ensure sufficient quality and correct execution throughout the process.
- Perform the rolled acquisition process of all fingers, beginning with the thumb of the right hand and ending with the little finger of the left hand. Enforce insufficient quality for at least one finger. Repeat producing bad quality fingerprints for that finger. Apart from that, ensure correct execution throughout the process.
- Examine the response returned by the IUT via the test interface.

Expected Result:

- The acquisition of the reference slaps is performed by the IUT.
- The IUT successively performs the acquisition of the rolled fingerprints.
- The IUT detects the insufficient quality and restarts the acquisition of the respective finger.
- The rolled acquisition of the finger with insufficient fingerprint quality is repeated twice (i.e. total of three performed attempts).
- The IUT finishes without any error messages.
- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response includes the information corresponding to the performed acquisition process (bio:FingerAcquisition and relevant sub-elements). The raw fingerprint data may be stored either as sub-element of this element or as descendant of the sibling element bio:Records
- The images contained in the response correspond to the fingers, that were captured.

Table 3.37 Test Case ID: TC-PAP-ACQ-FP10R-SV-1-001**Test Case ID: TC-PAP-ACQ-FP10R-SV-1-002***Scope*

Examination of the correct process for the acquisition of rolled fingerprints in case the IUTs hardware reports issues.

Test Case ID: TC-PAP-ACQ-FP10R-SV-1-002

- Preconditions*
- The IUT is in operation, required modules are loaded.
 - The fingerprint scanner providing rolled fingerprints used in the specific application scenario is connected
 - If HLBS is used by the system, the HLBS service definition for "Rolled Fingerprint Acquisition" specified in TR-03121-2.2 is implemented.
 - The following test resources are necessary:
 - One test subject is available to perform the acquisition with.
 - Product documentation of IUT is at hand (e.g. data sheet, manual).
- CTS Mode*
- interactive
 - no provision of pre-defined input data
 - HTTP method: GET
 - test case variants:

/TR03122/TC-PAP-ACQ-FP10R-SV-1-002/1

Table 3.40 TC-PAP-ACQ-FP10R-SV-1-002 Test Case Variants*Description*

Step	Description / Expected Result
1	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • Perform the flat/control acquisition process of all fingers in order to obtain the necessary reference data for the control verification. Ensure sufficient quality and correct execution throughout the process. Do not induce any errors. • Perform the rolled acquisition process of all fingers, beginning with the thumb of the right hand and ending with the little finger of the left hand. Enforce a hardware reported issue during the rolled capture process for at least one finger. Consult the documentation of the hardware in regards of supported issues (e.g. "finger shifted") • If a hardware reported issue was induced for a finger: Repeat the capture of that finger without enforcing any further errors and finalize the acquisition process. • Examine the response returned by the IUT via the test interface. <i>Expected Result:</i> • The acquisition of the reference slaps is performed by the IUT. • The IUT successively performs the acquisition of the rolled fingerprints. • The IUT detects the hardware issue and restarts the acquisition of the current finger. • The IUT finishes without any error message that is disrupting the flow of the PAP or that is unrelated to the induced hardware reported issue. • The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • The response includes the information corresponding to the performed acquisition process (elements <code>bio:FingerAcquisition</code> and relevant sub-elements). The raw fingerprint data may be stored either as sub-element of this element or as descendant of the sibling element <code>bio:Records</code>. • The <code>rejectionReason</code> element matches the reason chosen by the operator • The images contained in the response correspond to the fingers, that were captured.

Test Case ID: TC-PAP-ACQ-FP10R-SV-1-002

2

Description:

- Initiate test case by calling the IUT via the test interface.
- Perform the flat/control acquisition process of all fingers in order to obtain the necessary reference data for the control verification. Ensure sufficient quality and correct execution throughout the process. Do not induce any errors.
- Perform the rolled acquisition process of all fingers, beginning with the thumb of the right hand and ending with the little finger of the left hand. Enforce a hardware reported issue during the rolled capture process for at least one finger. Consult the documentation of the hardware in regards of supported issues (e.g. "finger shifted").
- If a hardware reported issue was induced for a capture: Repeat the capture of that finger enforcing bad quality.
- For each finger, where no hardware-reported issue was induced: Capture this finger enforcing good quality.
- If bad quality was enforced during a capture: Repeat the capture of that finger enforcing good quality.
- Examine the response returned by the IUT via the test interface.

Expected Result:

- The acquisition of the reference slaps is performed by the IUT.
- The IUT successively performs the acquisition of the rolled fingerprints.
- If a hardware issue was induced: The IUT detects the hardware issue and restarts the acquisition of the current finger.
- If insufficient quality was enforced: The IUT detects insufficient quality and restarts the acquisition of the current finger.
- If a hardware issue was induced: The acquisition of that finger is repeated twice in total (one time due to the hardware issue, one time due to the quality).
- The IUT finishes without any error message that is disrupting the flow of the PAP or that is unrelated to the induced hardware issue or insufficient quality.
- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response includes the information corresponding to the performed acquisition process (elements `bio:FingerAcquisition` and relevant sub-elements). The raw fingerprint data may be stored either as sub-element of this element or as descendant of the sibling element `bio:Records`.
- The IUT selected the fingerprint capture attempts with the best QA-score (via `@selected`).
- The `rejectionReason` element matches the reason chosen by the operator
- The images contained in the response correspond to the fingers, that were captured.

Test Case ID: TC-PAP-ACQ-FP10R-SV-1-002

3

Description:

- Initiate test case by calling the IUT via the test interface.
- Perform the flat/control acquisition process of all fingers in order to obtain the necessary reference data for the control verification. Ensure sufficient quality and correct execution throughout the process. Do not induce any errors.
- Perform the rolled acquisition process of all fingers, beginning with the thumb of the right hand and ending with the little finger of the left hand. Enforce a hardware reported issue during the rolled capture process for at least one finger. Consult the documentation of the hardware in regards of supported issues (e.g. "finger shifted").
- If a hardware reported issue was induced for a capture: Repeat the capture of that finger enforcing bad quality.
- If bad quality was enforced during a capture: Repeat the capture of that finger inducing a hardware reported issue.
- For the remaining fingers: Capture this finger enforcing good quality and no hardware reported issues.
- Examine the response returned by the IUT via the test interface.

Expected Result:

- The acquisition of the reference slaps is performed by the IUT.
- The IUT successively performs the acquisition of the rolled fingerprints.
- If a hardware issue was induced: The IUT detects the hardware issue and restarts the acquisition of the current finger.
- If insufficient quality was enforced: The IUT detects insufficient quality and restarts the acquisition of the current finger.
- If a hardware issue was induced on the third capture of a finger: The IUT detects the hardware issue and offers to either restart the capture or to continue with the current result.
- The IUT finishes without any error message that is disrupting the flow of the PAP or that is unrelated to the induced hardware issue or insufficient quality.
- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- The response includes the information corresponding to the performed acquisition process (elements `bio:FingerAcquisition` and relevant sub-elements). The raw fingerprint data may be stored either as sub-element of this element or as descendant of the sibling element `bio:Records`.
- The first repetition with insufficient QA score is marked as `@selected`.
- The `@rejectionReason` element matches the reason chosen by the operator.
- The images contained in the response correspond to the fingers, that were captured.

Table 3.39 Test Case ID: TC-PAP-ACQ-FP10R-SV-1-002

3.11 Test Cases PAP ACQ-FPS-USV-1: Unsupervised Acquisition Single Slap

Test Case ID: TC-PAP-ACQ-FPS-USV-1-001

Scope

Verification the correct acquisition process for a single slap with four fingers and missing finger(s)

Test Case ID: TC-PAP-ACQ-FPS-USV-1-001*Preconditions*

- The IUT is in operation, required modules are loaded.
- The multi fingerprint scanner used in the specific application scenario is connected
- If HLBS is used by the system, the HLBS service definition for "Fingerprint Acquisition" specified in TR-03121-2.2 is implemented.
- Possibility to configure minimum required quality thresholds
- Number of retries (in case the wrong slap is acquired) is configured to zero.
- The following test resources are necessary:
 - One test subject is available to perform the acquisition with.
 - Product documentation of IUT is at hand (e.g. data sheet, manual).

CTS Mode

- interactive
- no provision of pre-defined input data
- HTTP method: GET
- test case variants:

/TR03122/TC-PAP-ACQ-FPS-USV-1-001/1

Table 3.42 TC-PAP-ACQ-FPS-USV-1-001 Test Case Variants*Description*

Step	Description / Expected Result
1	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • Present slap with four fingers (either left or right hand). • Follow the instructions of the IUT for the slap placement on the scanner for the biometric test subject. • Check the capturing of surveillance images, segmentation, quality assessment, PAD, uniqueness check result and time required for the slap image acquisition. • Examine the response returned by the IUT via the test interface. <i>Expected Result:</i> • The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • PAD has been performed for each single fingerprint image. • The PAD system didn't report an error. • The PAD system did report no warning of an attack. • No Uniqueness check error was reported. • The response contains the correct number of elements based on the XPath expression <code>//bio:FingerAcquisition/bio:Records/bio:BinaryRecord[@purpose="surveillance"]</code>. At least 2 elements are expected in accordance to the acquisition process. • QA values have been calculated for each single fingerprint image. • The response contains the correct number of elements based on the XPath expression <code>//bio:FingerAcquisition/bio:FingerClassifierInformation/bio:ProbeFingerClassifierInfo/@classificationResult</code>. At least 1 element is expected in accordance to the acquisition process. • The overall finger slap capture process was performed in under ten seconds. • The response contains the correct number of elements based on the XPath expression <code>//bio:FingerAcquisition/bio:Records/bio:BinaryRecord[@purpose!="surveillance"]</code>. At least 5 elements are expected in accordance to the acquisition process.

Test Case ID: TC-PAP-ACQ-FPS-USV-1-001

2

Description:

- Increase the minimum required quality thresholds for each fingerprint to 100.

Expected Result:

- The configuration is successful.

3

Description:

- Initiate test case by calling the IUT via the test interface.
- Present slap with four fingers (either right or left hand).
- Follow the instructions of the IUT for the slap placement on the scanner for the biometric test subject.
- Examine the response returned by the IUT via the test interface.

Expected Result:

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- PAD has been performed for each single fingerprint image.
- The PAD system didn't report an error.
- The PAD system did report no warning of an attack.
- No Uniqueness check error was reported.
- The response contains the correct number of elements based on the XPath expression `//bio:FingerAcquisition/bio:Records/bio:BinaryRecord[@purpose="surveillance"]`. At least 2 elements are expected in accordance to the acquisition process.
- QA values have been calculated for each single fingerprint image.
- The response contains the correct number of elements based on the XPath expression `//bio:FingerAcquisition/bio:FingerQuality/bio:OverallQualityValue`. At least 1 element is expected in accordance to the acquisition process.
- The overall QA result in `"/bio:FingerAcquisition/bio:FingerQuality/bio:OverallQualityValue"` indicates that the quality of the captured fingerprints was below the configured threshold.
- The response contains the correct number of elements based on the XPath expression `//bio:FingerAcquisition/bio:FingerClassifierInformation/bio:ProbeFingerClassifierInfo/@classificationResult`. At least 1 element is expected in accordance to the acquisition process.
- The capturing process is restarted until the configured timeout of 10 seconds is reached.
- The response contains the correct number of elements based on the XPath expression `//bio:FingerAcquisition/bio:Records/bio:BinaryRecord[@purpose!="surveillance"]`. At least 5 elements are expected in accordance to the acquisition process.
- After the timeout is reached the captured slap with best quality is identified. Evidence may be collected from the following sources: Developer-Documentation.

4

Description:

- Initiate test case by calling the IUT via the test interface.
- Present slap with at least one finger missing (either left or right hand).
- Follow the instructions of the IUT for the slap placement on the scanner for the biometric test subject.
- Examine the response returned by the IUT via the test interface.

Test Case ID: TC-PAP-ACQ-FPS-USV-1-001

	<i>Expected Result:</i> • The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • The response contains the correct number of elements based on the XPath expression //bio:FingerAcquisition/bio:Records. No elements are expected to appear in accordance to the acquisition process. • The process ended without capturing the slap.
5	<i>Description:</i> • Increase the number of retries for each slap to a number larger than zero. <i>Expected Result:</i> • The configuration is successful.
6	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • Present wrong slap (i.e. left hand slap instead of right hand slap) with four fingers, follow the instructions of the IUT for the slap placement on the scanner for the biometric test subject. • Repeat presenting the wrong slap until process ends. • Examine the response returned by the IUT via the test interface. <i>Expected Result:</i> • The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • For every retry a message is shown, asking the biometric test subject to present the correct slap. (I.e. if the right hand slap is presented, the message asks for the left hand slap.) • A warning is sent to the operator, that the slap classifier detected the presentation of the wrong slap. • The classification result indicates that acquired slap mismatches the requested slap.
7	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • Follow the instructions of the IUT for the placement of the slap on the scanner for the biometric test subject. • Present the same slap a second time, following the instructions of the IUT for the placement of the slap on the scanner for the biometric test subject. • Present the same slap a third time, following the instructions of the IUT for the placement of the slap on the scanner for the biometric test subject. • Examine the response returned by the IUT via the test interface. <i>Expected Result:</i> • The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • After the second presentation of the same slap, a warning message about the detected sequence error (e.g. "This slap has already been captured.") is shown to the biometric subject. • After the third presentation of the same slap, the process ends without capturing a slap. • The response contains the correct number of elements based on the XPath expression //bio:FingerAcquisition/bio:Records. No elements are expected to appear in accordance to the acquisition process.

Table 3.41 Test Case ID: TC-PAP-ACQ-FPS-USV-1-001

3.12 Test Cases PAP ACQ-FP2P-USV-1: Unsupervised Acquisition of Two Plain Fingerprints on Multi-Finger Hardware for Enrolment

Test Case ID: TC-PAP-ACQ-FP2P-USV-1-001

Scope	Verify the correct acquisition process for a slap consisting of two fingers: Acquisitions are made compliantly, with qa-thresholds set too high, no fingers presented at all and an induced sequence error.
Preconditions	• The IUT is in operation, required modules are loaded. • The multi fingerprint scanner used in the specific application scenario is connected • If HLBS is used by the system, the HLBS service definition for "Fingerprint Acquisition" specified in TR-03121-2.2 is implemented. • Possibility to configure minimum required quality thresholds • The following test resources are necessary: • One test subject is available to perform the acquisition with. • Product documentation of IUT is at hand (e.g. data sheet, manual).
CTS Mode	• interactive • no provision of pre-defined input data • HTTP method: GET • test case variants:

/TR03122/TC-PAP-ACQ-FP2P-USV-1-001/1

Table 3.44 TC-PAP-ACQ-FP2P-USV-1-001 Test Case Variants

Description	Step	Description / Expected Result
	1	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • Present two index fingers simultaneously i.e. as a slap. • Follow the instructions of the IUT for the biometric test subject regarding slap placement on the scanner. • Examine the response returned by the IUT via the test interface. <i>Expected Result:</i> • The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • The captured slap image is segmented into two single fingerprint images. • PAD has been performed for both fingerprint images. • The PAD system didn't report an error. • The PAD system did report no warning of an attack. • No Uniqueness check error or warning was reported. • QA values have been calculated for both fingerprint images individually. • The overall finger capture process was performed in under ten seconds. • The response contains two encoded fingerprint images for the index fingers. • The images contained in the response correspond to the fingers, that were captured.
	2	<i>Description:</i> • Increase the minimum required quality thresholds for each fingerprint to 100.

Test Case ID: TC-PAP-ACQ-FP2P-USV-1-001

3	<i>Expected Result:</i> • The configuration is successful. <i>Description:</i> • Initiate test case by calling the IUT via the test interface. • Present two index fingers simultaneously i.e. as a slap. • Follow the instructions of the IUT for the biometric test subject regarding slap placement on the scanner. • Examine the response returned by the IUT via the test interface. <i>Expected Result:</i> • The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • The captured slap image is segmented into two single fingerprint images. • PAD has been performed for both fingerprint images. • The PAD system didn't report an error. • The PAD system did report no warning of an attack. • No Uniqueness check error or warning was reported. • QA values have been calculated for both fingerprint images. • The individual QA results and (if it exists) the overall QA result indicate that the quality of the captured fingerprints was below the configured threshold. • The capturing process is restarted until the configured timeout of 10 seconds is reached. • The response contains two encoded fingerprint images for the index fingers. • After the timeout is reached the captured fingerprint with best quality is identified. Evidence may be collected from the following sources: Developer-Documentation. • Check that the images contained in the response correspond to the fingers, that were captured.
4	<i>Description:</i> • Reset the minimum required quality thresholds to the default values. <i>Expected Result:</i> • The configuration is successful.
5	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • Present a index finger slap with at least one finger missing. <i>Expected Result:</i> • The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • Acquired finger images (and associated surveillance images) are discarded. • The process ended without capturing any fingerprint.

Table 3.43 Test Case ID: TC-PAP-ACQ-FP2P-USV-1-001

3.13 Test Cases PAP ACQ-FP2P-USV-2: Unsupervised Acquisition of Two Plain Fingerprints on Single-Finger Hardware for Enrolment

Test Case ID: TC-PAP-ACQ-FP2P-USV-2-001

Scope	Verify the correct acquisition process for two fingers: Acquisitions are made compliantly, with qa-thresholds set too high, no fingers presented at all and an induced sequence error.
Preconditions	• The IUT is in operation, required modules are loaded. • Hardware capable of capturing fingerprint images is connected. • If HLBS is used by the system, the HLBS service definition for "Fingerprint Acquisition" specified in TR-03121-2.2 is implemented. • Possibility to configure minimum required quality thresholds • The following test resources are necessary: • A test person to perform the biometric acquisition with. • Product documentation of IUT is at hand (e.g. data sheet, manual).
CTS Mode	• interactive • no provision of pre-defined input data • HTTP method: GET • test case variants:

/TR03122/TC-PAP-ACQ-FP2P-USV-2-001/1

Table 3.46 TC-PAP-ACQ-FP2P-USV-2-001 Test Case Variants

Description	Step	Description / Expected Result
	1	<i>Description:</i> • Initiate test case by calling the IUT via the test interface. • Present two index fingers subsequently. • Follow the instructions of the IUT for the biometric test subject regarding single finger placement on the scanner. • Check the quality assessment, PAD, uniqueness check result and time required for the fingerprint image acquisition. <i>Expected Result:</i> • The IUT delivers a response that is conformant to the schema file trbio6v1.xsd. • PAD has been performed for both fingerprint images. • The PAD system didn't report an error. • The PAD system did report no warning of an attack. • No Uniqueness check error was reported. • QA values have been calculated for both fingerprint images. • The overall finger capture process was performed in under ten seconds. • The response contains two encoded fingerprint images for the index fingers.
	2	<i>Description:</i> • Increase the minimum required quality thresholds for each fingerprint to 100. <i>Expected Result:</i> • The configuration is successful.

Test Case ID: TC-PAP-ACQ-FP2P-USV-2-001

3

Description:

- Initiate test case by calling the IUT via the test interface.
- Present two index fingers subsequently.
- Follow the instructions of the IUT for the biometric test subject regarding single finger placement on the scanner.
- Check the quality assessment, PAD, uniqueness check result and time required for the fingerprint image acquisition.

Expected Result:

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- PAD has been performed for both fingerprint images.
- The PAD system didn't report an error.
- The PAD system did report no warning of an attack.
- No Uniqueness check error was reported.
- QA values have been calculated for both fingerprint images.
- The overall QA result exists.
- The overall QA result in /bio:FingerAcquisition/bio:FingerQuality/bio:OverallQualityValue indicates that the quality of the captured fingerprints was below the configured threshold.
- The capturing process is restarted until the configured timeout of 10 seconds is reached.
- The response contains two encoded fingerprint images for the index fingers.
- After the timeout is reached the captured fingerprint with best quality is identified. Evidence may be collected from the following sources: Developer-Documentation.

4

Description:

- Initiate test case by calling the IUT via the test interface.
- Present no finger at all.

Expected Result:

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- Acquired finger images and associated surveillance images are discarded.
- The process ended without capturing any fingerprint.

5

Description:

- Initiate test case by calling the IUT via the test interface.
- Present two index fingers subsequently.
- Follow the instructions of the IUT for the biometric test subject regarding single finger placement on the scanner.
- Present the same finger a second time, following the instructions of the IUT for the biometric test subject regarding single finger placement on the scanner.
- Present the same finger a third time, following the the instructions of the IUT for the biometric test subject regarding single finger placement on the scanner.

Test Case ID: TC-PAP-ACQ-FP2P-USV-2-001*Expected Result:*

- The IUT delivers a response that is conformant to the schema file trbio6v1.xsd.
- After the second presentation of the same finger, a warning message about the detected sequence error (e.g. "This finger has already been captured.") is shown to the biometric subject.
- After the third presentation of the same finger, the process ends without capturing a finger.
- Acquired finger images and associated surveillance images are discarded.

Table 3.45 Test Case ID: TC-PAP-ACQ-FP2P-USV-2-001

List of Abbreviations

Abbreviation	Description
BDB	Biometric Data Block
BHT	Biometric Header Template
CBEFF	Common Biometric Exchange Formats Framework
CTF	contrast transfer function
FMR	false-match-rate
FNMR	false-non-match-rate
GUI	graphical user interface
IUT	Implementation Under Test
PAD	Presentation Attack Detection
QA	Quality Assessment
SNR	signal-to-noise ratio
WSQ	Wavelet Scalar Quantisation

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